

Musculoskeletal

and

Integumentary system module!

- Introduction to the musculoskeletal system

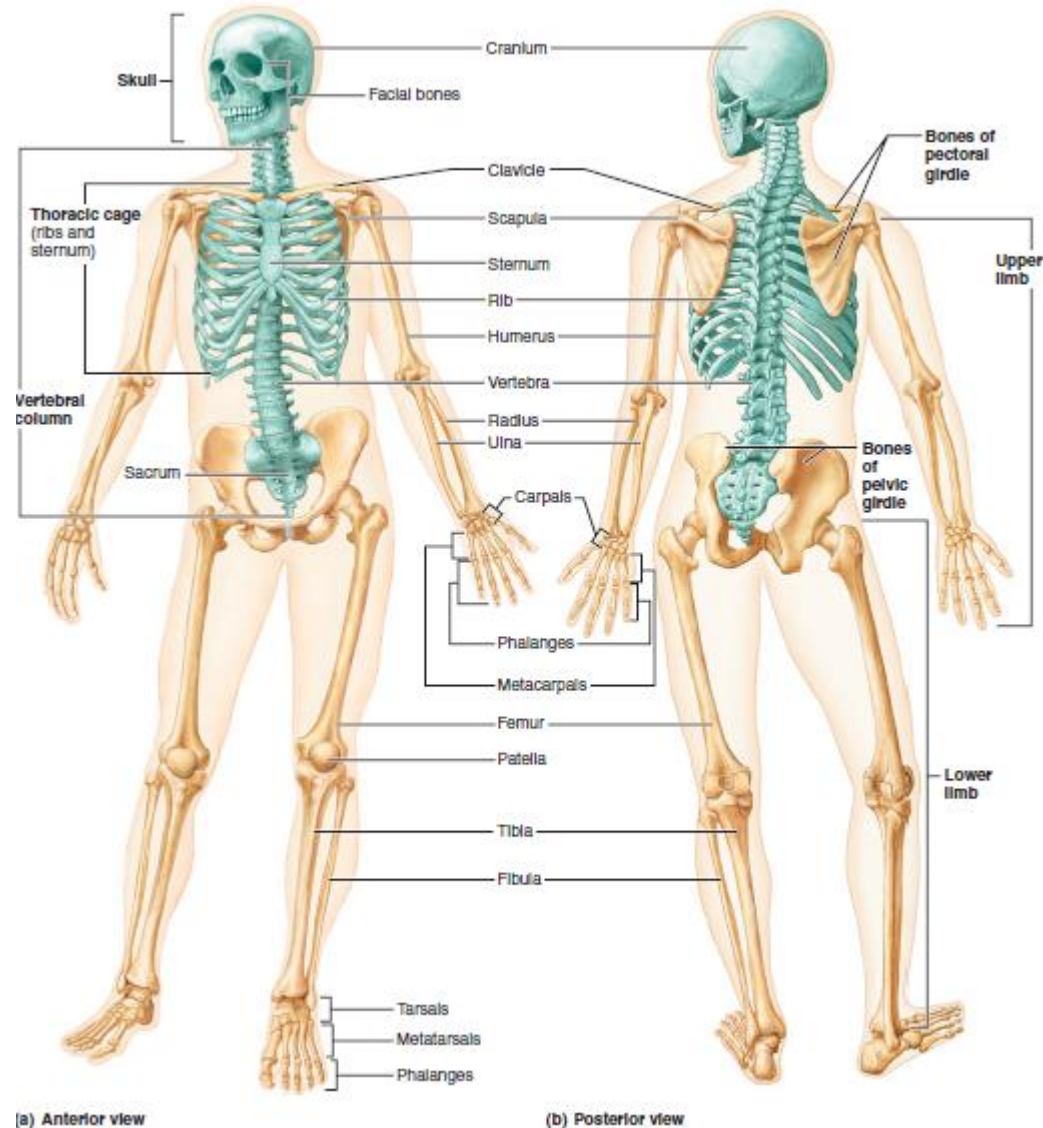
❖ **Anatomy of upper limb**

- ✓ Introduction to the upper limb
- ✓ Bones of the upper limb
- ✓ Superficial structures of upper limb
- ✓ Cubital fossa
- Axio - appendicular Muscles
- Axilla
- Brachial plexus
- Arm
- Forearm
- Hand
- Surface anatomy the upper limb
- Joints of the upper limb

❑ **Anatomy of Lower limb**

- ✓ Bones of the lower limb
- ✓ Superficial structures of the lower limb
- ✓ Thigh and gluteal region
- ✓ Lumbar and Sacral plexuses
- ✓ Foot
- ✓ Joints of lower limb
- Development of the skeletal system
- Development of the muscular system
- Development of limbs
- Gross structure and functions of the skin
- Development of the Integumentary system

Musculoskeletal and Integumentary system module!



Skeletal muscle

Temporalis

Masseter

Sternocleidomastoid

Pectoralis major

Biceps brachii

Abdominal muscles

Sartorius

Quadriceps femoris

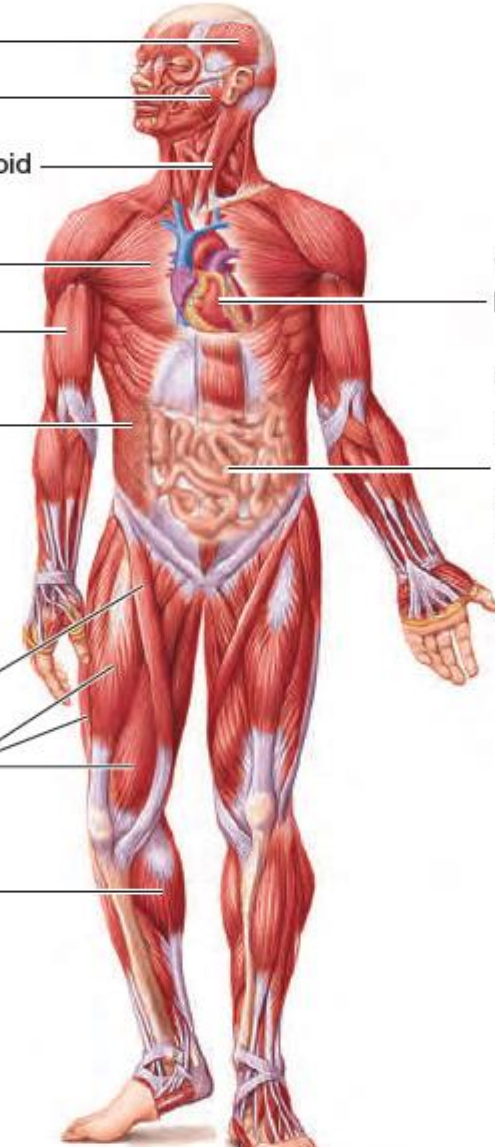
Gastrocnemius

Cardiac muscle

Heart

Smooth muscle

Muscle of the intestines and other internal organs and blood vessels



Introduction to the musculoskeletal system

- The musculoskeletal system (locomotor system) is a human body system that provides our body with movement, stability, shape, and support.
- It is subdivided into two broad systems:
 - ✓ **Muscular system**, which includes all types of voluntary muscles of body.
 - ✓ **Skeletal system**, whose main component is the bone.
- *The branch of medical* science concerned with the prevention or correction of disorders of the musculoskeletal system is called **orthopedics**.

Introduction to the musculoskeletal system,,,

- The muscular system is an organ system composed of specialized contractile tissue called the muscle tissue.

□ There are three types of muscle tissue:

- I. Cardiac muscle, which forms the muscular layer of the heart (myocardium).
 - II. Smooth muscle, which comprises the walls of blood vessels and hollow organs.
 - III. Skeletal muscle, which attaches to the bones and provides voluntary movement.
- Based on their histological appearance, these types are classified into striated and non-striated muscles; with the skeletal and cardiac muscles being grouped as striated, while the smooth muscle is non-striated..

Introduction to the musculoskeletal system,,

- **The skeletal system** is composed of the bones, cartilages, and joints that form the internal framework of the body.
- **Although the term *skeleton* is derived from a Greek word meaning dried, but the skeletal system consists of dynamic, living tissues that are able to grow, detect pain stimuli, adapt to stress, and undergo repair after injury.**
- ❖ **Bones provide support and shape** to the body, attachment sites for muscle, and a storage depot for essential minerals.

Introduction to the musculoskeletal system,,

- Knowledge of skeletal anatomy provides a point of reference for studying the gross anatomy of other organ systems because many organs are named for their relationships to nearby bones..
- Many organs are named for their relationships to nearby bones, example **subclavian vessels, temporalis muscle, the ulnar nerve and radial artery, and the frontal, parietal, temporal, and occipital lobes of the brain.**
- Understanding how the muscles produce body movements also depends on knowledge of skeletal anatomy.

Introduction to the musculoskeletal system,,,

The **Skeletal System** has many important functions:

- Providing support for the body
- Storing minerals (calcium, phosphate)
- Producing red blood cells
- Protecting the organs and tissues
- Allowing movement (the bones act as levers)

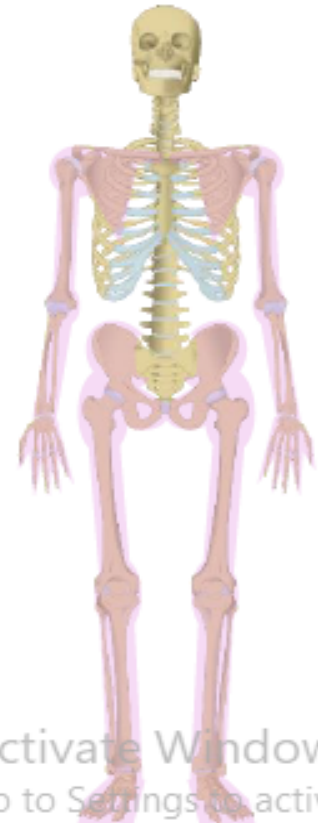
The skeleton can be subcategorized into two divisions:

The ***Axial Skeleton*** (left, in blue)

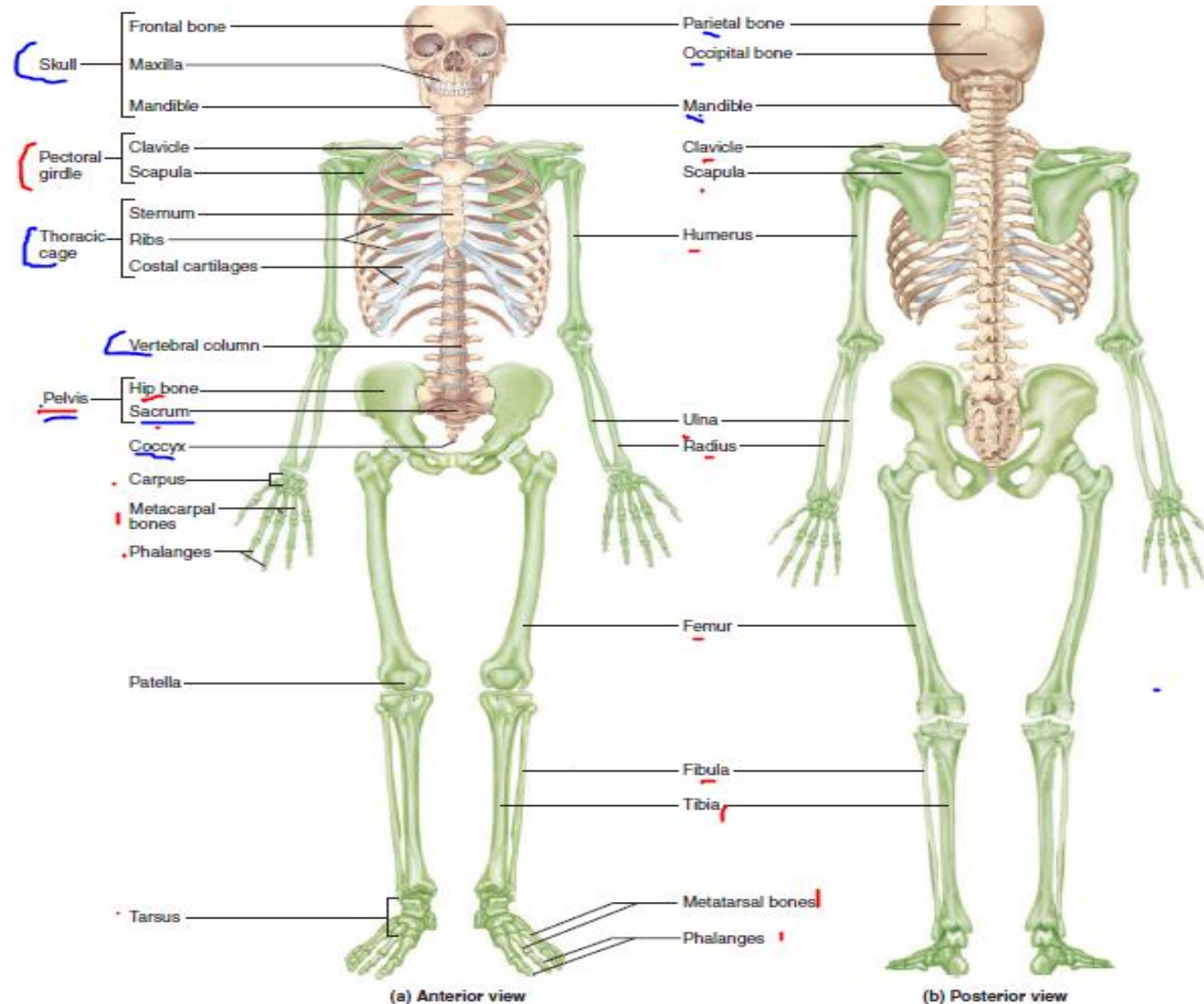
Includes: Bones of the skull, vertebrae, sternum, ribs, and sacrum

The ***Appendicular Skeleton*** (right, in pink)

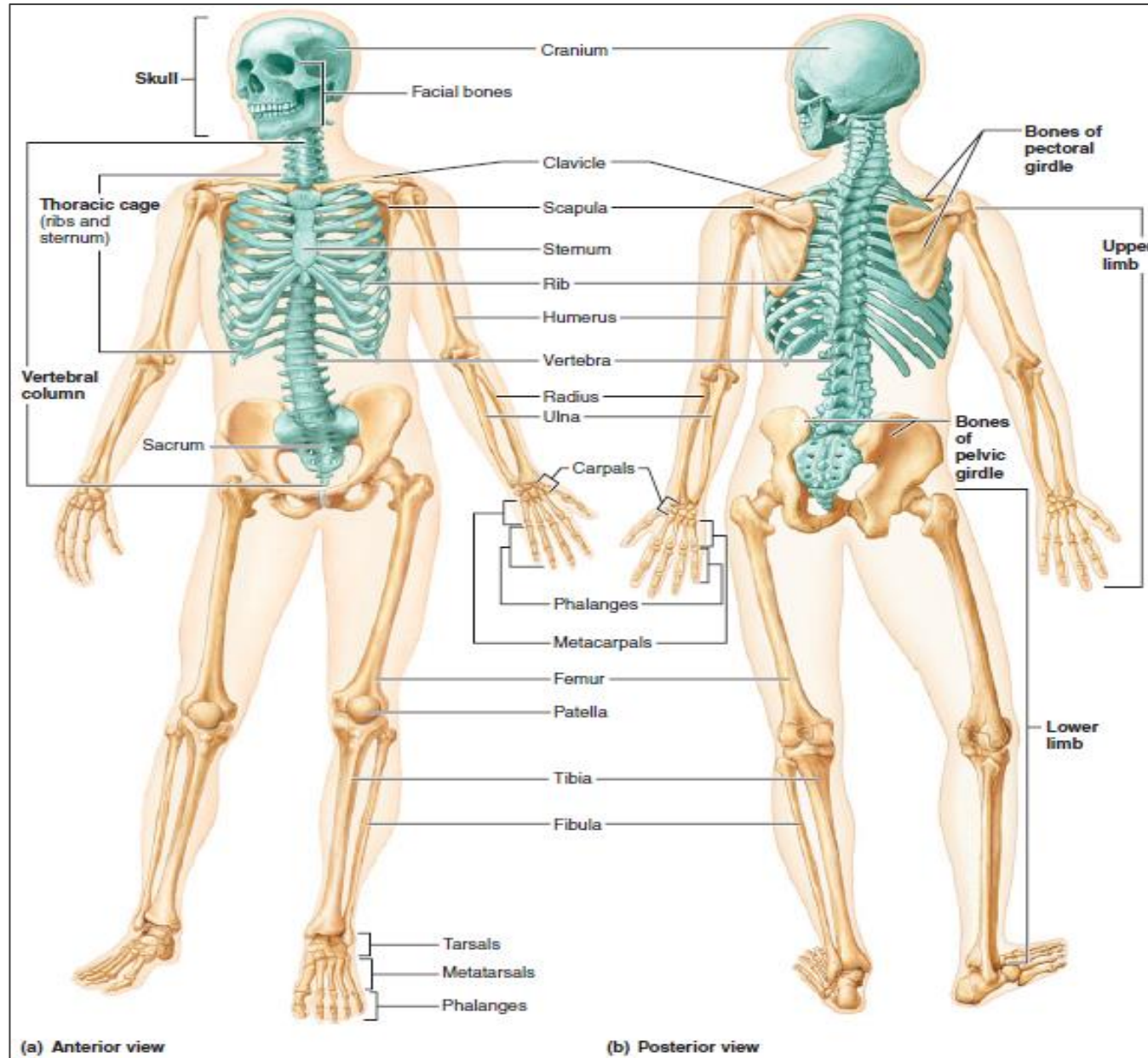
Includes: Bones of the upper and lower limbs, scapula, clavicles, and hip bones,



Introduction to the musculoskeletal system,,,



Introduction to the musculoskeletal system,,,



❑ The human skeleton.

- ✓ Bones of the axial skeleton are colored **green**.
- ✓ Bones of the appendicular skeleton **are tan**.

Introduction to the musculoskeletal system,,,

- The adult human skeleton consists of 206 named bones, most of which are paired, with one member of each pair on the right and left sides of the body.
- ❖ **The skeletons of infants and children** have more than 206 bones because some of their bones (sacrum and coccyx) fuse later in life.
- **So it is traditional to list 206 bones in the average adult skeleton, although the actual number varies from person to person and decreases with age as some bones fuse.**
- Bones of the adult skeleton are grouped into **two principal divisions:**
 - ❖ The **axial skeleton** and the **appendicular skeleton** presents the 80 bones of the axial skeleton and the 126 bones of the appendicular skeleton.

Introduction to the musculoskeletal system...

The Bones of the Adult Skeletal System

DIVISION OF THE SKELETON	STRUCTURE	NUMBER OF BONES
--------------------------	-----------	-----------------

Axial skeleton



Skull	
Cranium	8
Face	14
Hyoid	1
Auditory ossicles	6
Vertebral column	26
Thorax	
Sternum	1
Ribs	24

Number of bones = 80

DIVISION OF THE SKELETON	STRUCTURE	NUMBER OF BONES
--------------------------	-----------	-----------------

Appendicular skeleton



Pectoral (shoulder) girdles	
Clavicle	2
Scapula	2
Upper limbs	
Humerus	2
Ulna	2
Radius	2
Carpals	16
Metacarpals	10
Phalanges	28
Pelvic (hip) girdle	
Hip, pelvic, or coxal bone	2
Lower limbs	
Femur	2
Patella	2
Fibula	2
Tibia	2
Tarsals	14
Metatarsals	10
Phalanges	28

Number of bones = 126

Total bones in an adult skeleton = 206

Introduction to the upper limb

❑ Upper limb consists of 4 segments

✚ Shoulder (pectoral girdle)

✚ Arm

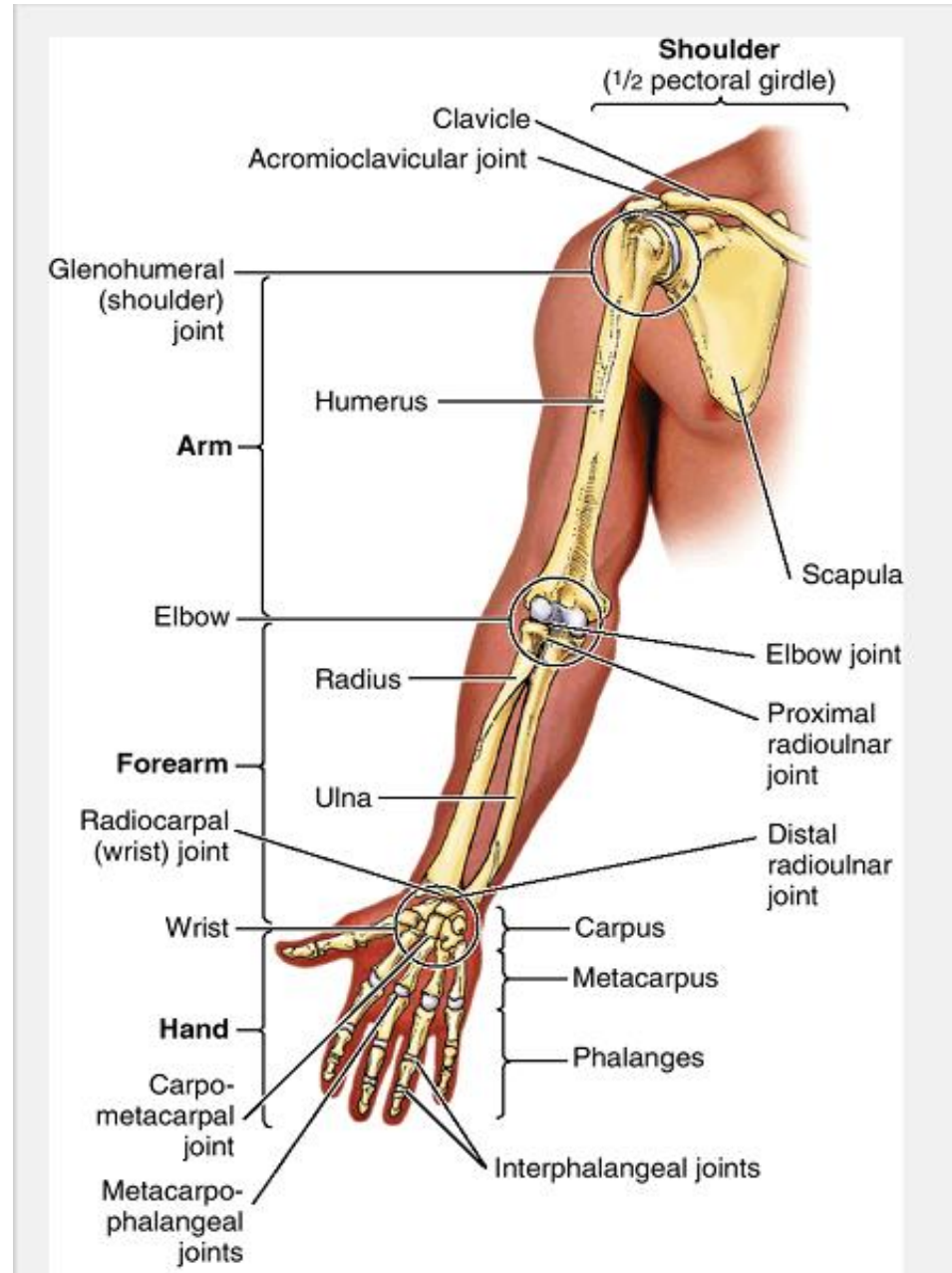
✚ Forearm

✚ Hand

❖ The upper limb is characterized by its mobility and ability to grasp, strike, and conduct fine motor skills (*manipulation*).

❑ The **pectoral girdle** (shoulder girdle) supports the arm and links it to the axial skeleton.

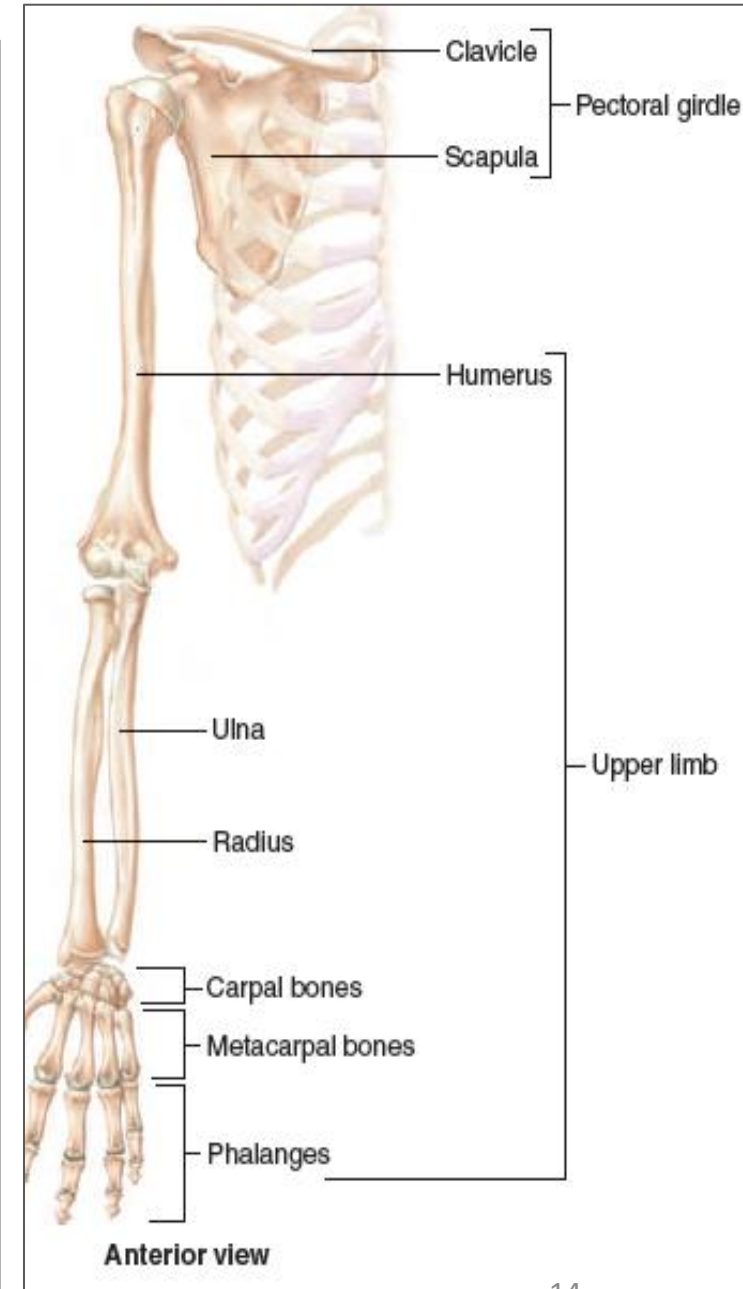
❑ It consists of 2 bones on each side of the body: **the clavicle & scapula**.



Bones of the upper limb

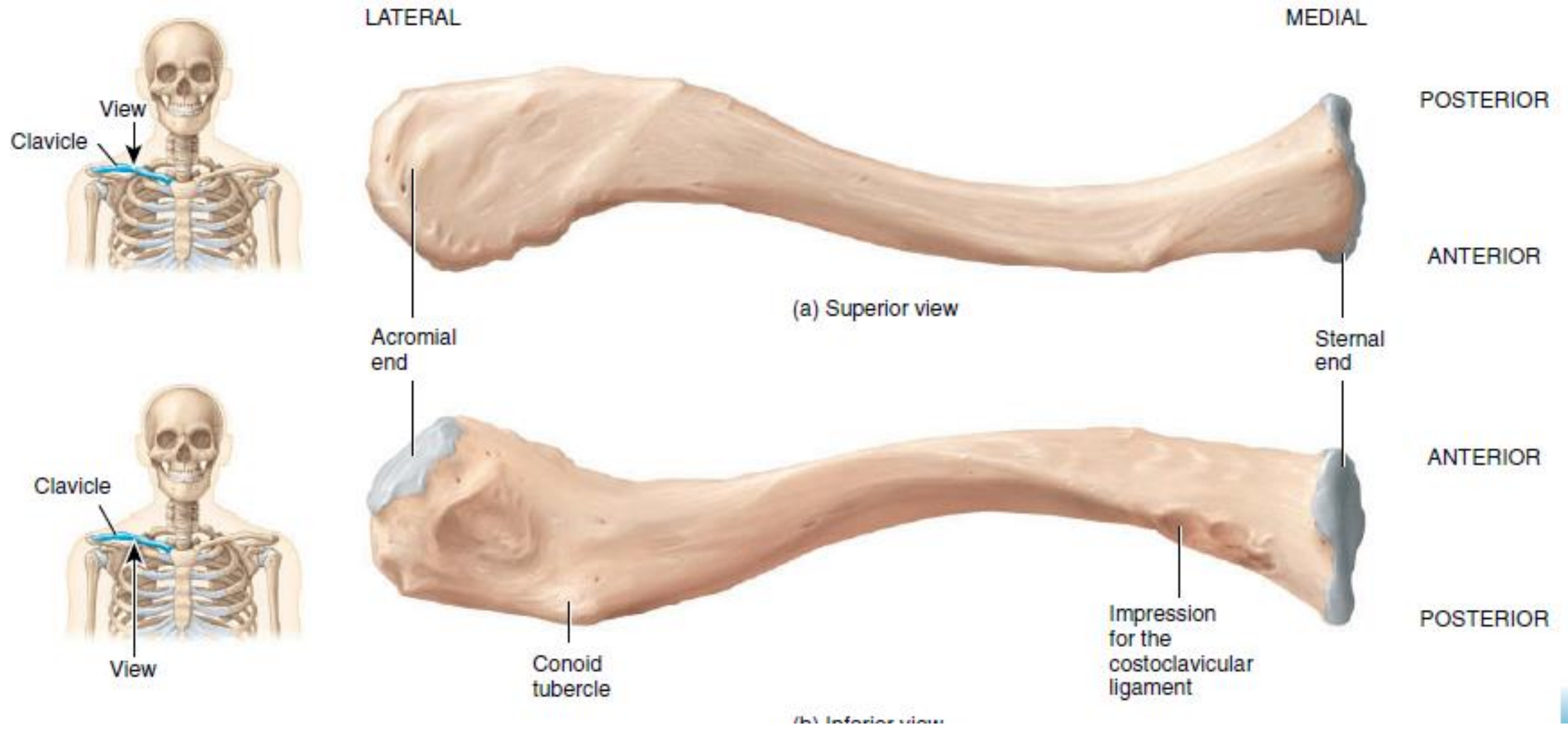
❑ The skeleton of each upper limb is composed of:

- ✓ clavicle and a scapula in the **pectoral** or **shoulder girdles**
- ✓ humerus in the arm;
- ✓ ulna and radius in the forearm;
- ✓ 8 carpals in the carpus (wrist),
- ✓ 5 metacarpals in the metacarpus (palm), and
- ✓ 14 phalanges (bones of the digits) in the hand.



Clavicle (collar bone)

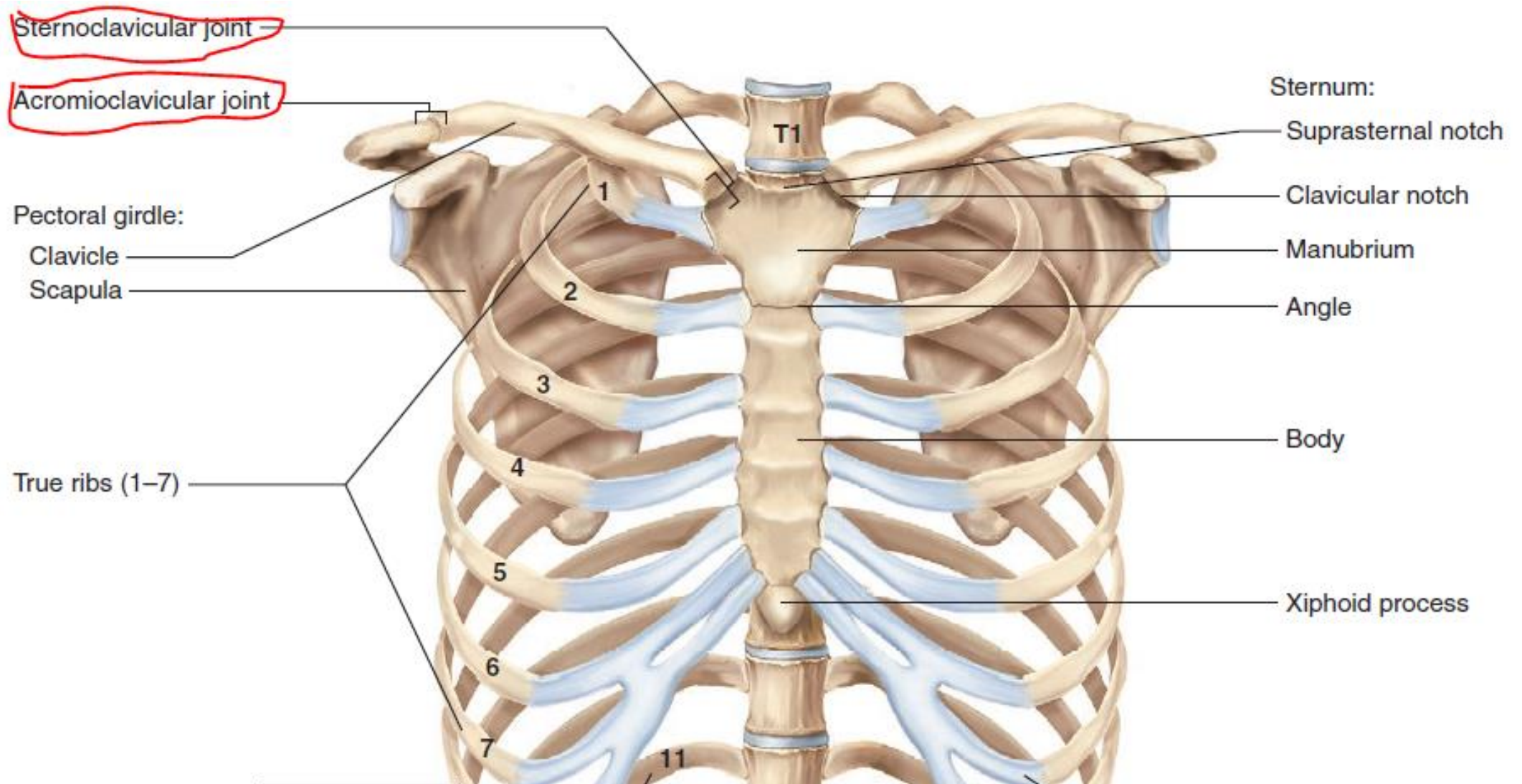
- Each slender, S-shaped **clavicle**(*collarbone*), lies horizontally across the anterior part of the thorax superior to the first rib



Clavicle (collar bone),,,

- The medial two thirds of the shaft of the clavicle are convex anteriorly, whereas the lateral third is flattened and concave anteriorly.
- **sternal end:** The medial end, called the *sternal end*, is rounded and articulates with the manubrium of the sternum to form the ***sternoclavicular joint***.
- **acromial end:** Is broad, flat, articulates with the acromion of the scapula to form the *acromioclavicular (AC) joint*
- **The *conoid tubercle*** on the inferior surface of the lateral end of the bone is a point of attachment for the conoid ligament, which attaches the clavicle and scapula.

Clavicle (collar bone),,,

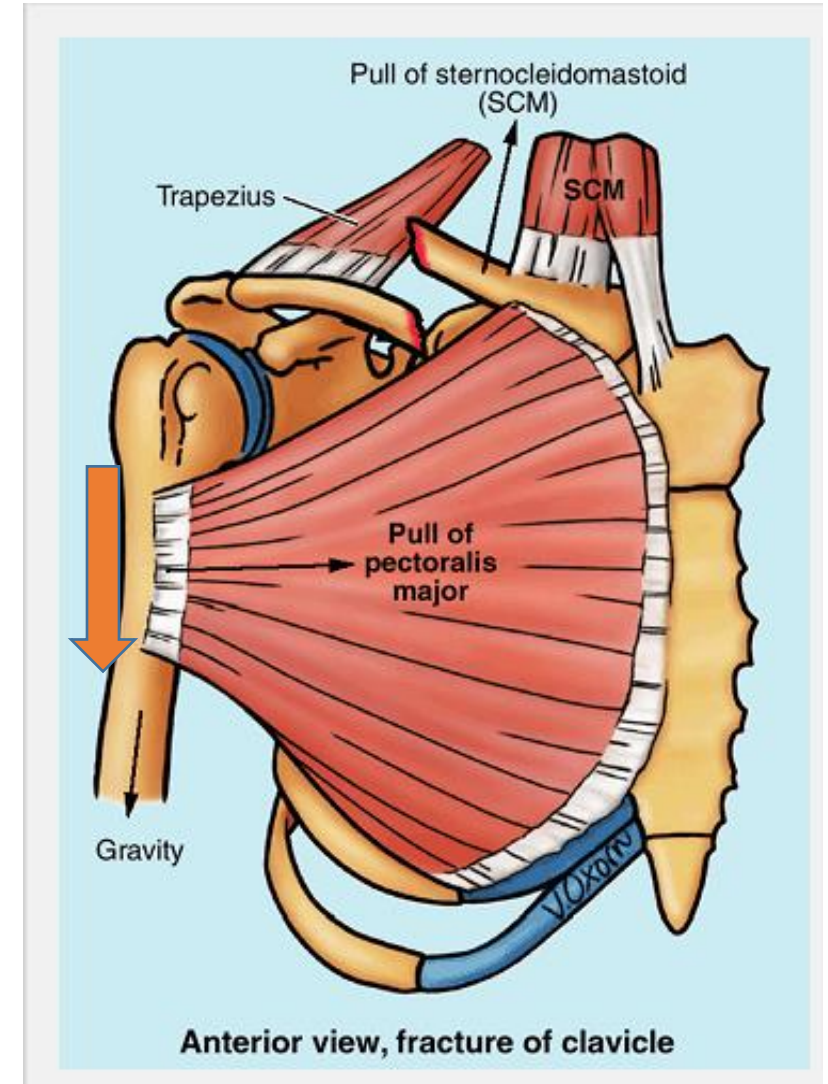


Fracture of clavicle

- The weakest part is at the junction of its middle and lateral thirds.

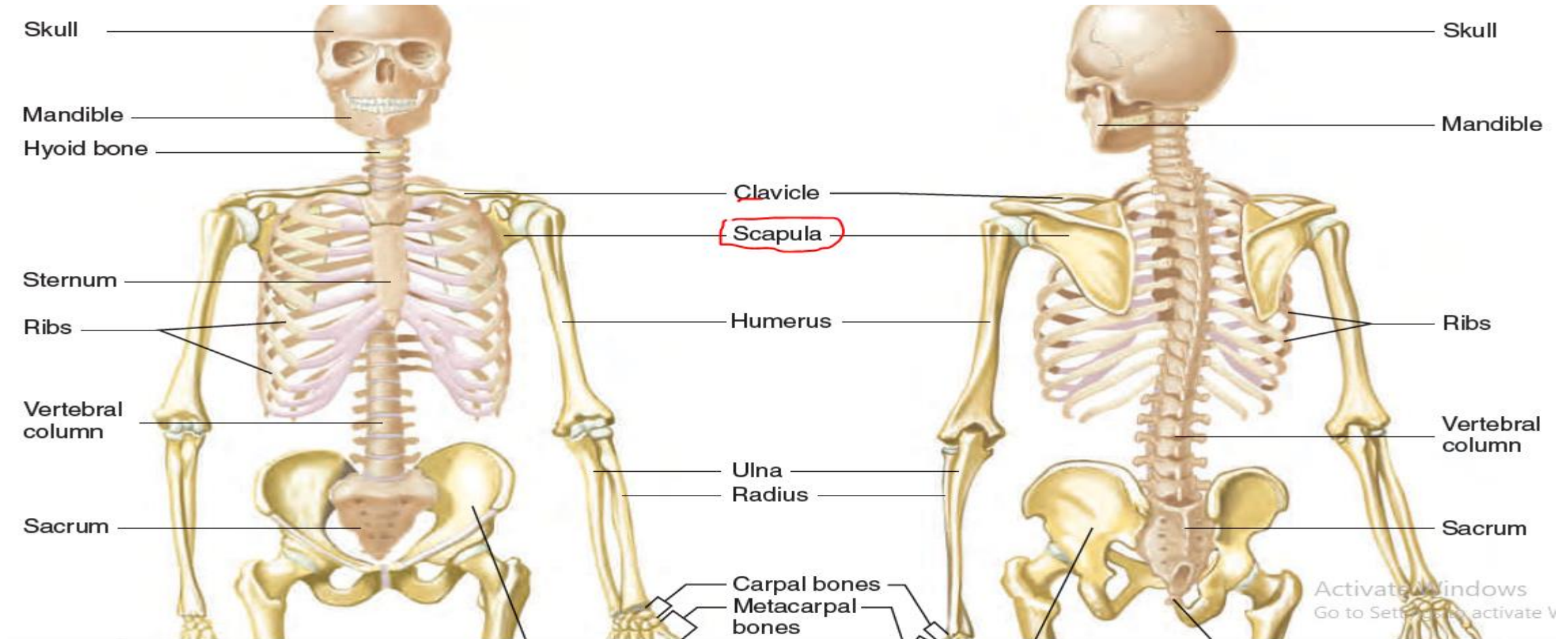
□ After fracture:

- sternocleidomastoid (SCM) muscle elevates the medial fragment of bone
- trapezius muscle is unable to hold up the lateral fragment owing to the weight of the upper limb results the **shoulder drops**.



Scapula (shoulder blade)

❑ **Scapula** is triangular flat bone lies on the posterolateral aspect of the thorax, overlying the **2nd-7th** ribs.



Scapula (shoulder blade)

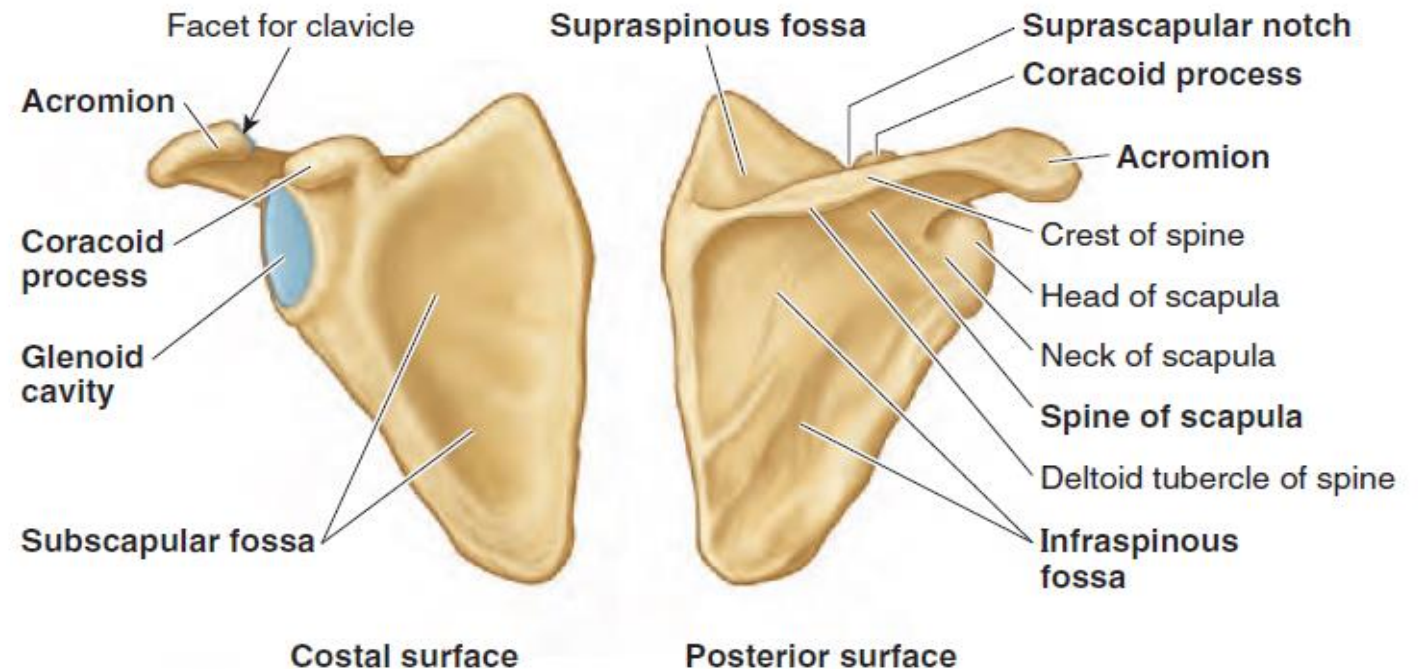
❑ The scapula has two surfaces:

❑ The convex **posterior surface** is divided by the **spine of the scapula**, into a small **supraspinous** and a much larger **infraspinous fossa**.

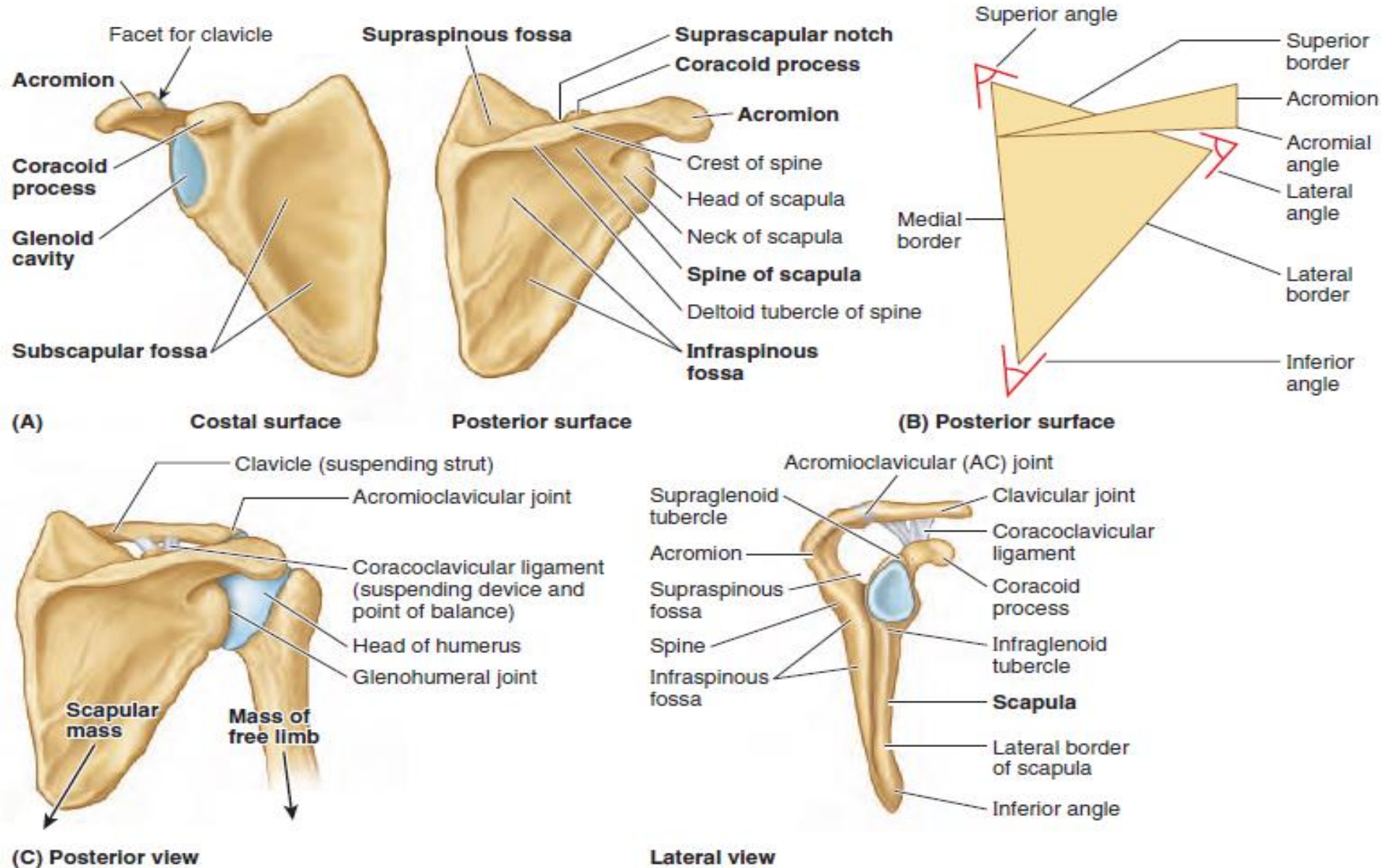
❑ The concave **costal surface** forms a large **subscapular fossa**.

✓ The scapula has 3 borders, superior, medial (vertebral), and lateral (axillary) borders, and it has 3 angles are the **superior, inferior, and lateral angles**.

❖ The spine divided posterior surface into **supraspinous fossa**, and **infraspinous fossa**.



Scapula (shoulder blade),,,

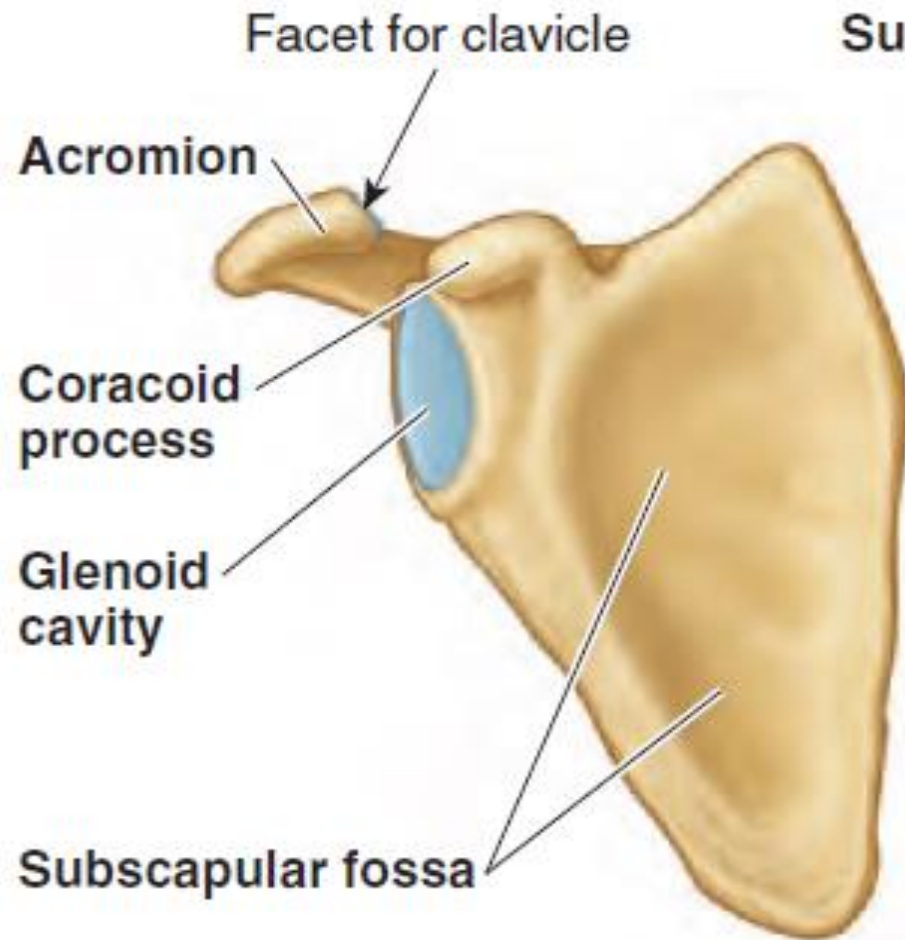


Scapula (shoulder blade),,,

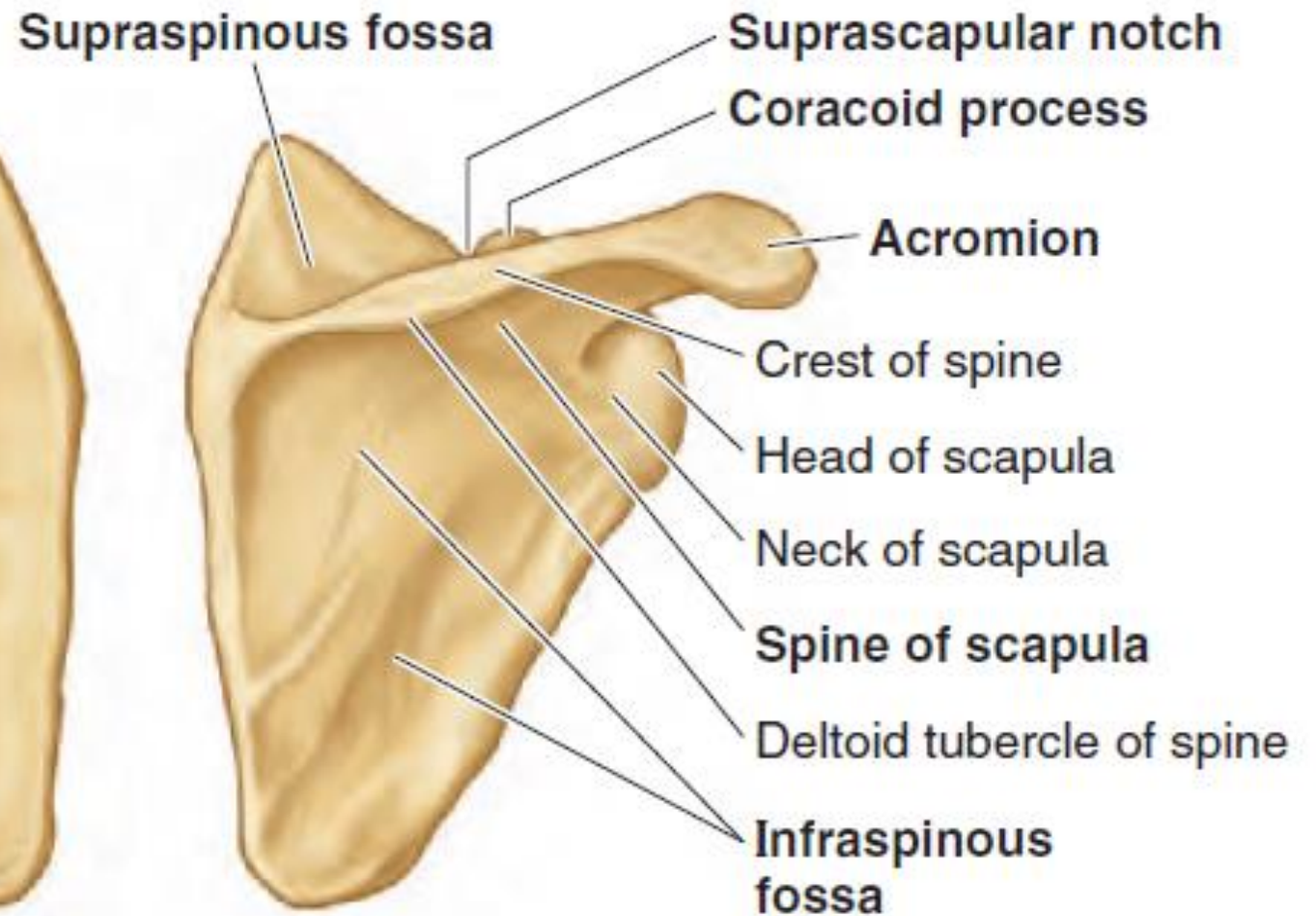
❑ The most complex region of the scapula is its lateral angle, which has three main features:

- ✓ The **acromion** is a platelike extension of the scapula which articulates with the **clavicle**, which forms the sole bridge from the appendicular to the axial skeleton.
- ✓ The **coracoid process** is shaped like a bent finger provides attachment for tendons of the *biceps brachii* and other muscles of the arm.
- ✓ The **glenoid cavity** is a shallow socket that articulates with the head of the humerus, forming the glenohumeral joint.

Scapula (shoulder blade),..



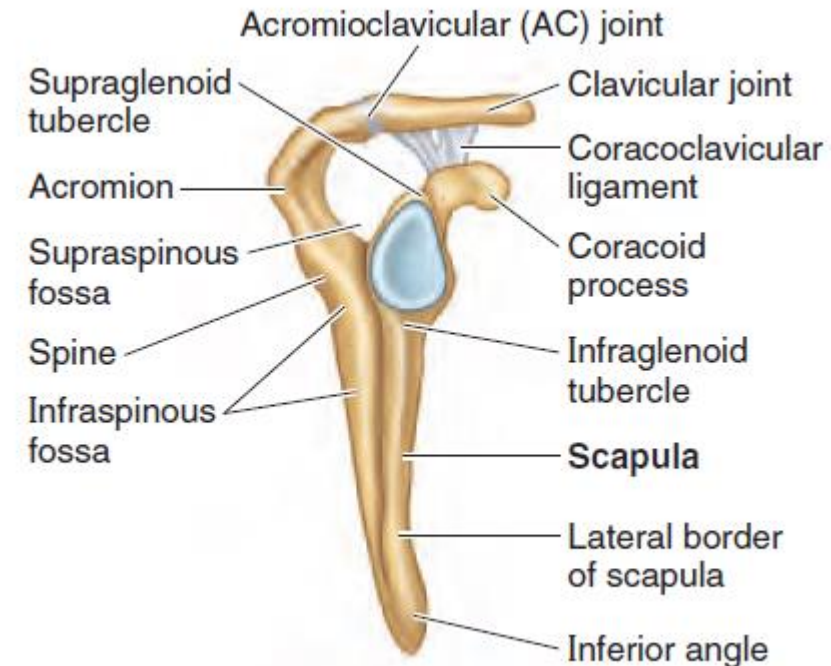
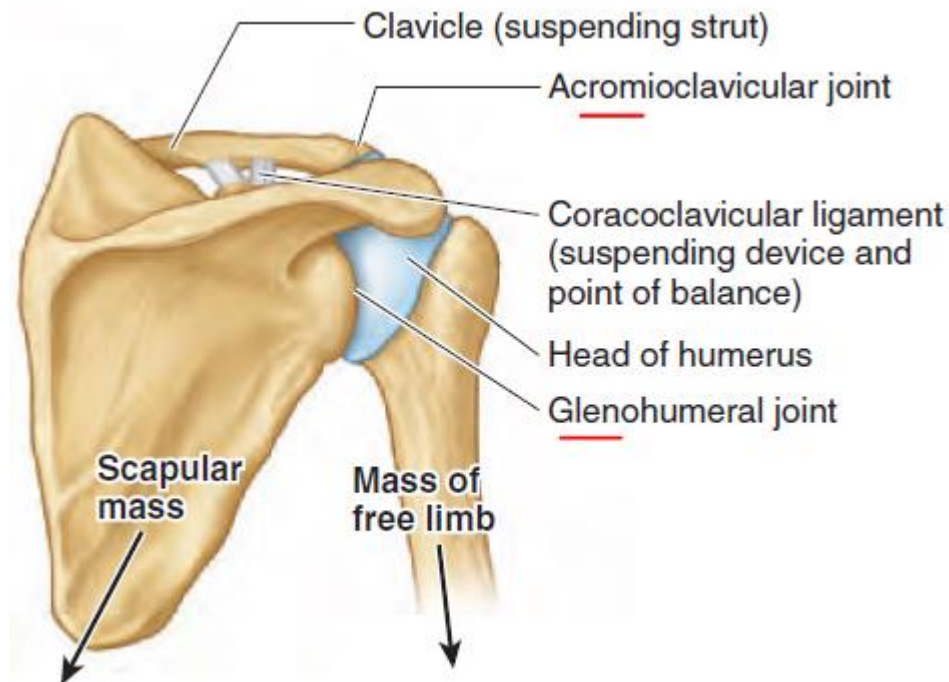
Costal surface



Posterior surface

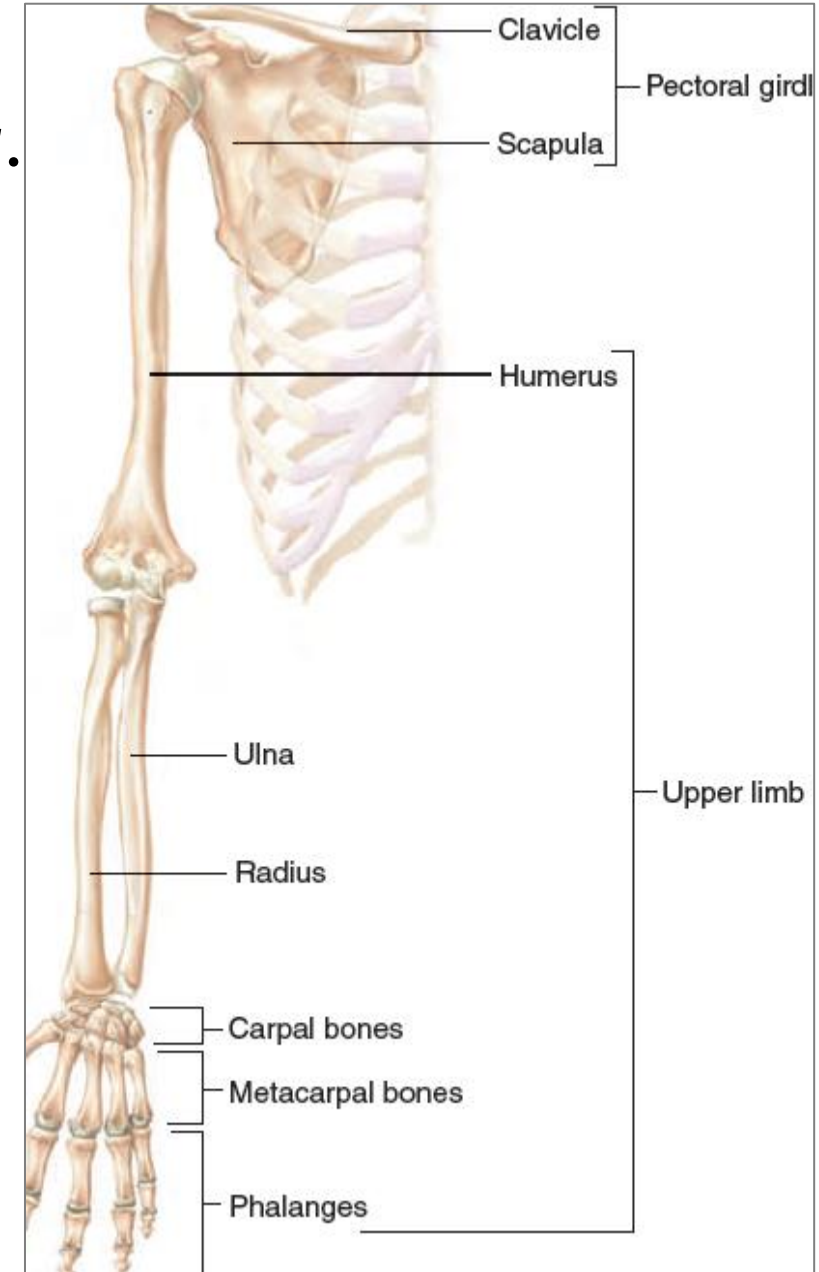
Scapula (shoulder blade),,,

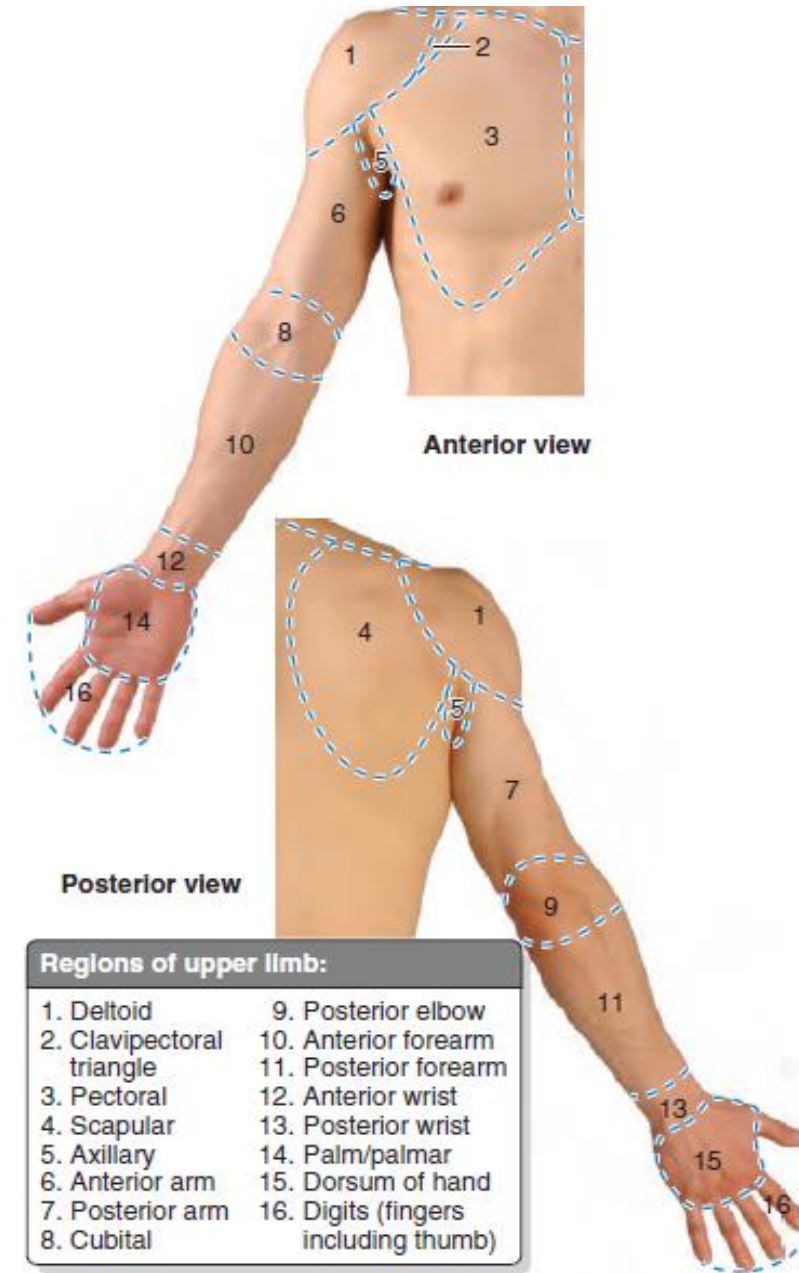
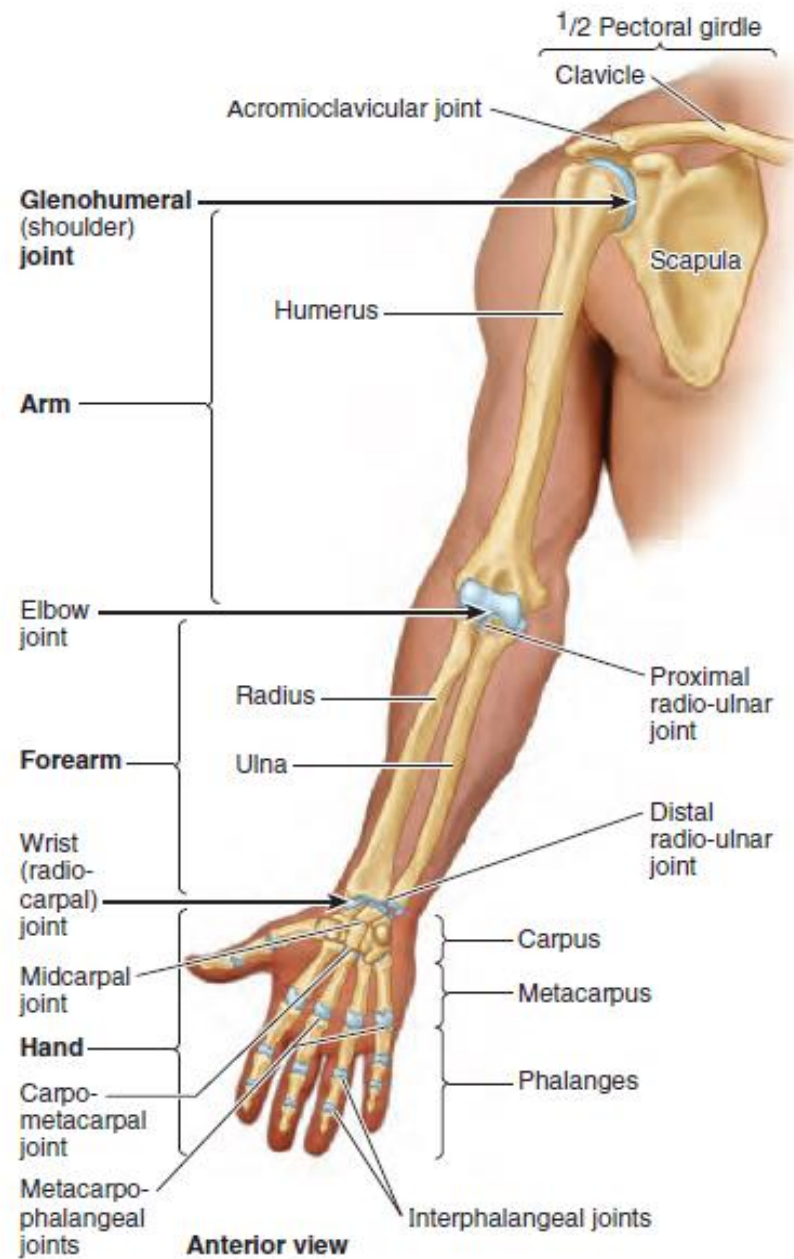
- The two major joints are on the lateral sides of the scapula, **glenoid cavity**, which articulates with the head of the arm bone (humerus) to form the ***glenohumeral (shoulder) joint*** and the acromion articulates with the acromial end of the clavicle to form the ***acromioclavicular joint***.



The free upper limb has:

- ❑ **The brachium or arm: extends** from shoulder to elbow. It contains only one bone, **the *humerus***.
- ❑ **The antebrachium or forearm:** extends from elbow to wrist and contains two bones:
 - ✓ Ulna
 - ✓ Radius
- ❑ **The carpus or wrist:** contains 8 small *carpal bones* arranged in two rows.
- ❑ **The manus or hand:** contains 19 bones in two groups:
 - ✓ Palm (5 metacarpal bones).
 - ✓ Fingers (14 phalanges).



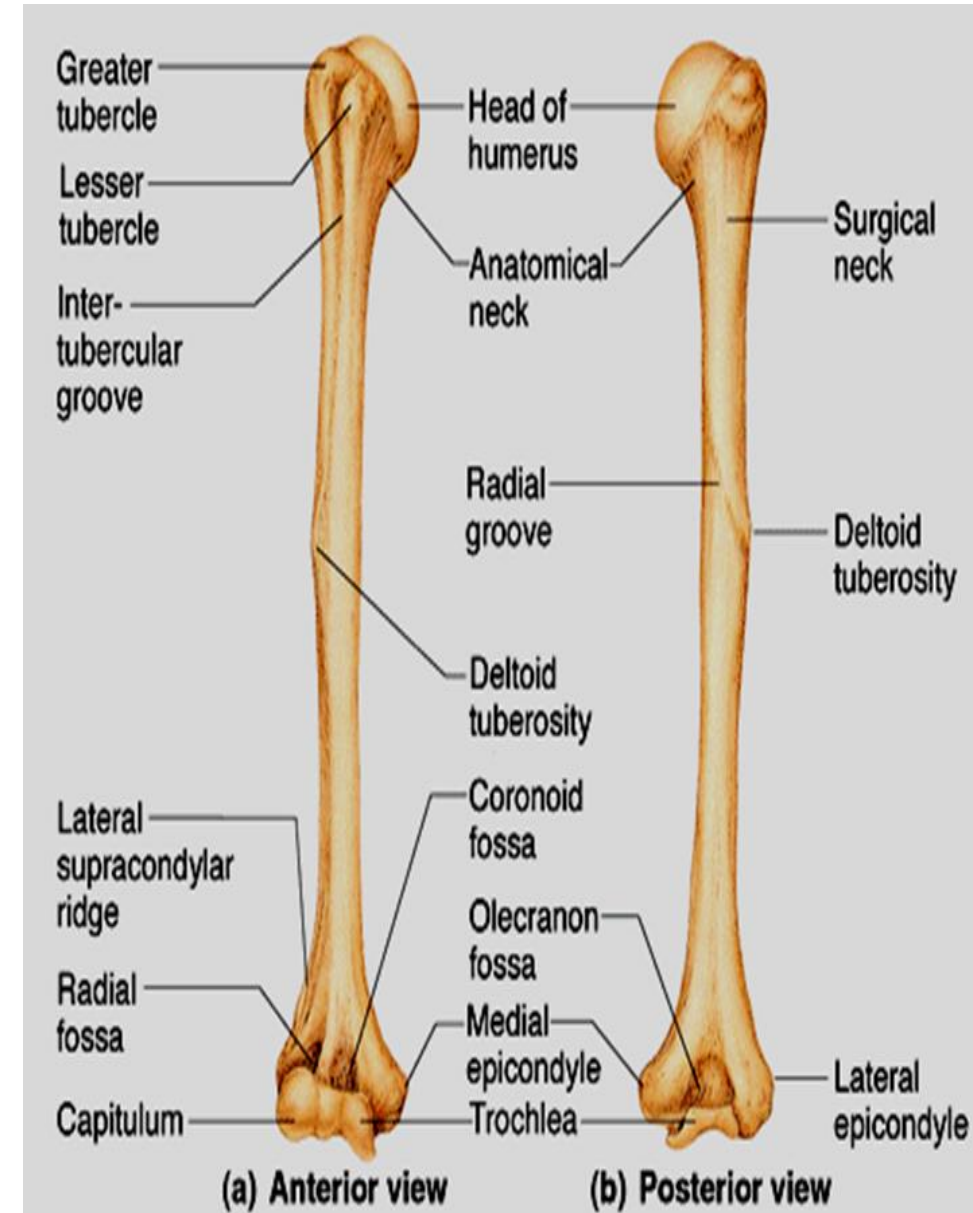


Humerus

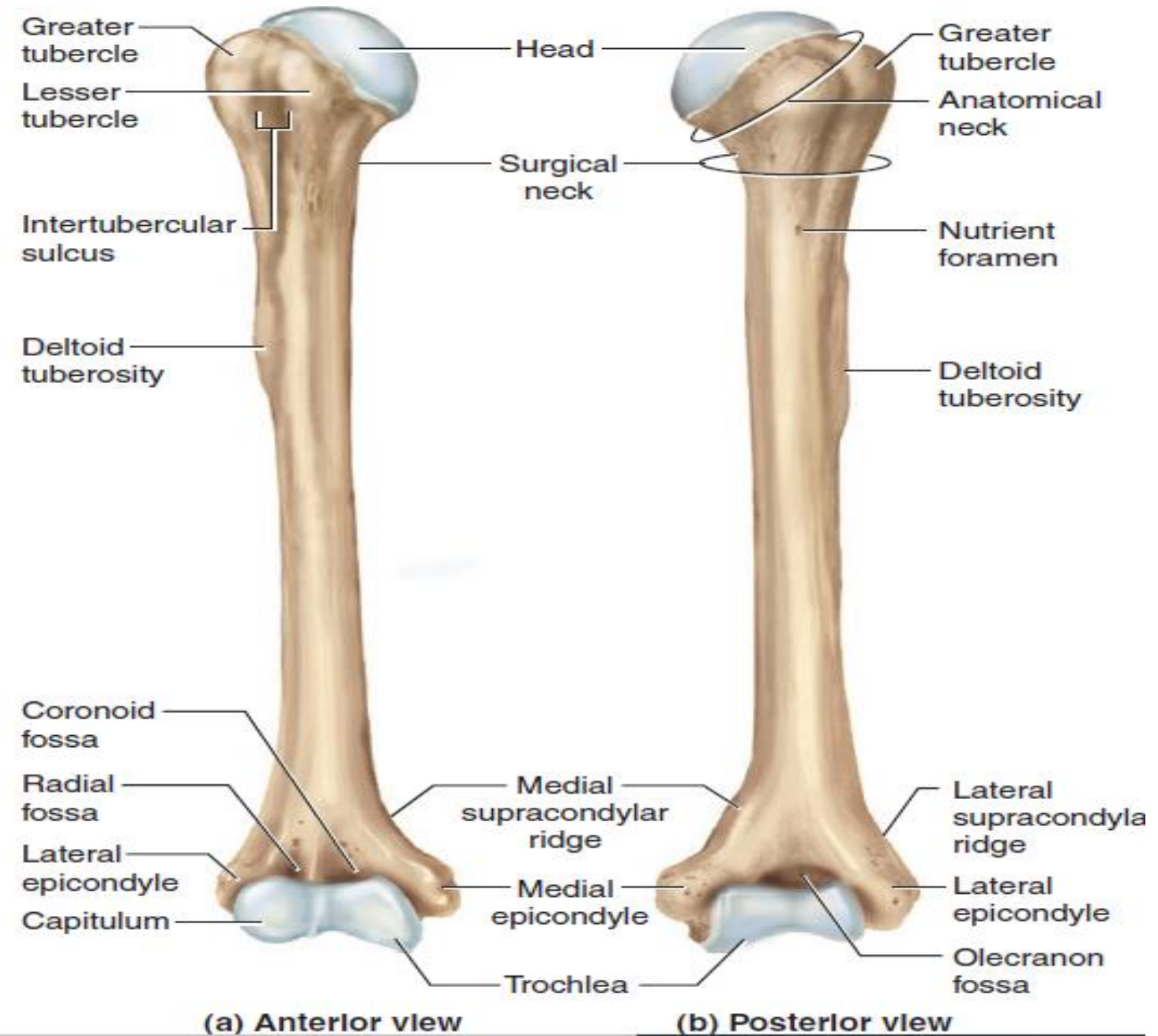
- The **humerus** is the only bone of the Arm.
- It is the largest & longest bone in the upper limb,
- It articulates with the scapula at the glenohumeral joint(proximally) & the radius and ulna at the elbow joint(distally).

❑ **The proximal end of the humerus has features like;** head, shaft (body), anatomical & surgical necks, tubercles (greater & lesser)

❑ **The distal end of the humerus features including;** the trochlea, capitulum, olecranon fossa, coronoid fossa, and radial fossa, two smooth epicondyles(medial and lateral).

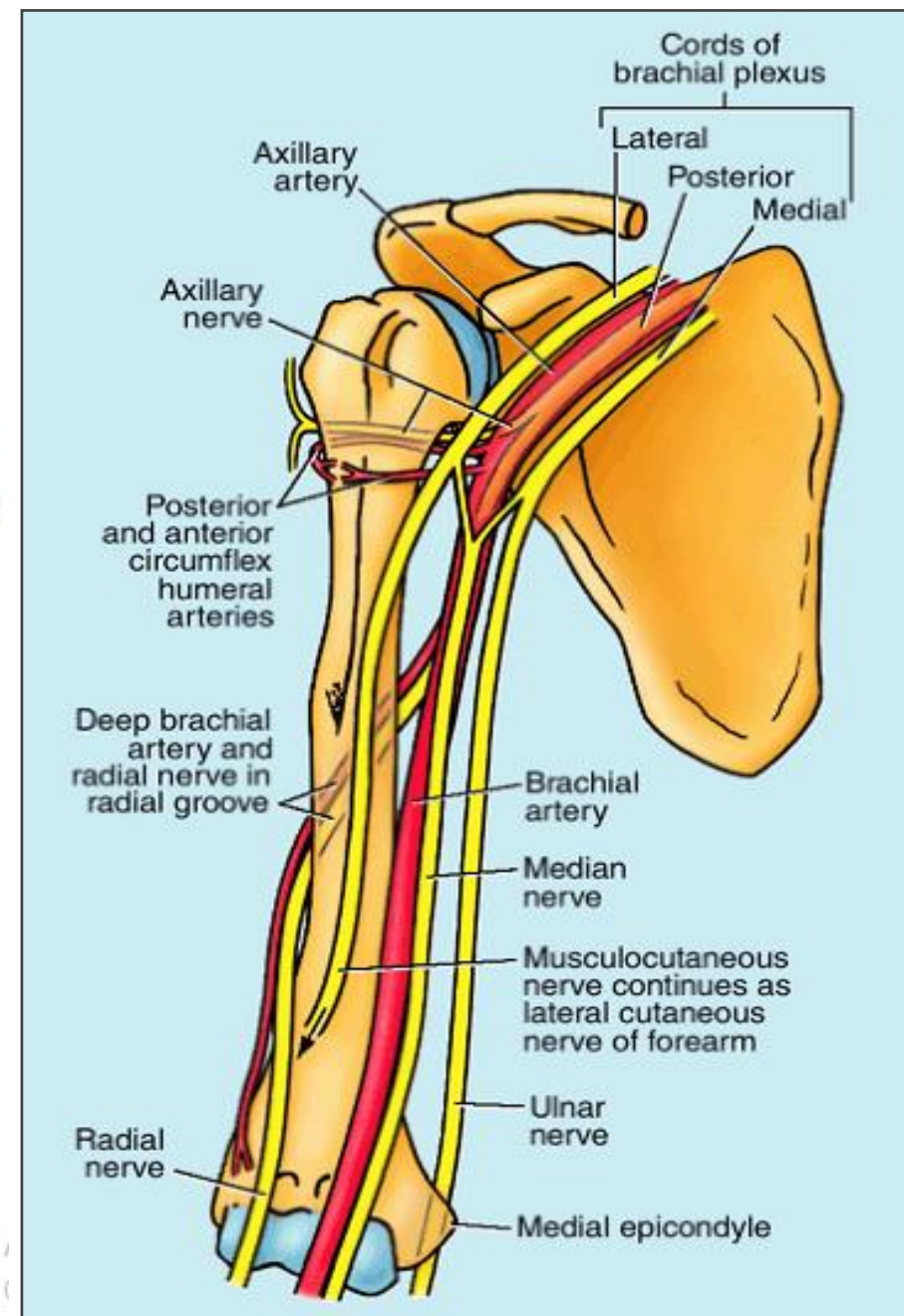
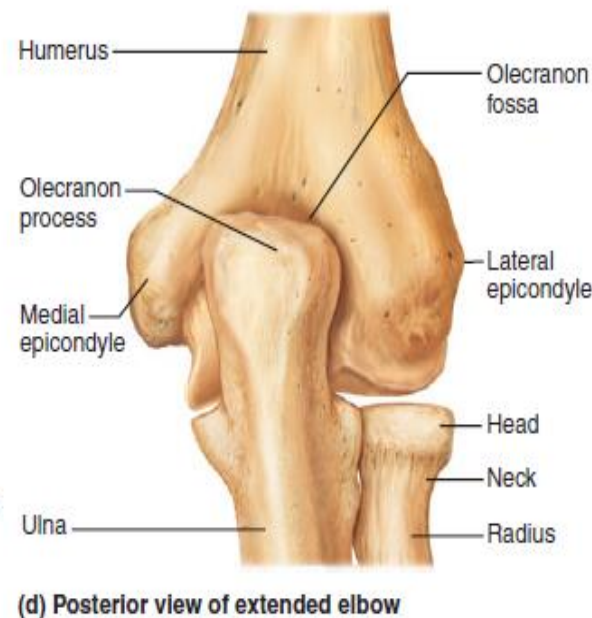
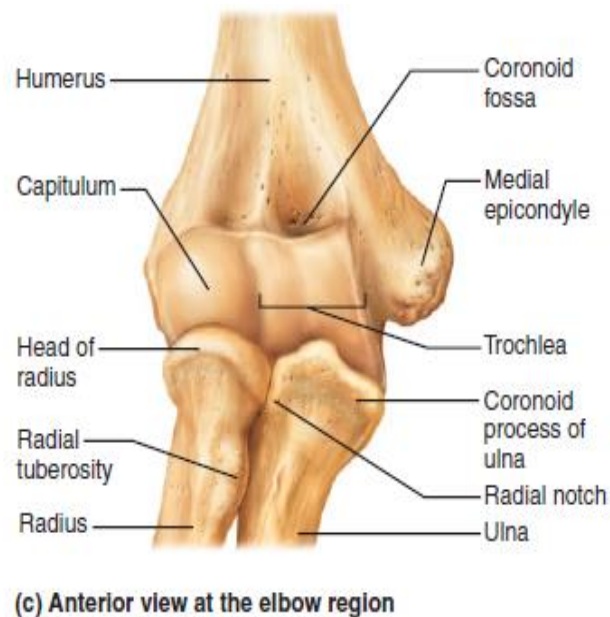
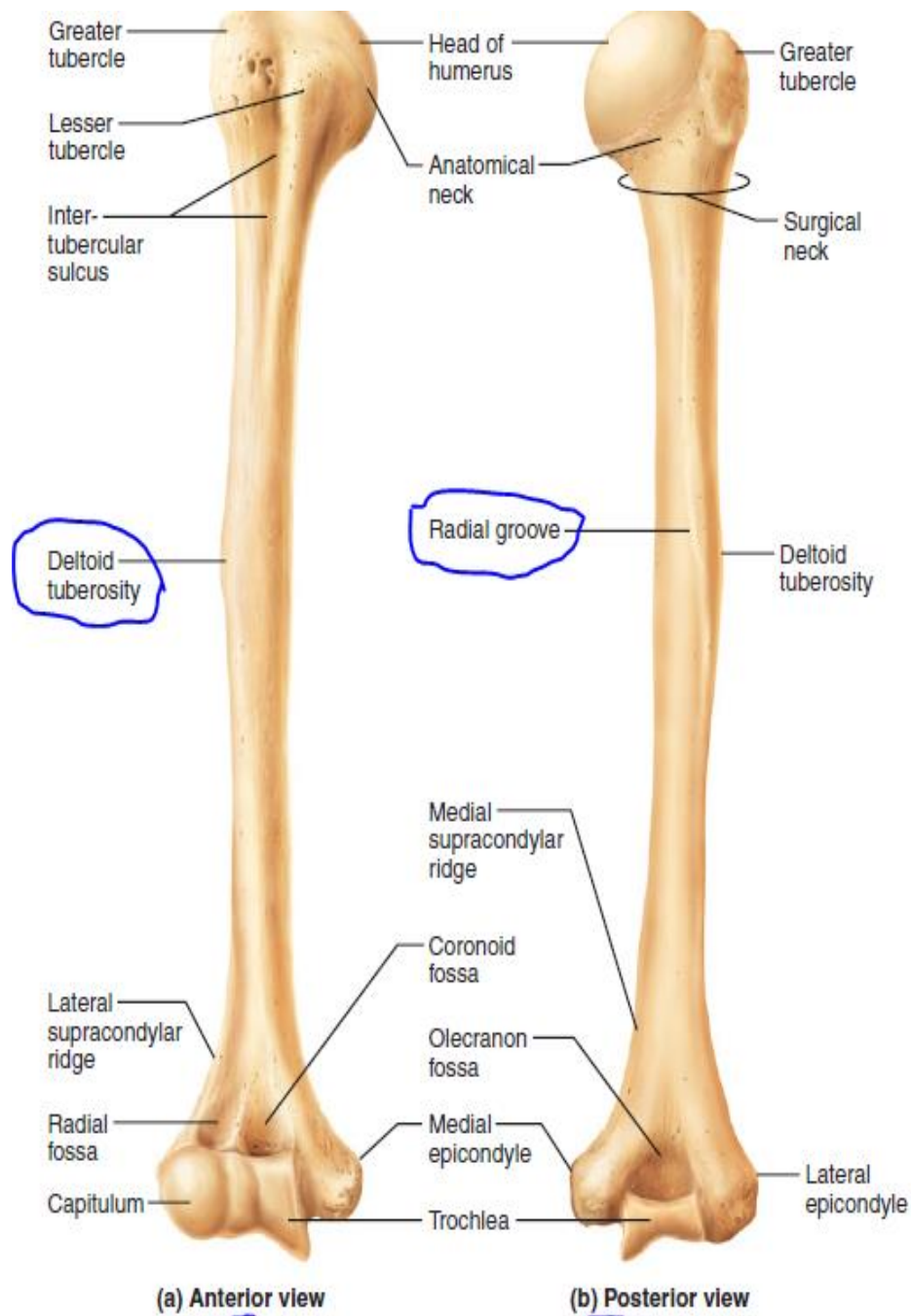


Humerus,,



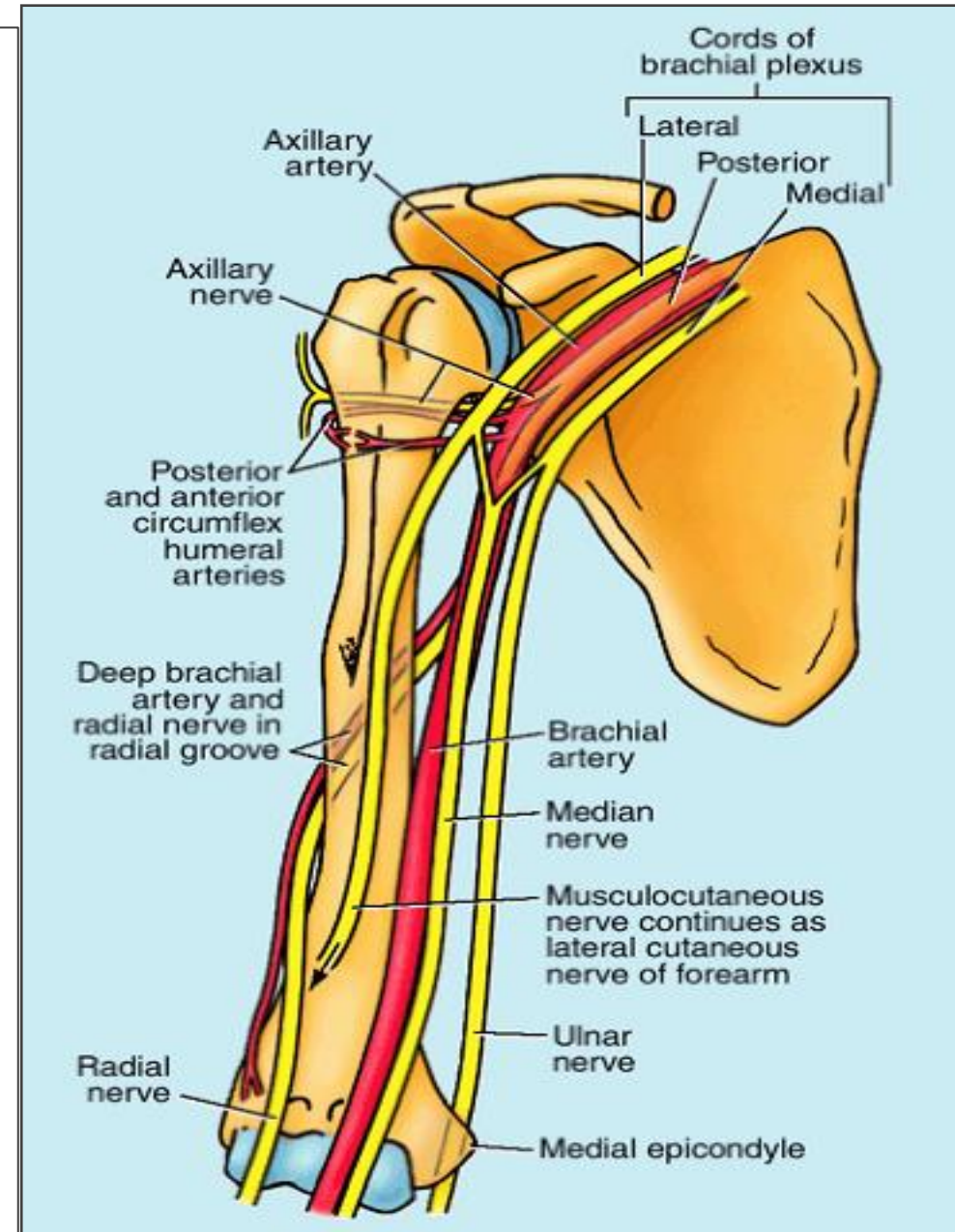
Humerus,,,

- The **surgical neck of the humerus**, a common site of fracture, is the narrow part distal to the head and tubercles.
- **The shaft of the humerus has 2 prominent features:**
 - ✓ the **deltoid tuberosity** laterally, for attachment of the deltoid muscle, and
 - ✓ the oblique **radial groove (groove for radial nerve, spiral groove)** posteriorly, in which the **radial nerve and profunda brachii artery** lie.
- The inferior end of the humeral shaft widens as the sharp medial and lateral **supra-epicondylar (supracondylar) ridges** form, and then end distally in the especially prominent **medial epicondyle** and the **lateral epicondyle**.

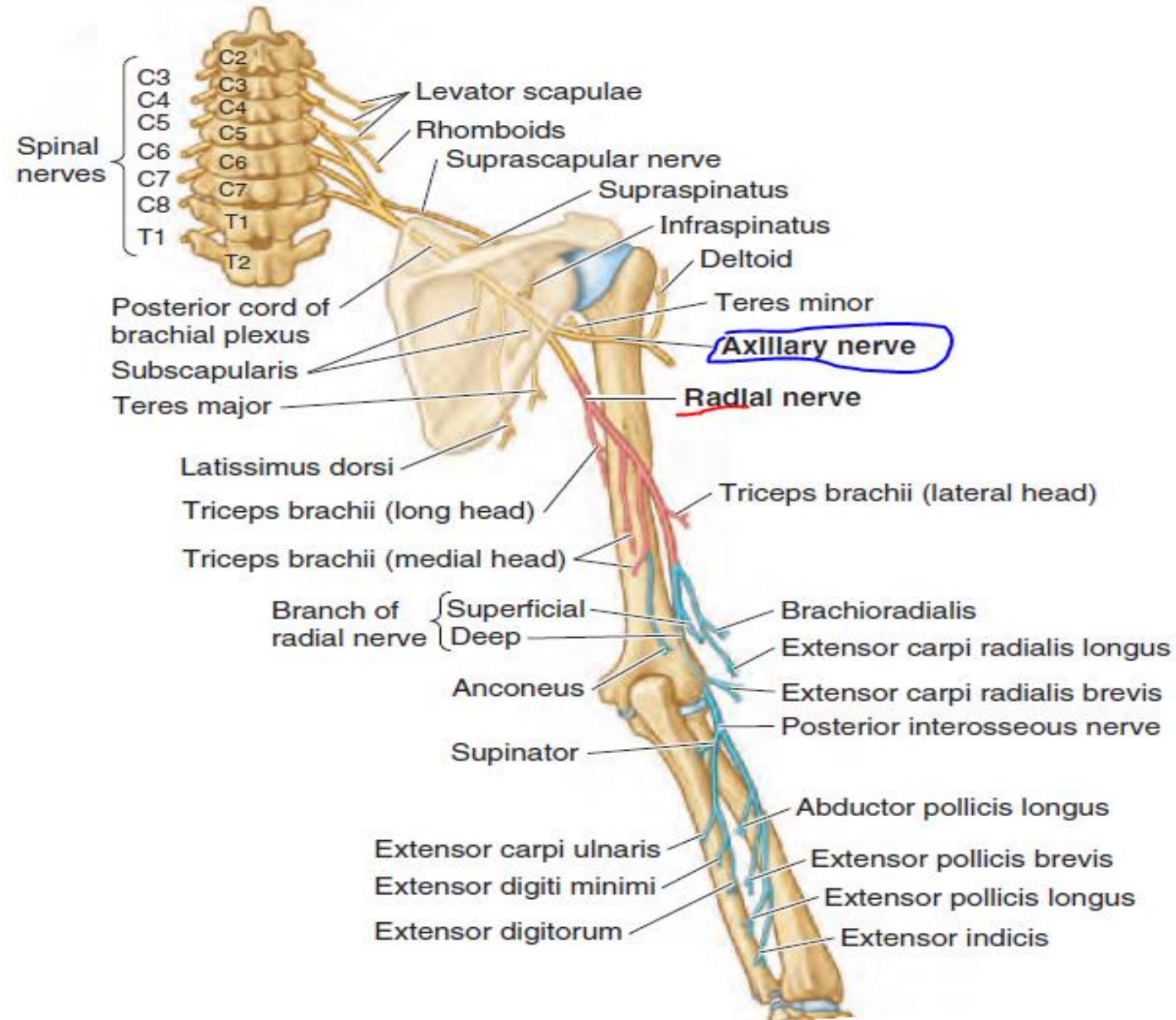


Fracture of the humerus

- Fractures of the **surgical neck** of the humerus
 - common in elderly people
 - usually result from a fall on the hand
- Because nerves are in contact with the humerus, they may be injured when the associated part of the humerus is fractured.
 - **Surgical neck** – axillary nerve
 - **Radial groove** – radial nerve
 - **Distal humerus** – median nerve
 - **Medial epicondyle** – ulnar nerve

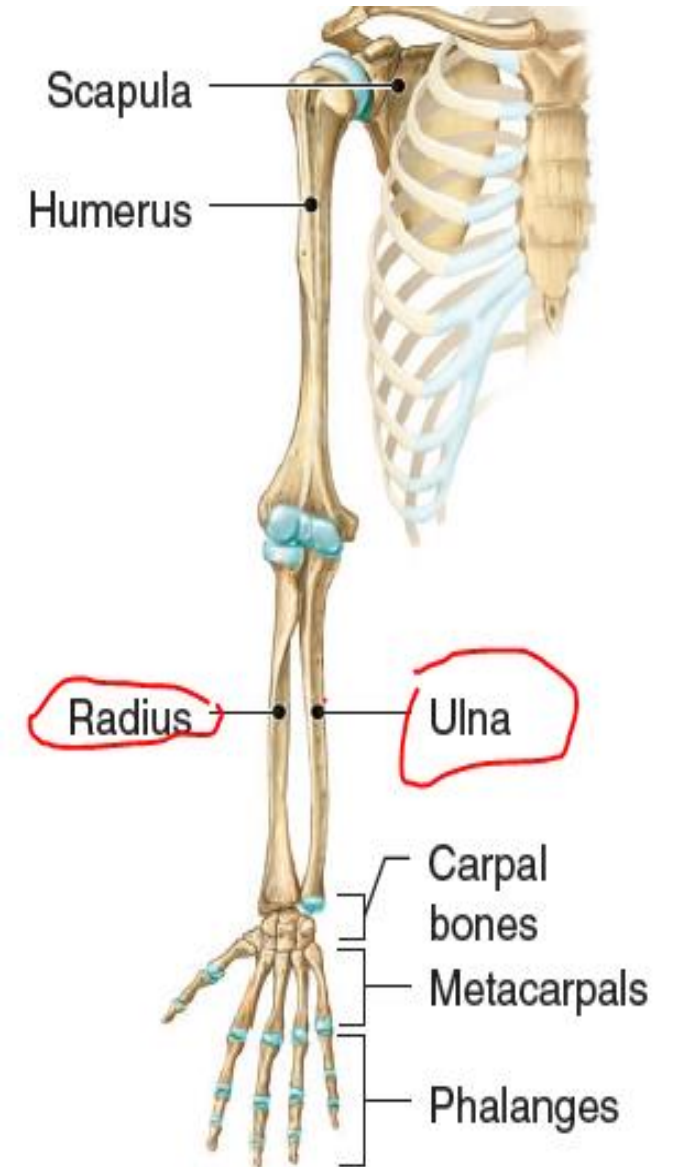


Fracture of the humerus,,,

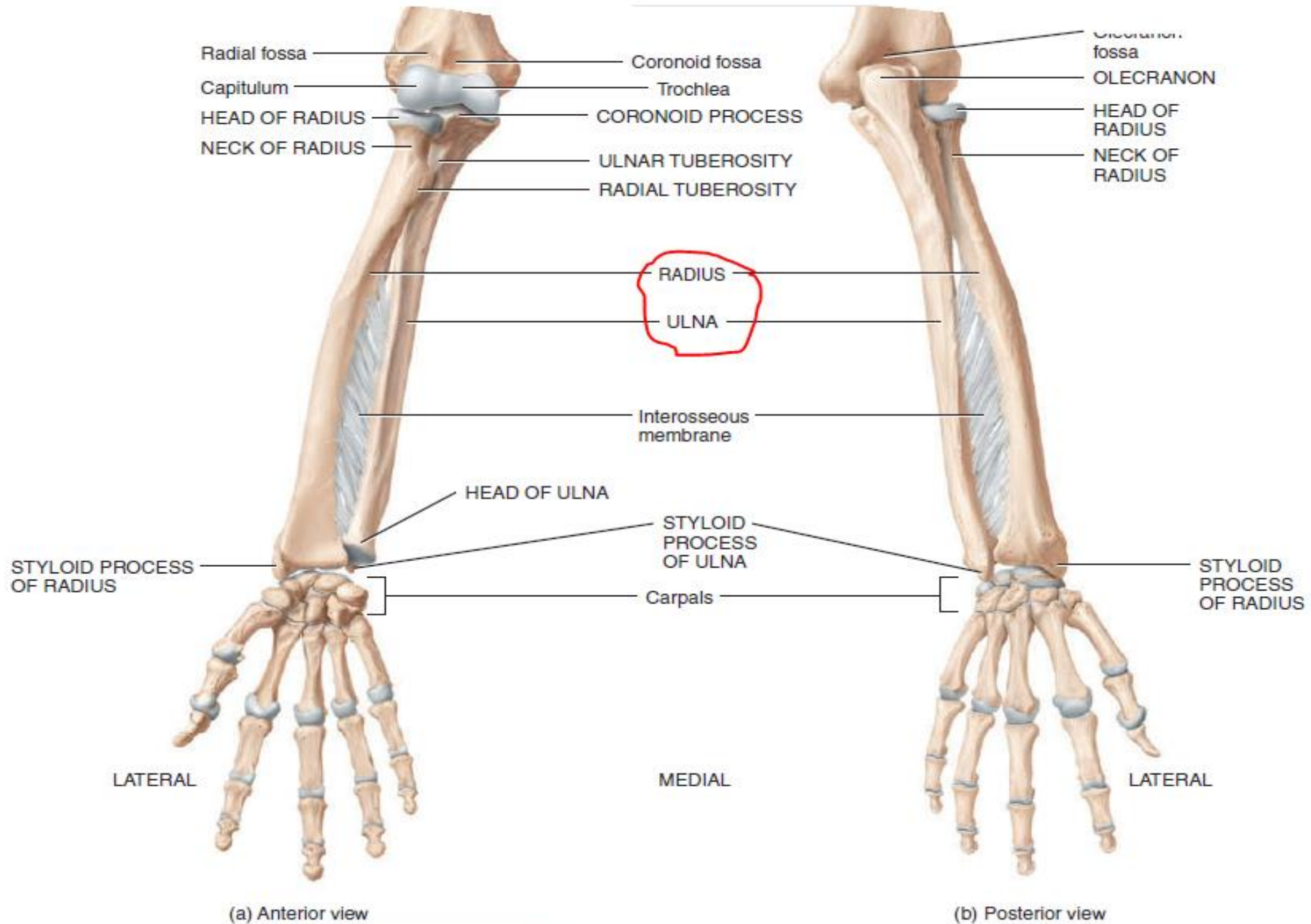


Bones of Forearm

- The **forearm** has 2 bones: the **ulna**, on the medial (little finger) side and the **radius** on the lateral (thumb) side.
- These 2 parallel long bones form the skeleton of the ***forearm or antebrachium***
- The radius and ulna that articulate with the humerus proximally and the bones of the wrist distally.
- **The radius and ulna also articulate with each other both proximally and distally at the small *radioulnar joints*.**
- They are interconnected along their entire length by a flat ligament called the **interosseous membrane**.



Bones of Forearm,,,



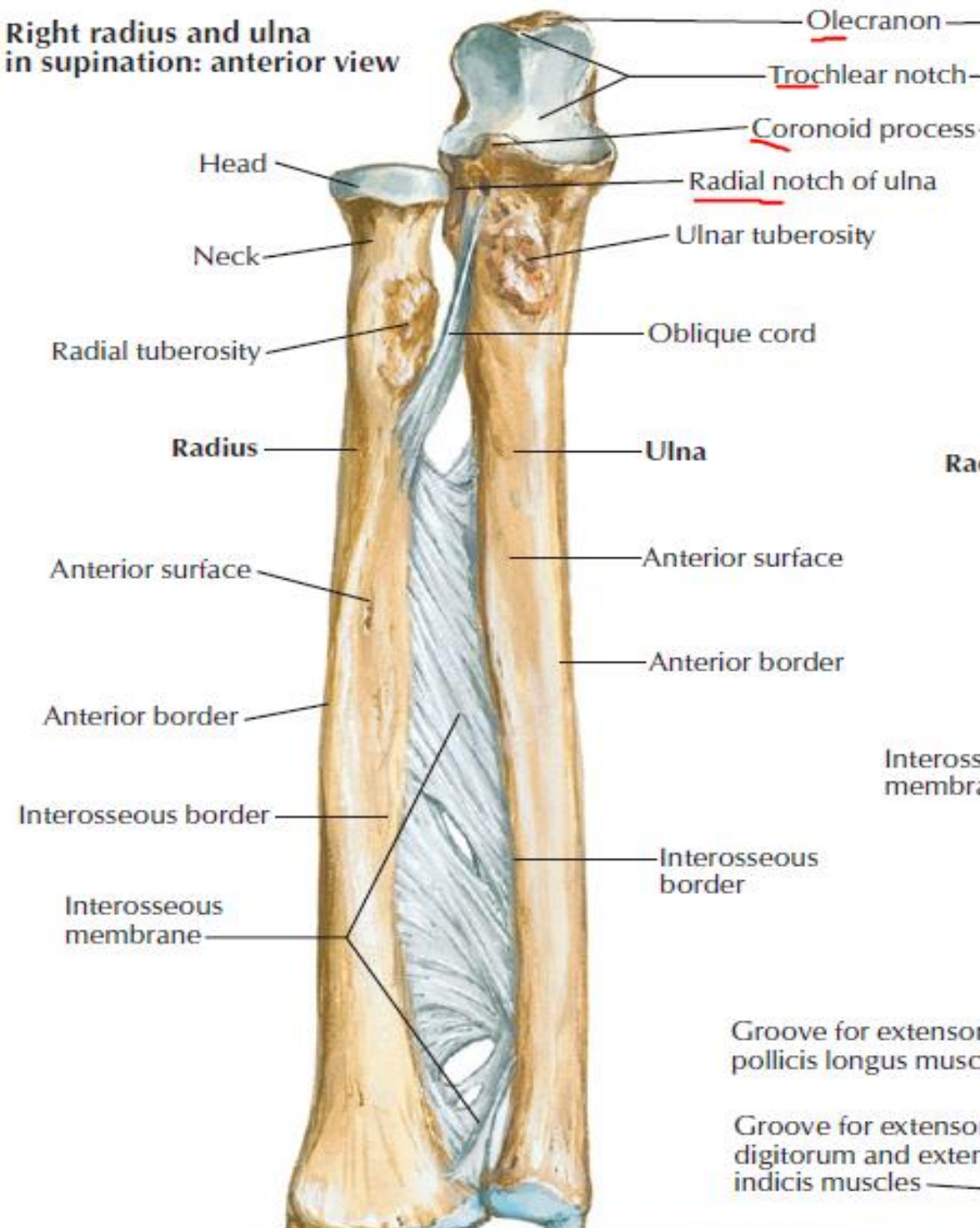
Ulna

- The **ulna** is the stabilizing bone of the forearm and is the medial and longer of the two forearm bones.
- The **ulna** is the main bone forming the elbow joint.
- Its more massive proximal end is specialized for articulation with the humerus proximally, and the head of the radius laterally.

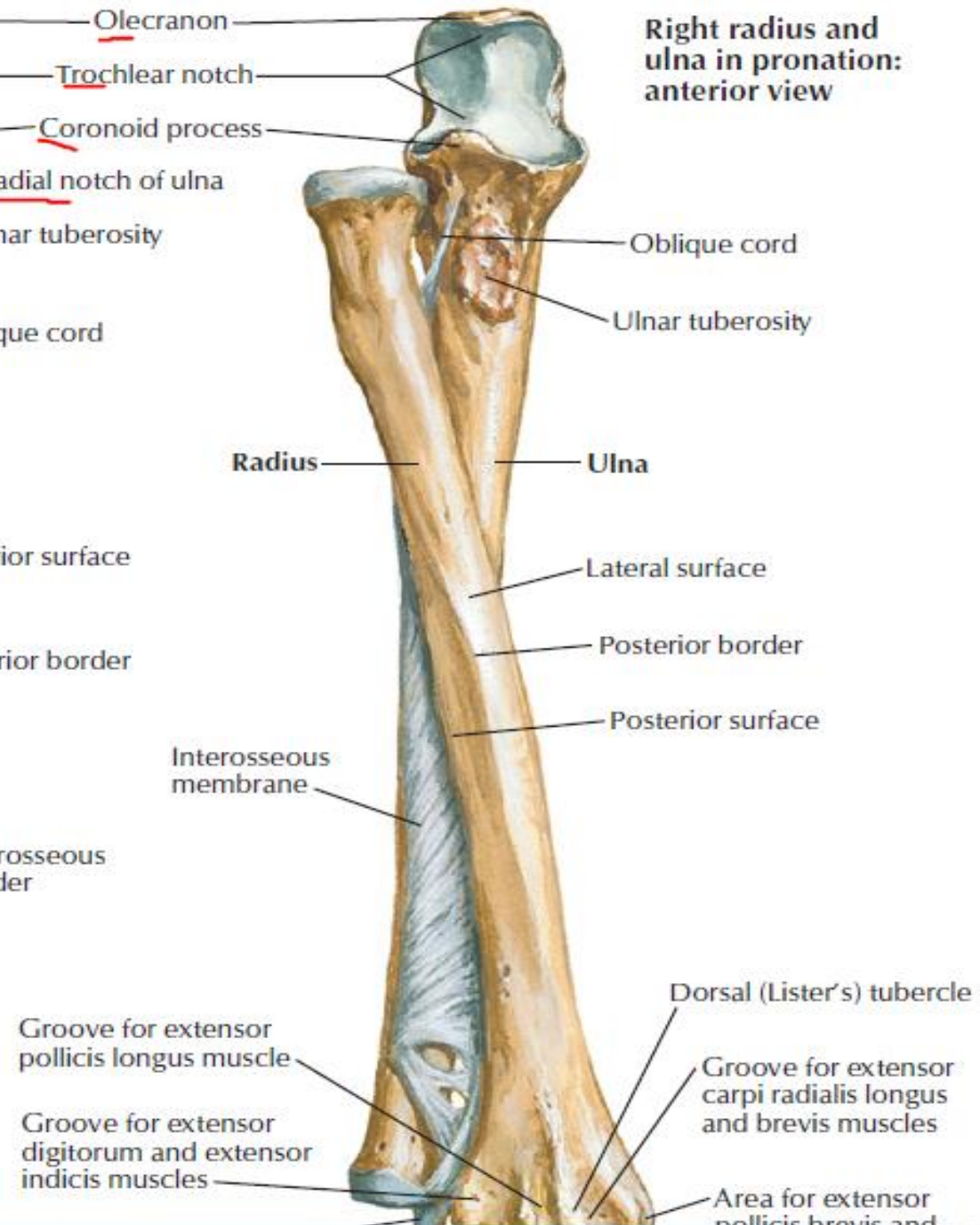
□ Proximally, the ulna has the following features:

- ✓ the **olecranon process**, posteriorly
- ✓ the **coronoid process**, which projects anteriorly
- ✓ The olecranon and coronoid processes form the walls of the **trochlear notch**,
- ✓ **Radial notch**; on the lateral side of the coronoid process articulates with the head of radius .

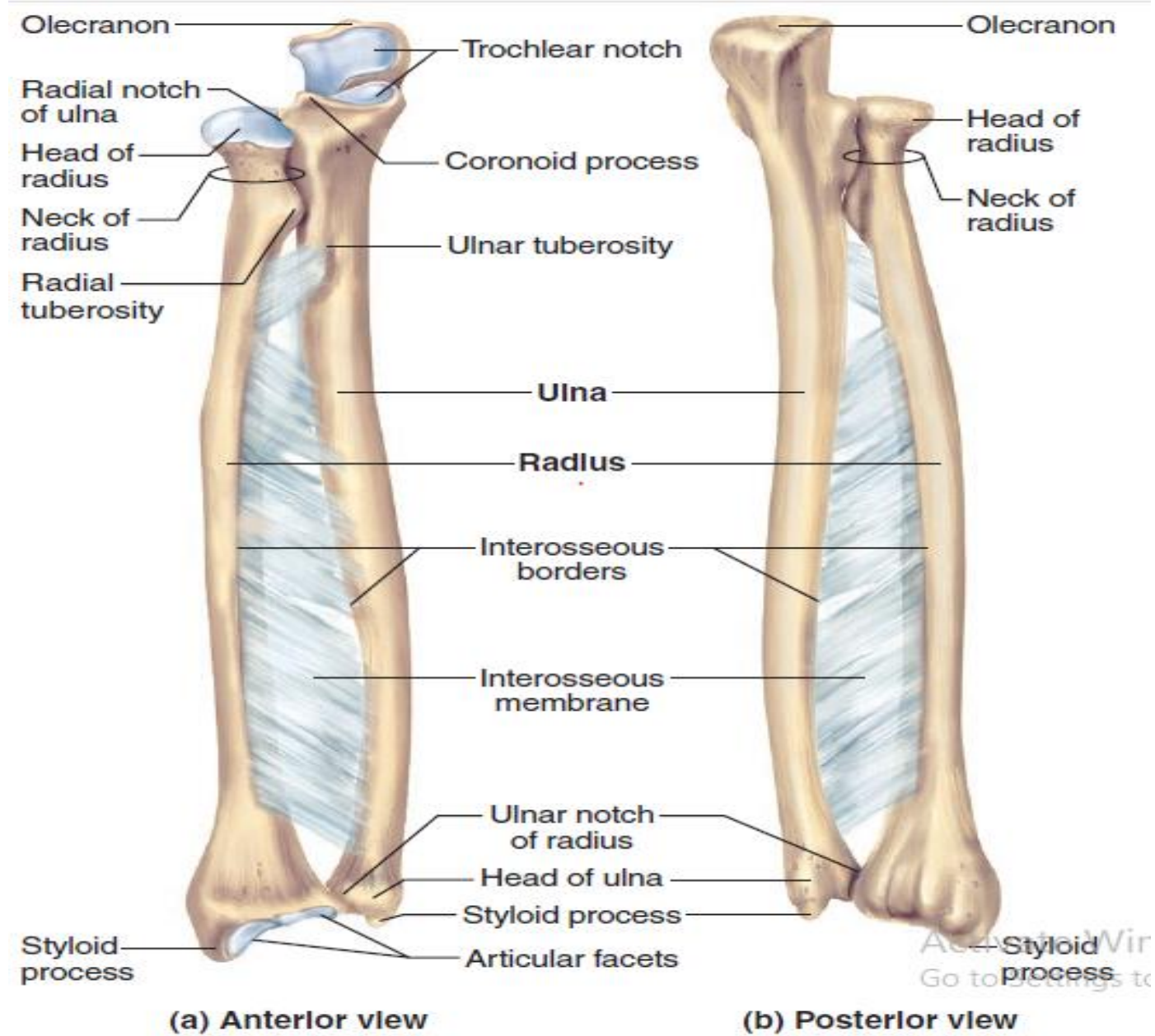
Right radius and ulna
in supination: anterior view



Right radius and
ulna in pronation:
anterior view



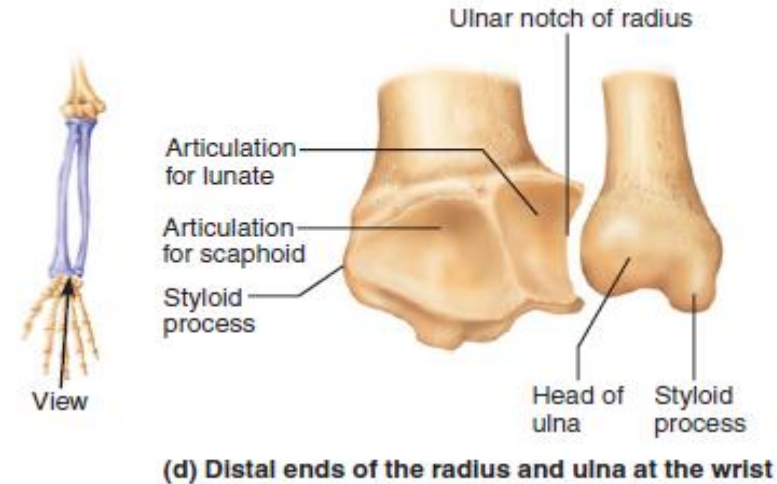
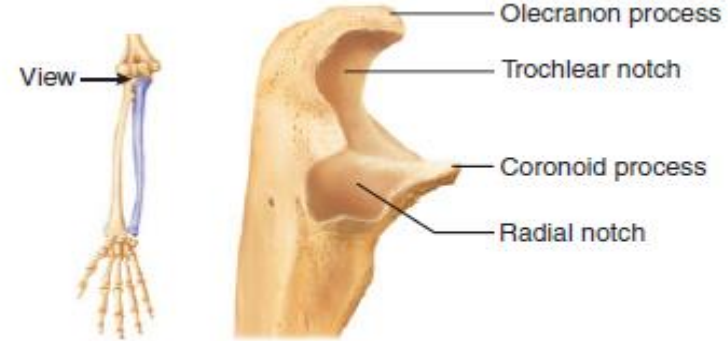
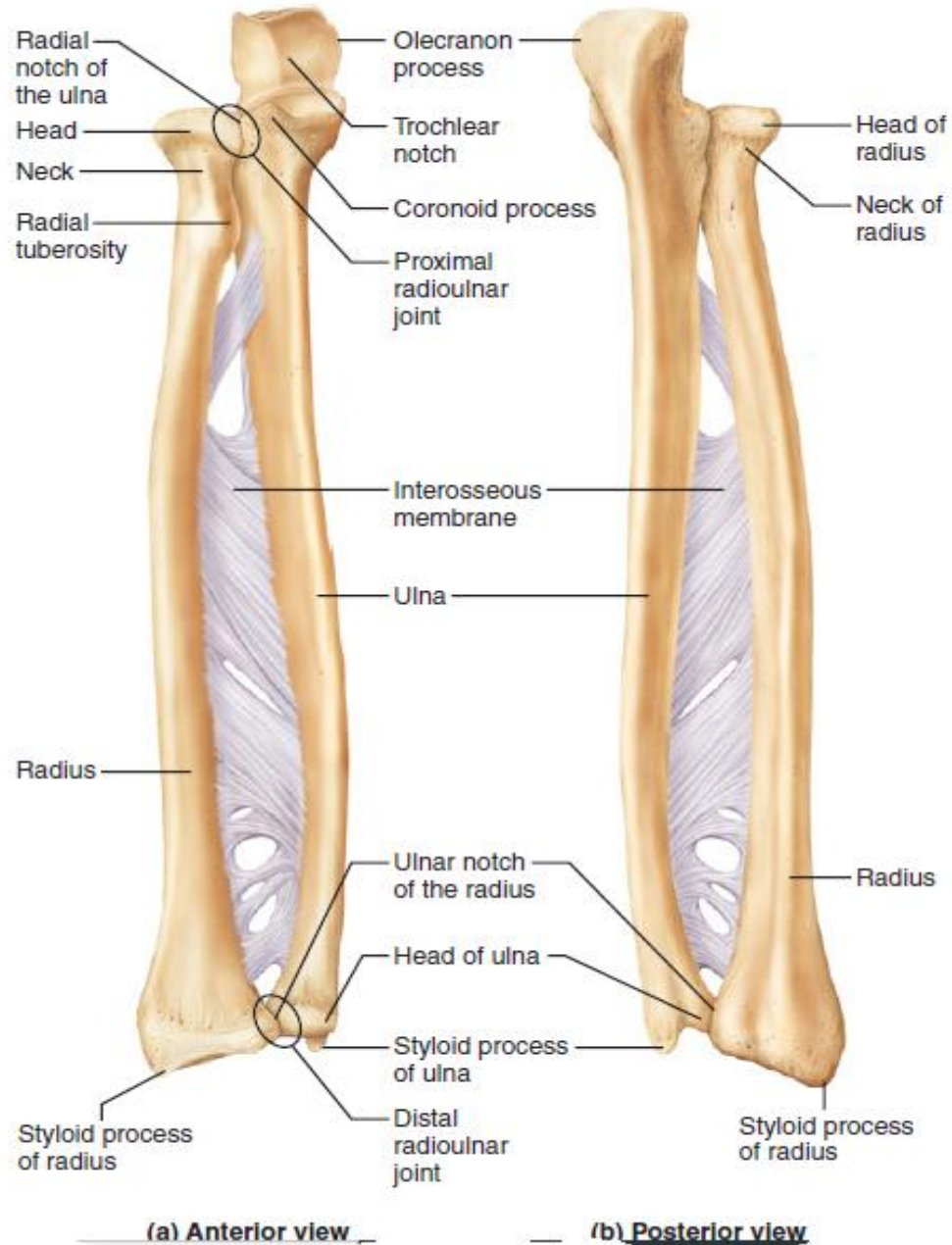
Ulna,,,



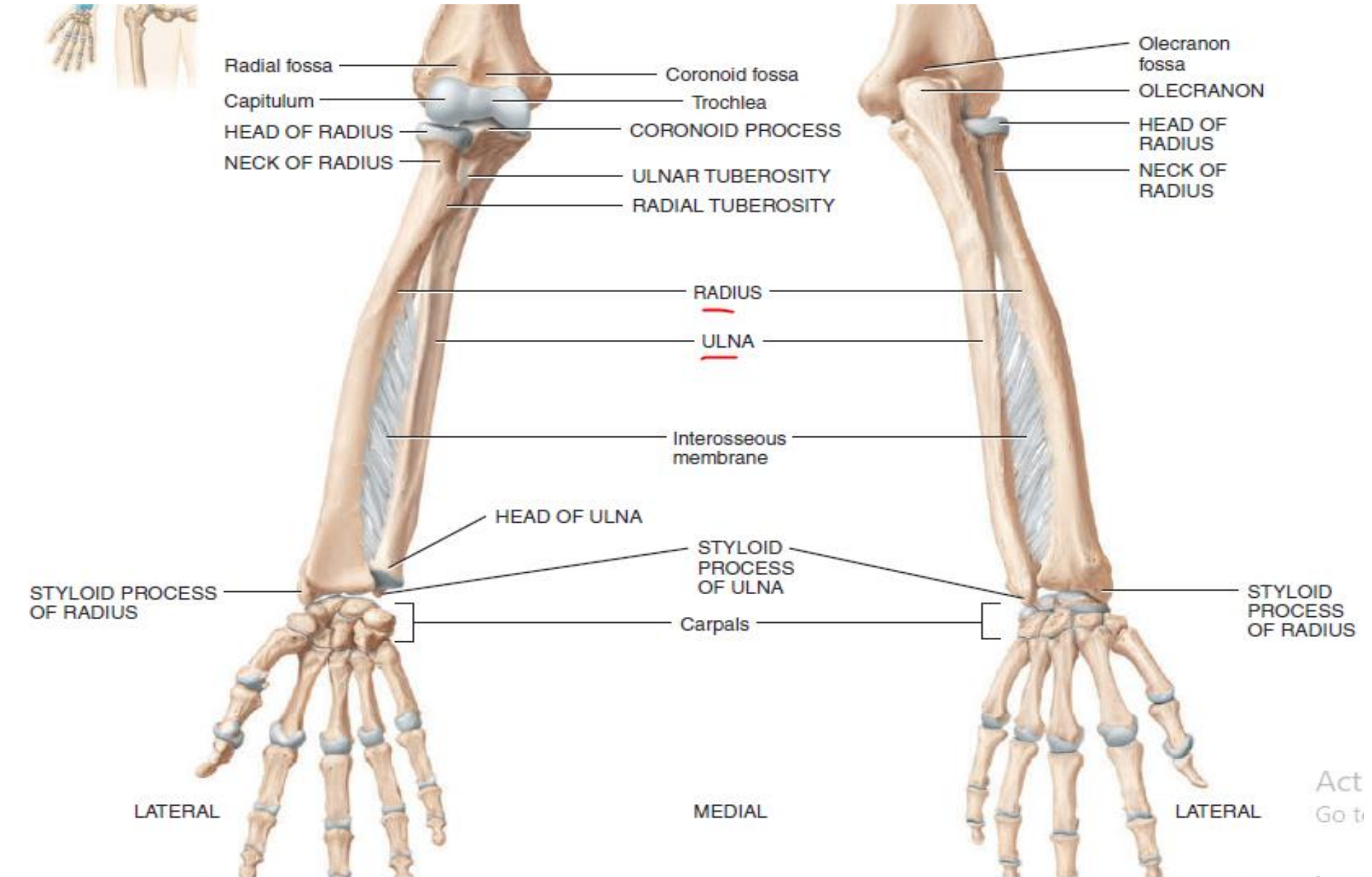
Ulna,,,

- The **shaft of the ulna** is thick and cylindrical proximally, but it tapers, diminishing in diameter, as it continues distally.
 - At the narrow distal end of the ulna is a small but abrupt enlargement, the disc-like **head of the ulna** with a small, conical **ulnar styloid process**.
- ❑ *The ulna does not participate in the wrist (radiocarpal) joint.*

Ulna,,,

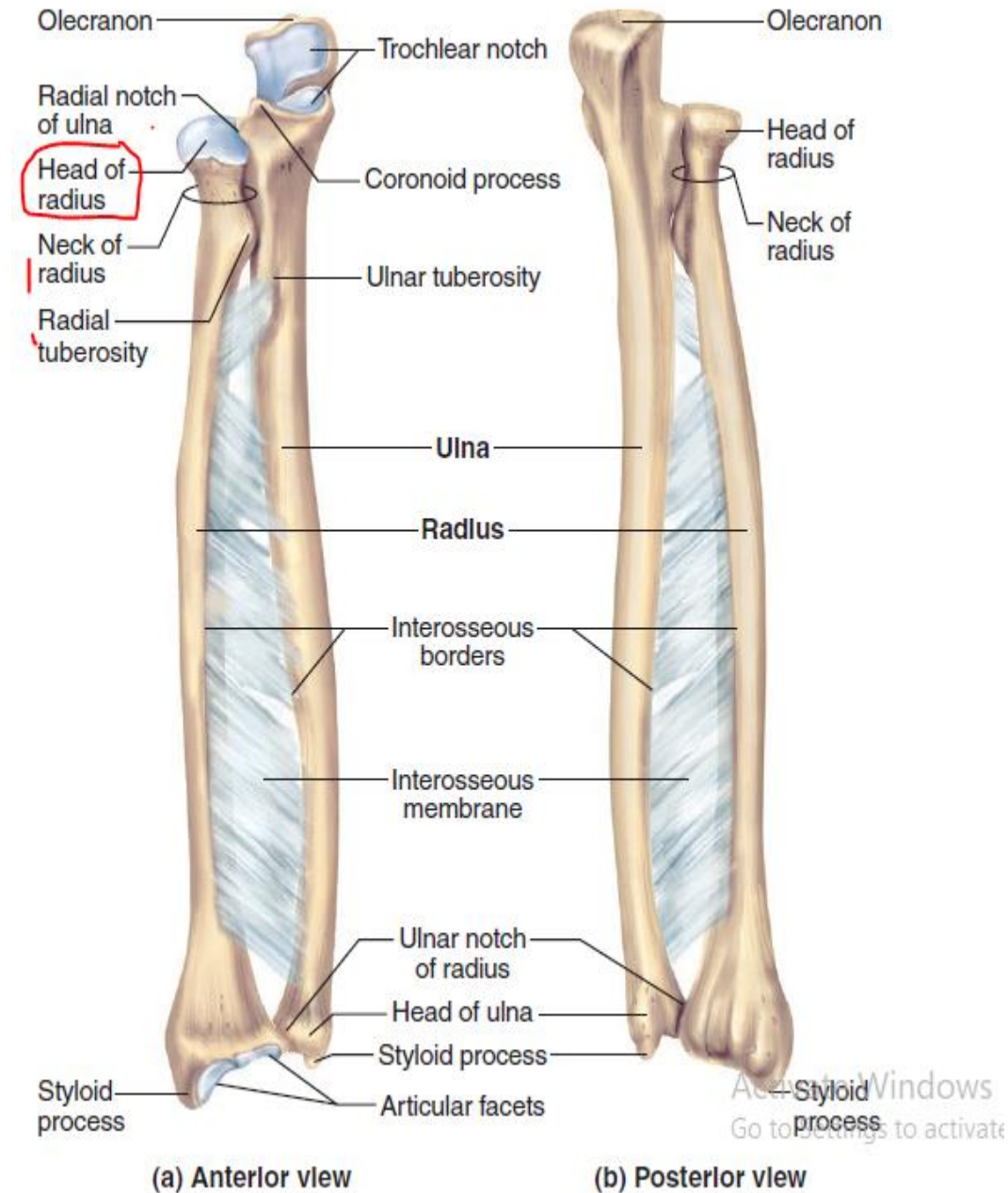


Ulna,,,

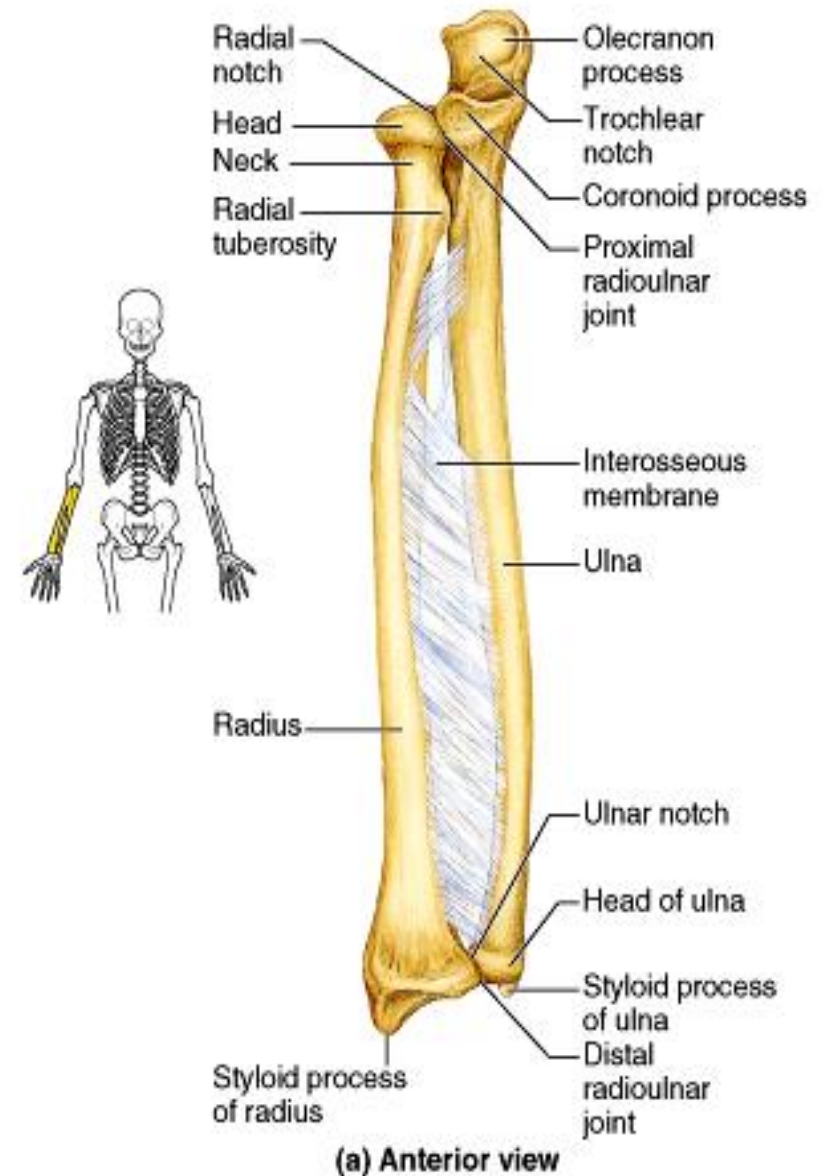
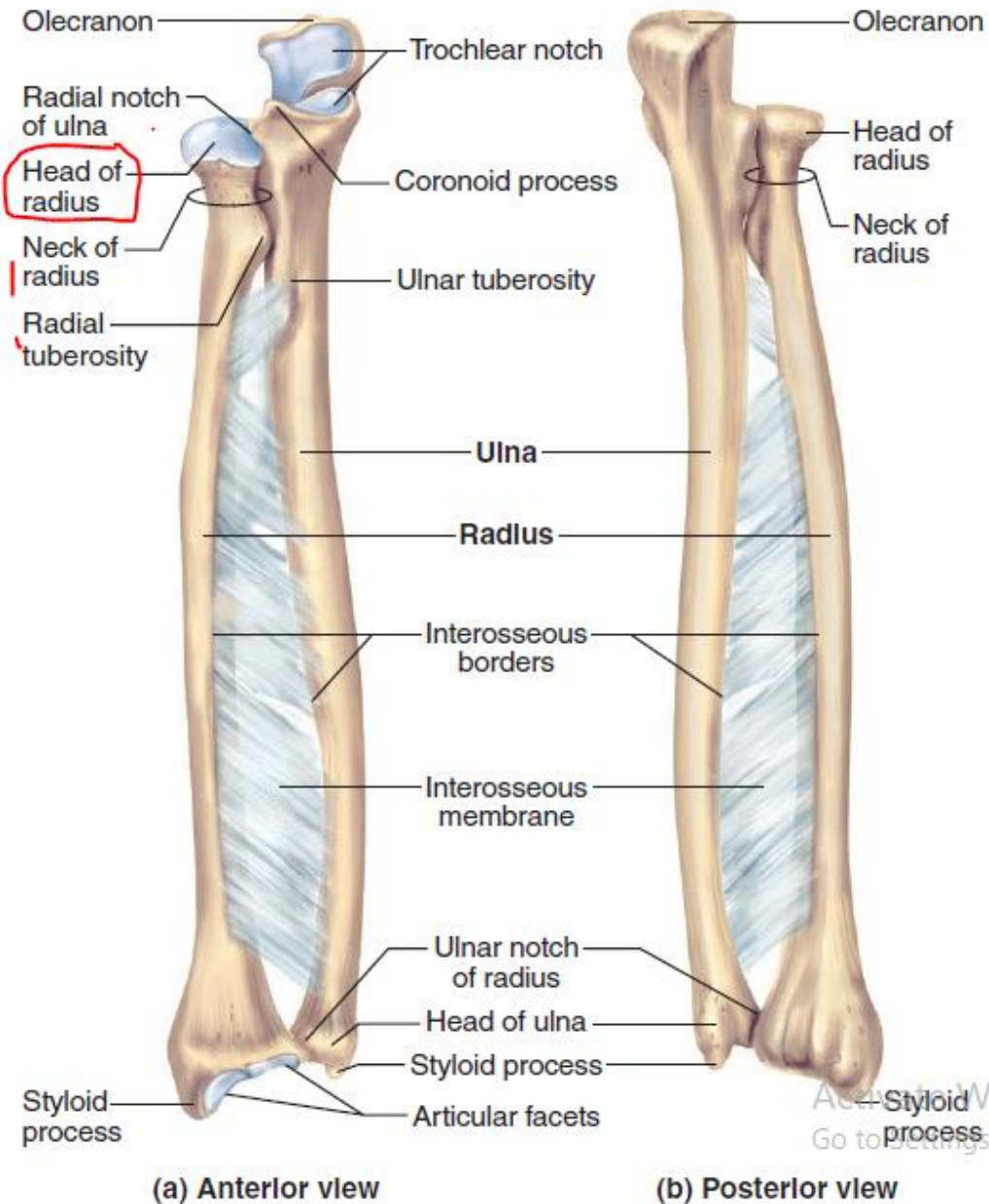


Radius

- The **radius** is the lateral and shorter of the two forearm bones.
- Its proximal end includes a short head, neck, and medially directed tuberosity.
- Proximally, the smooth superior aspect of the discoid **head of the radius** is concave for articulation with the capitulum of the humerus during flexion and extension of the elbow joint.
- **The distal end of the radius** has the following features, from lateral to medial: **styloid process**, articular facets and **ulnar notch**,

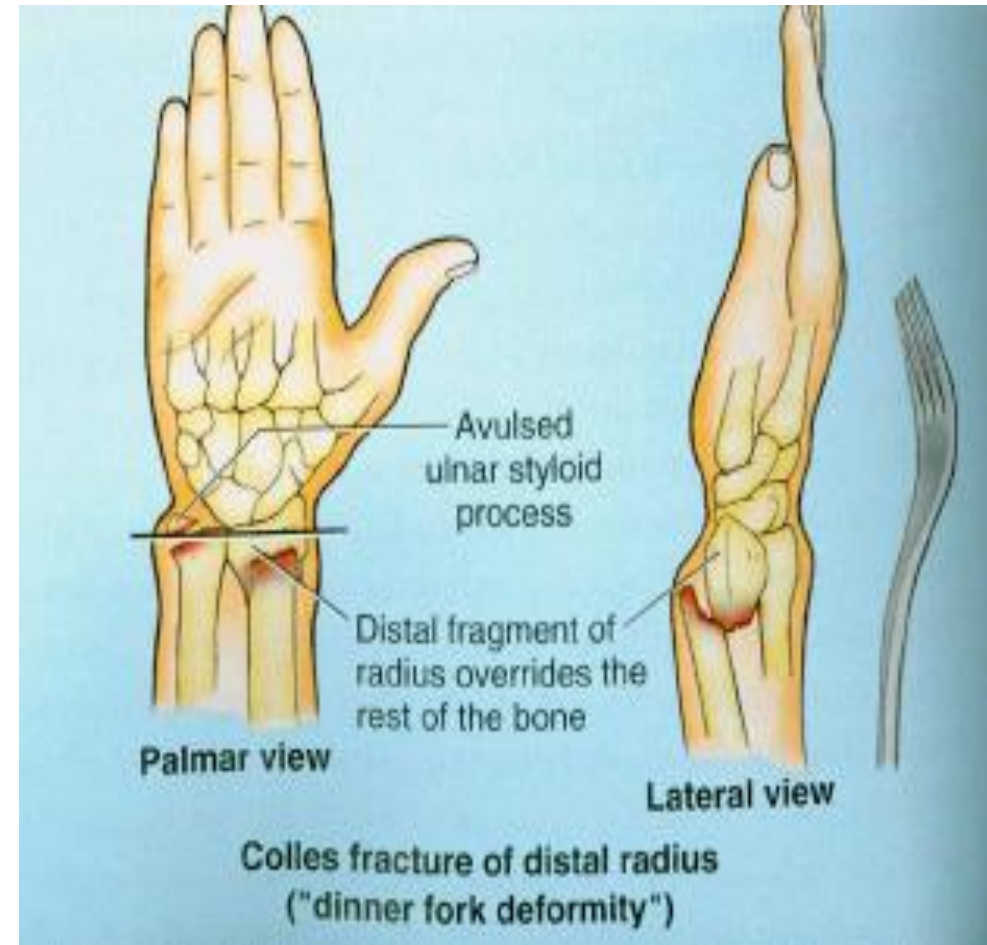


Radius,,



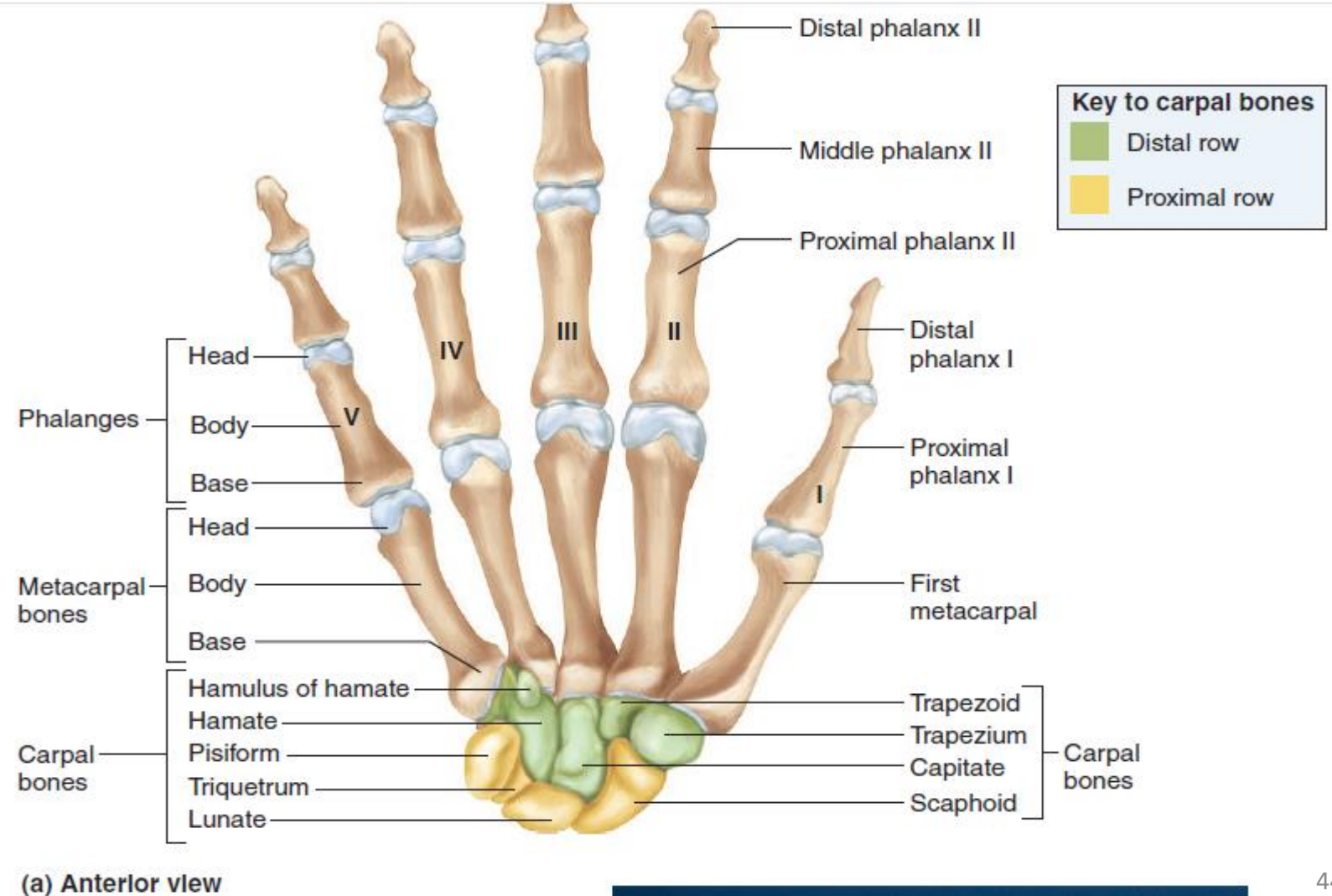
Fracture of ulna and radius

- **Colles fracture**
 - A complete fracture of the distal end of the **radius**
 - Results from forced dorsiflexion of the hand
 - The distal fragment of the radius is displaced dorsally and often broken into pieces



Bones of Hand

- The skeleton of the hand consists of the carpal bones, metacarpals, and phalanges



The carpal bones

- The **carpal bones**, or wrist bones, consist of 8 small bones that are arranged in two transverse rows of four bones each.
- They are joined by ligaments that allow limited gliding movement.

❑ **The carpal bones of the proximal row**, starting at the lateral (thumb) side, are the **scaphoid, lunate, triquetrum, and pisiform**.

- Unlike the other carpal bones, the pisiform is a sesamoid bone; it is not present at birth but develops around the age of 9 to 12 years within the tendon of the *flexor carpi ulnaris muscle*.

❑ **The bones of the distal row**, again starting on the lateral side, are the **trapezium, trapezoid, capitate, and hamate**.

The carpal bones,,,

- ❖ Proximal row articulate with radius and articular disc of wrist joint proximally and distal row of carpals distally.
- ❖ Distal row articulate with proximal row of carpals proximally and metacarpals distally

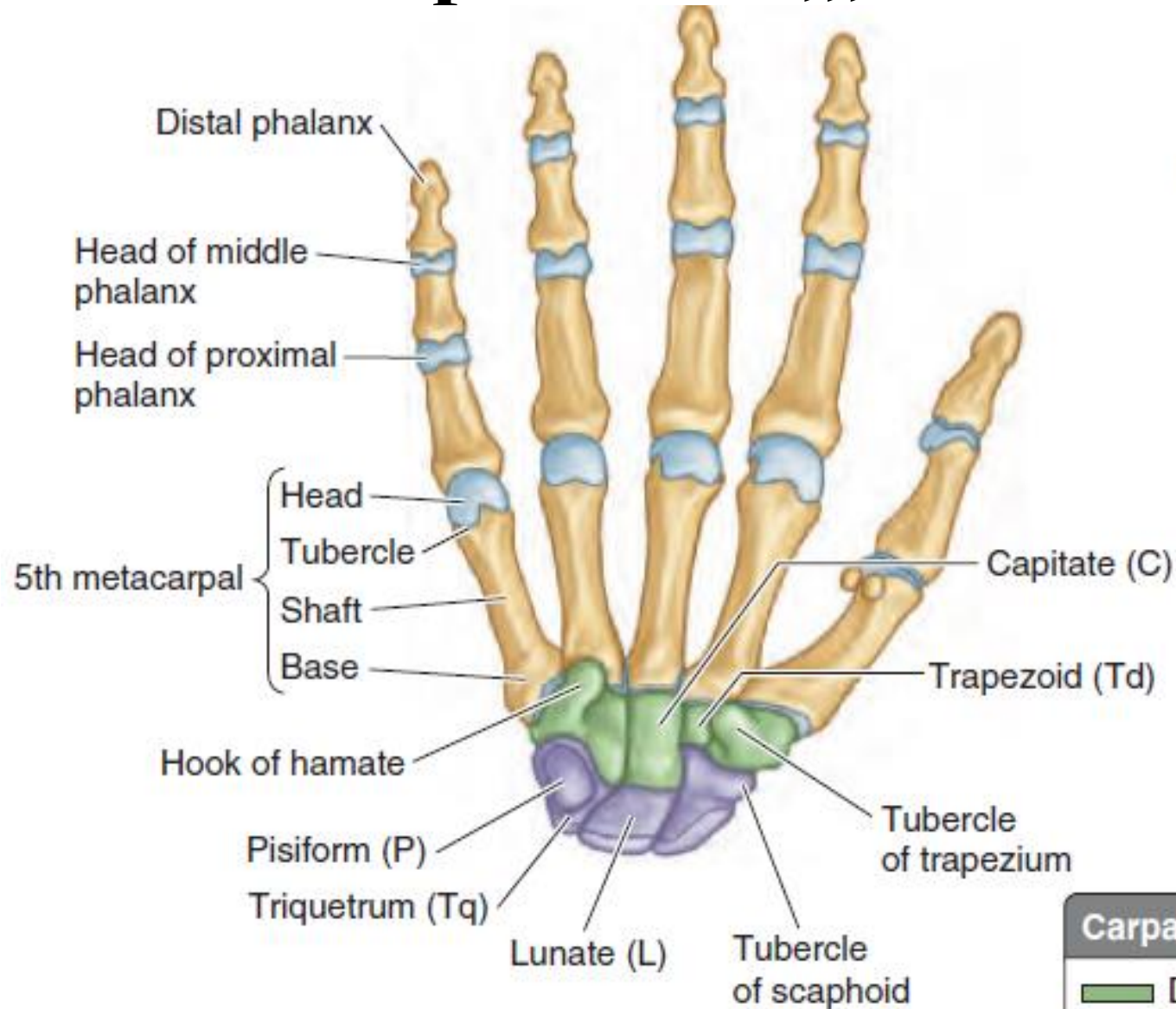
❑ Proximal row from lateral to medial:

- ✓ Scaphoid: boat-shaped with scaphoid tubercle
- ✓ Lunate: moon-shaped
- ✓ Triquetrum: pyramidal
- ✓ Pisiform: pea-shaped; lies on palmar surface of triquetrum

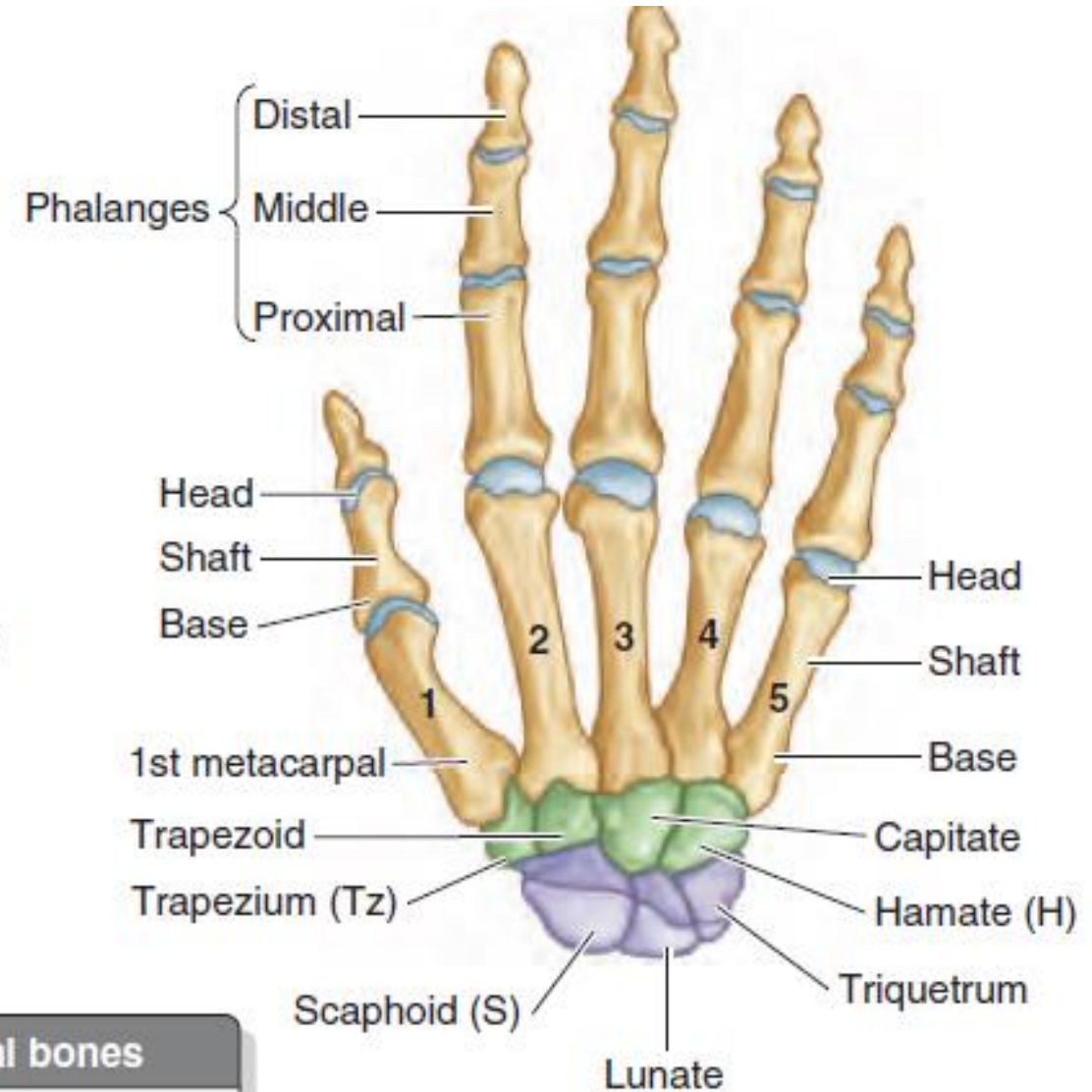
❑ Distal row from lateral to medial:

- ✓ Trapezium: four sided
- ✓ Trapezoid: wedge-shaped
- ✓ Capitate: head like; largest
- ✓ Hamate: wedge-shaped, has hooked process (hook of hamate)

The carpal bones,,,



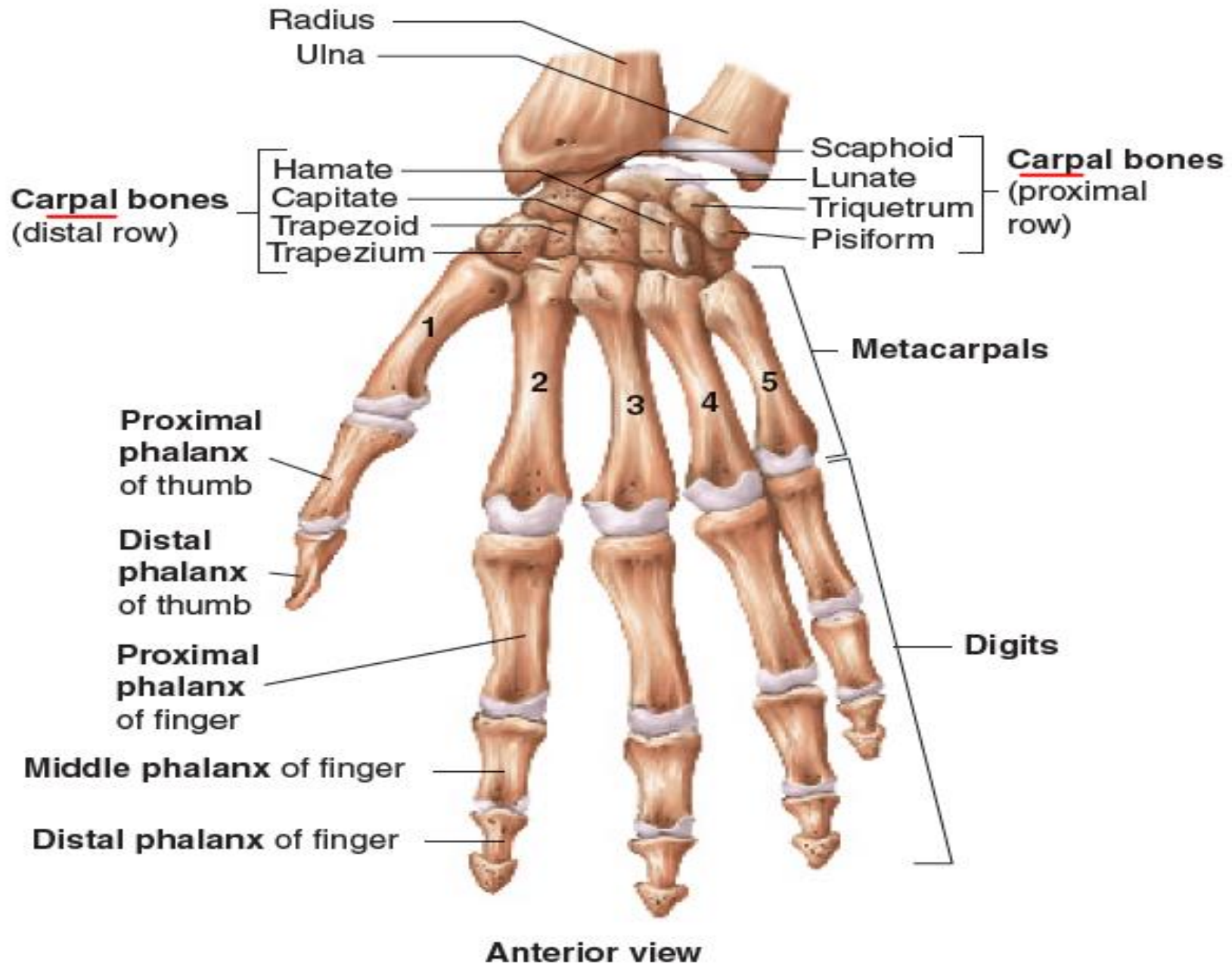
(A) Anterior (palmar) view



(B) Posterior (dorsal) view

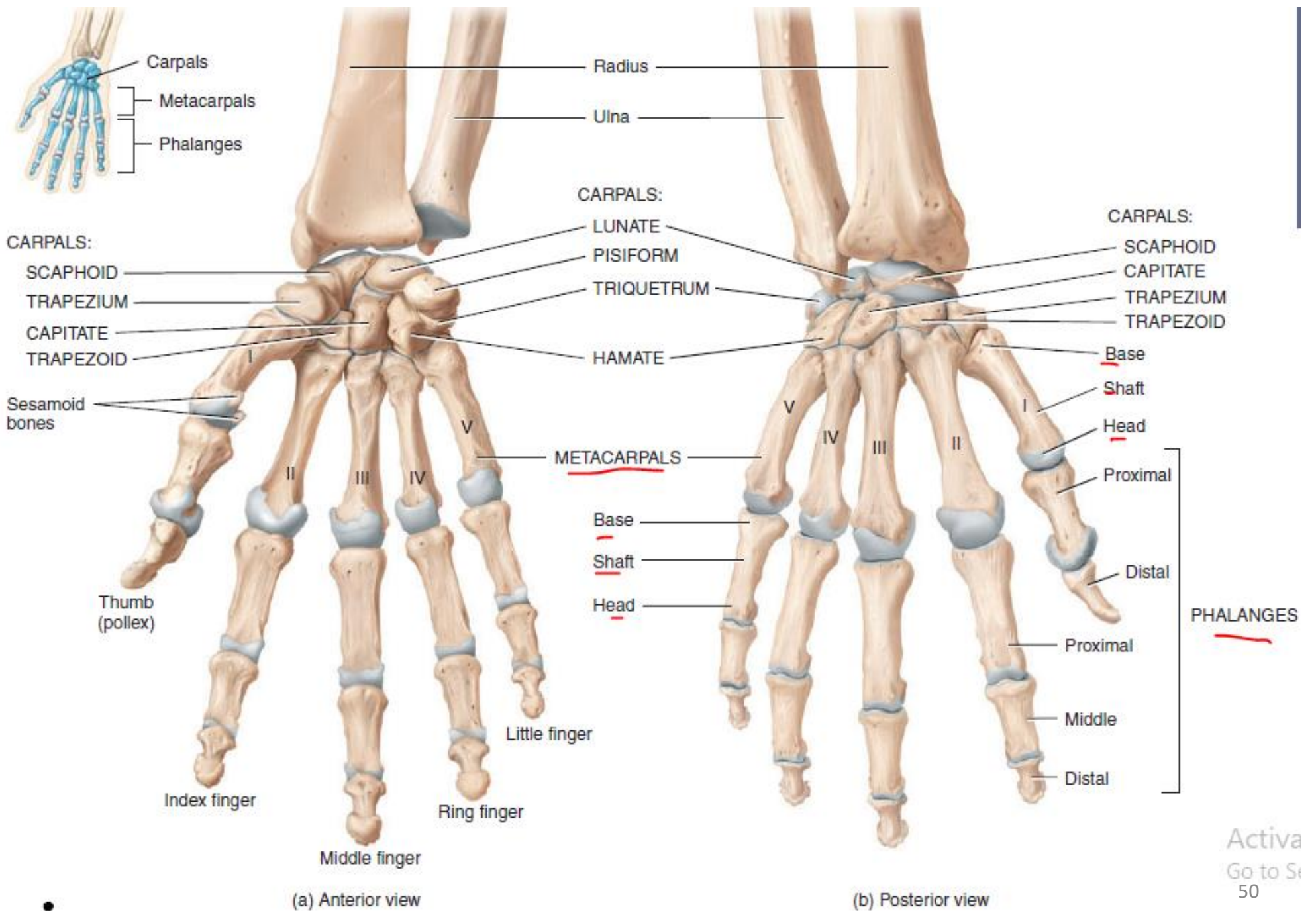
Carpal bones	
	Distal row
	Proximal row

The carpal bones,,,



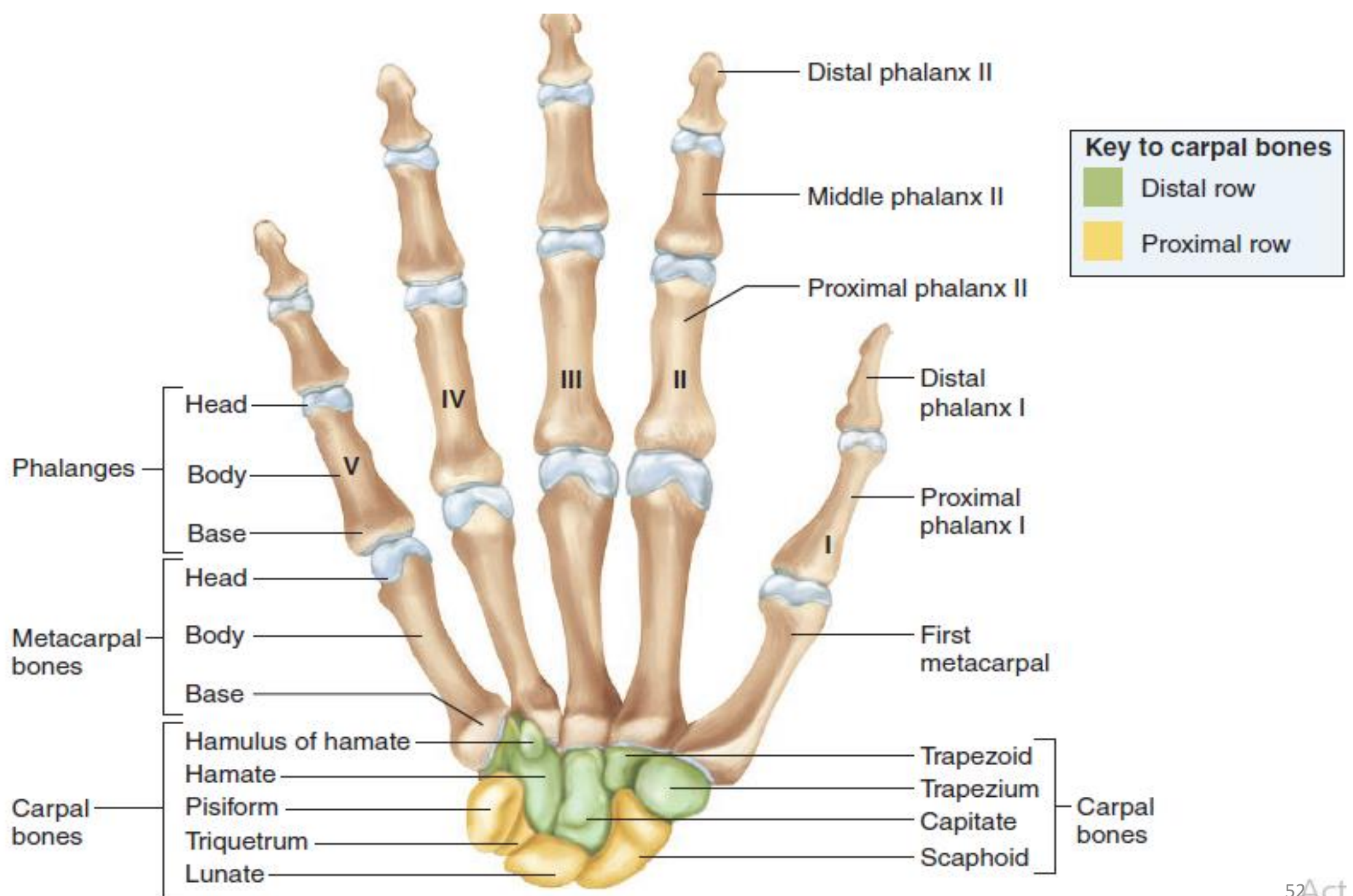
Metacarpal bones

- The **metacarpus** forms the skeleton of the palm of the hand between the carpus and the phalanges.
- Composed of **five** metacarpal bones:
- Each metacarpal consists of a base, shaft, and head.
- The proximal **bases of the meta carpals** articulate with the carpal bones, and the distal **heads of the metacarpals** articulate with the proximal phalanges, and form the **knuckles** of the hand.
- The 1st metacarpal (of the thumb) is the thickest and shortest of these bones.
- The 3rd metacarpal is distinguished by a **styloid process** on the lateral side of its base.



The Phalanges

- The bones of the fingers are called **phalanges**.
- Each digit (finger) has three **phalanges** except for the first (the thumb), which has only two; however, the phalanges of the first digit are stouter than those in the other fingers.
- Composed of **five 14** bones
- Each phalanx has a **base** proximally, a **shaft** (body), and a **head** distally.
- The proximal phalanges are the largest, the middle ones are intermediate in size, and the distal ones are the smallest.
- The terminal phalanges are flattened and expanded at their distal ends, which underlie the nail beds



Anatomy of Joints

- The rigid elements of the skeleton meet at sites called **joints**, or **articulations**.
- *The Greek root arthro means “joint,”* and the scientific study of joints is called *arthrology*.
- A joint (articulation) is a junction b/n 2 or more rigid parts of the skeleton
- Joints, or articulations, link the bones of the skeletal system into a functional whole a system that supports the body, permits effective movement, and protects the softer organs.
- A joint is usually considered movable, but that is not always the case. **Many** joints exhibit limited movement, & others are completely, or almost completely, immovable.

Classification of Joints

- Joints are **classified structurally**, based on their anatomical characteristics, and **functionally**, based on the type of movement they permit.
- ❖ **The structural classification of joints is based on two criteria:**
 - The presence or absence of a space between the articulating bones, called **a synovial cavity**, and
 - The type of connective tissue that binds the bones together.

Classification of Joints,,

❑ The functional classification of joints relates to the degree of movement they permit.

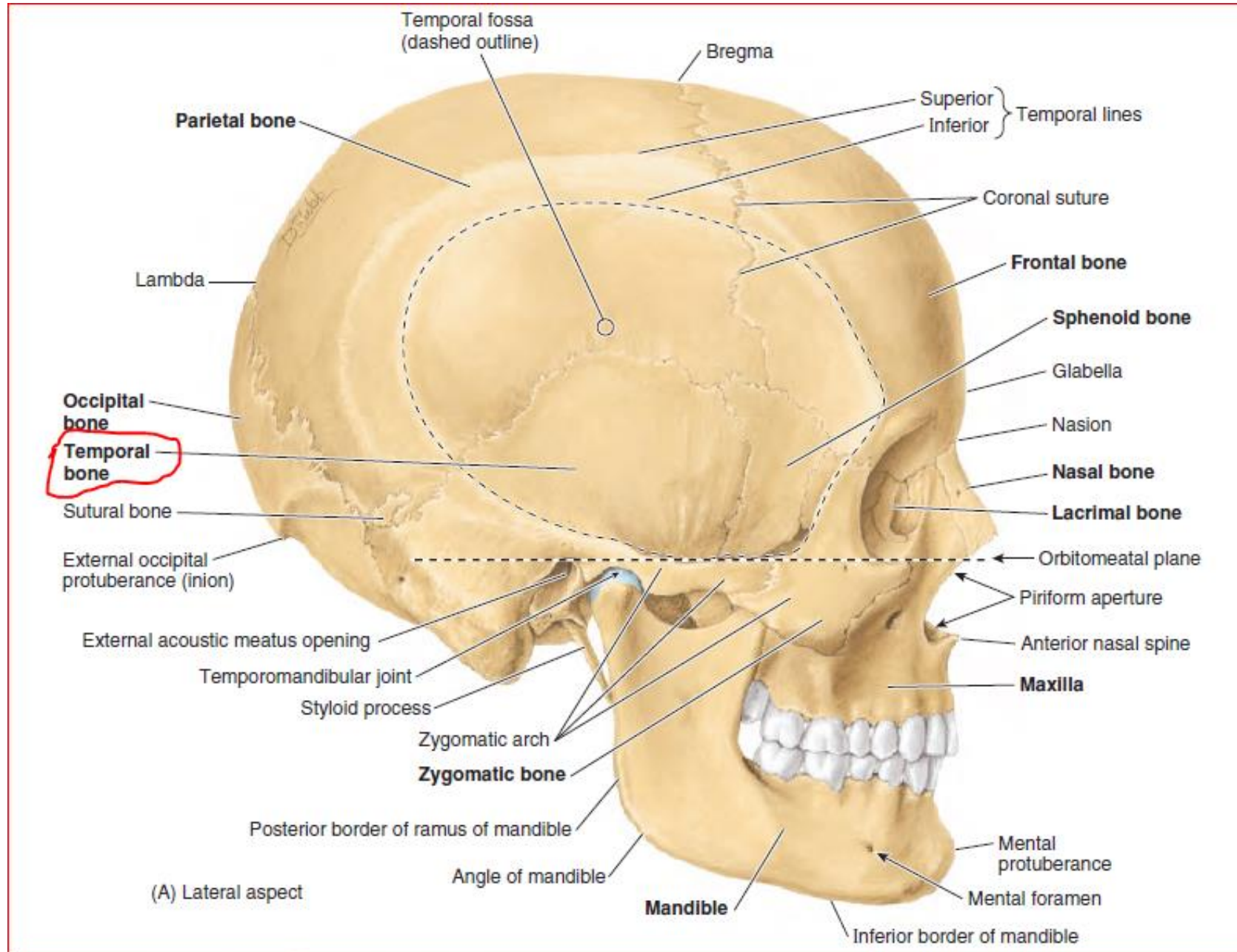
❖ Functionally, joints are classified as one of the following types:

- I. **Synarthrosis:** An immovable joint. The plural is *synarthroses*.
- II. **Amphiarthrosis:** A slightly movable joint. The plural is *amphiarthroses*.
- III. **Diarthrosis:** A freely movable joint. The plural is *diarthroses*.
 - All diarthroses are synovial joints.
 - They have a variety of shapes and permit several different types of movements.

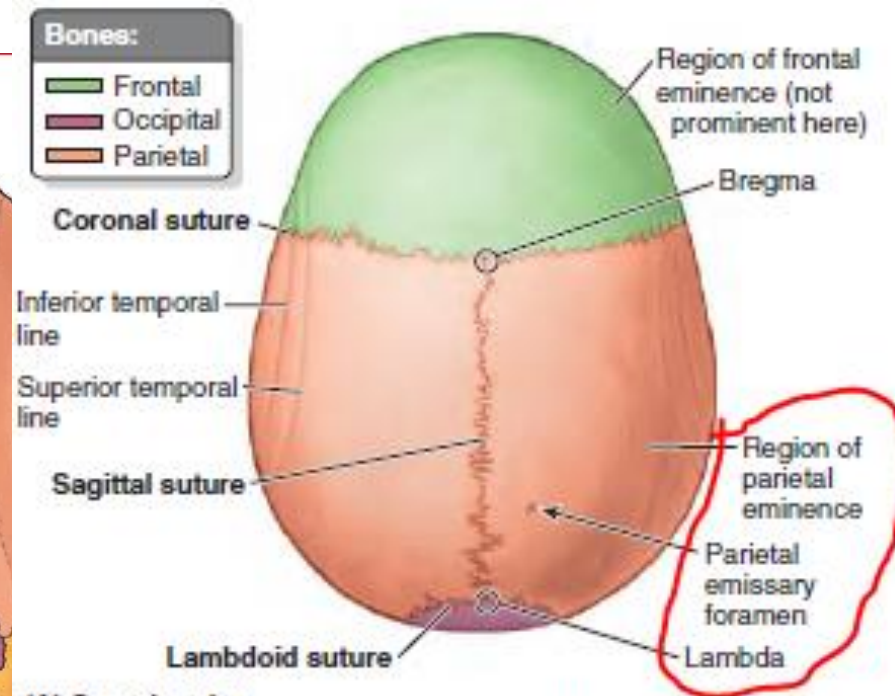
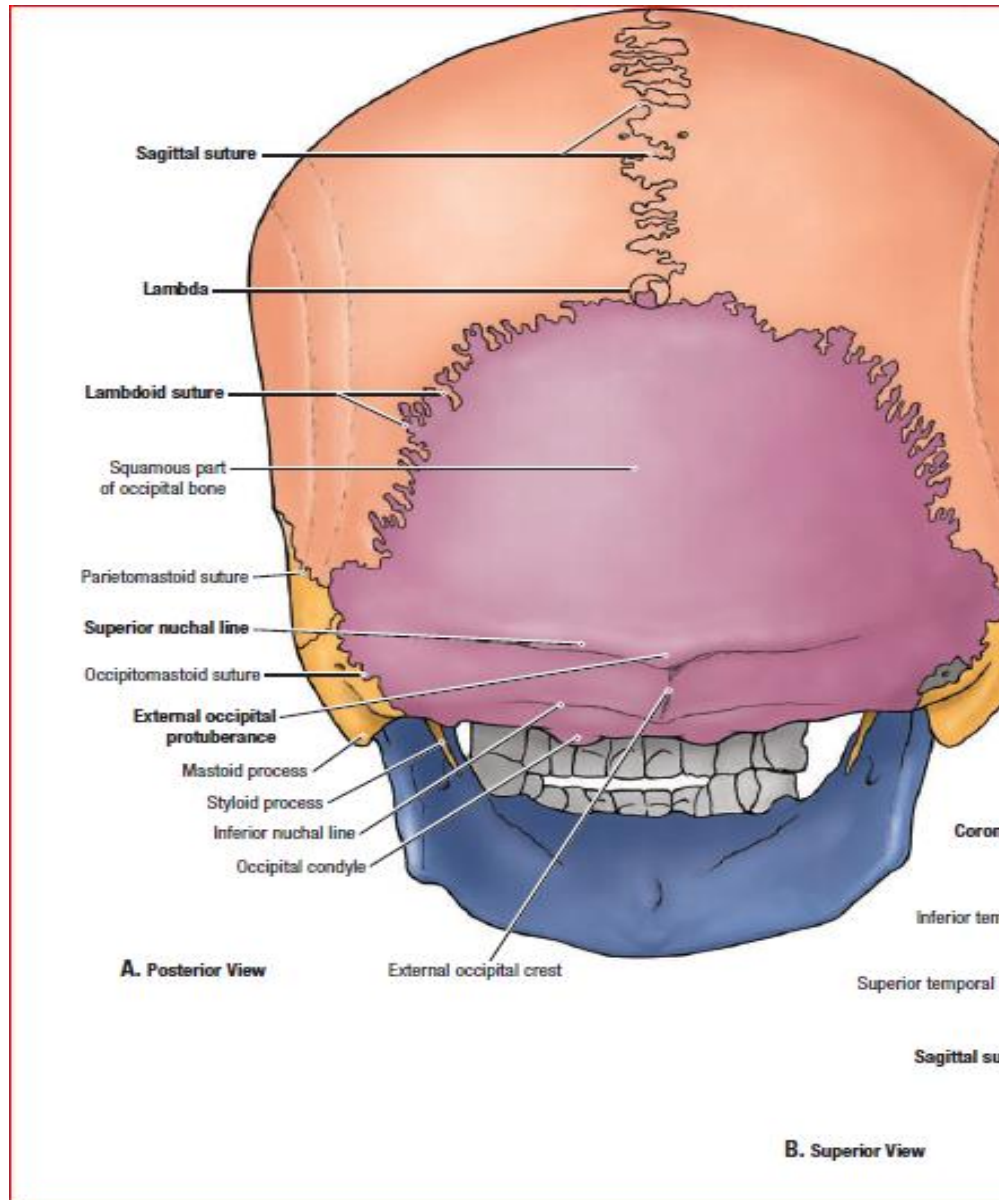
Immovable Joints

- Bones forming an immovable joint, or **synarthrosis** , are tightly joined and are separated by a thin band of dense connective tissue or a thin layer of hyaline cartilage.
- For example, **skull bones**, except the **mandible**, are joined by dense connective tissue **called sutures because** they resemble stitches.
- The joints between bones and teeth are also immovable joints separated by dense connective tissue.
- The epiphysial plates in growing bones, composed of hyaline cartilage, are also immovable joints

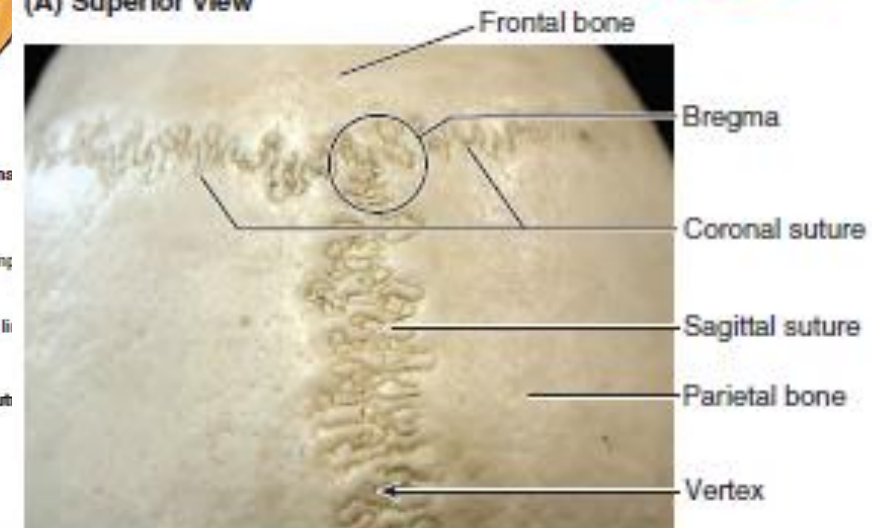
Immovable Joints,,



Immovable Joints,,,



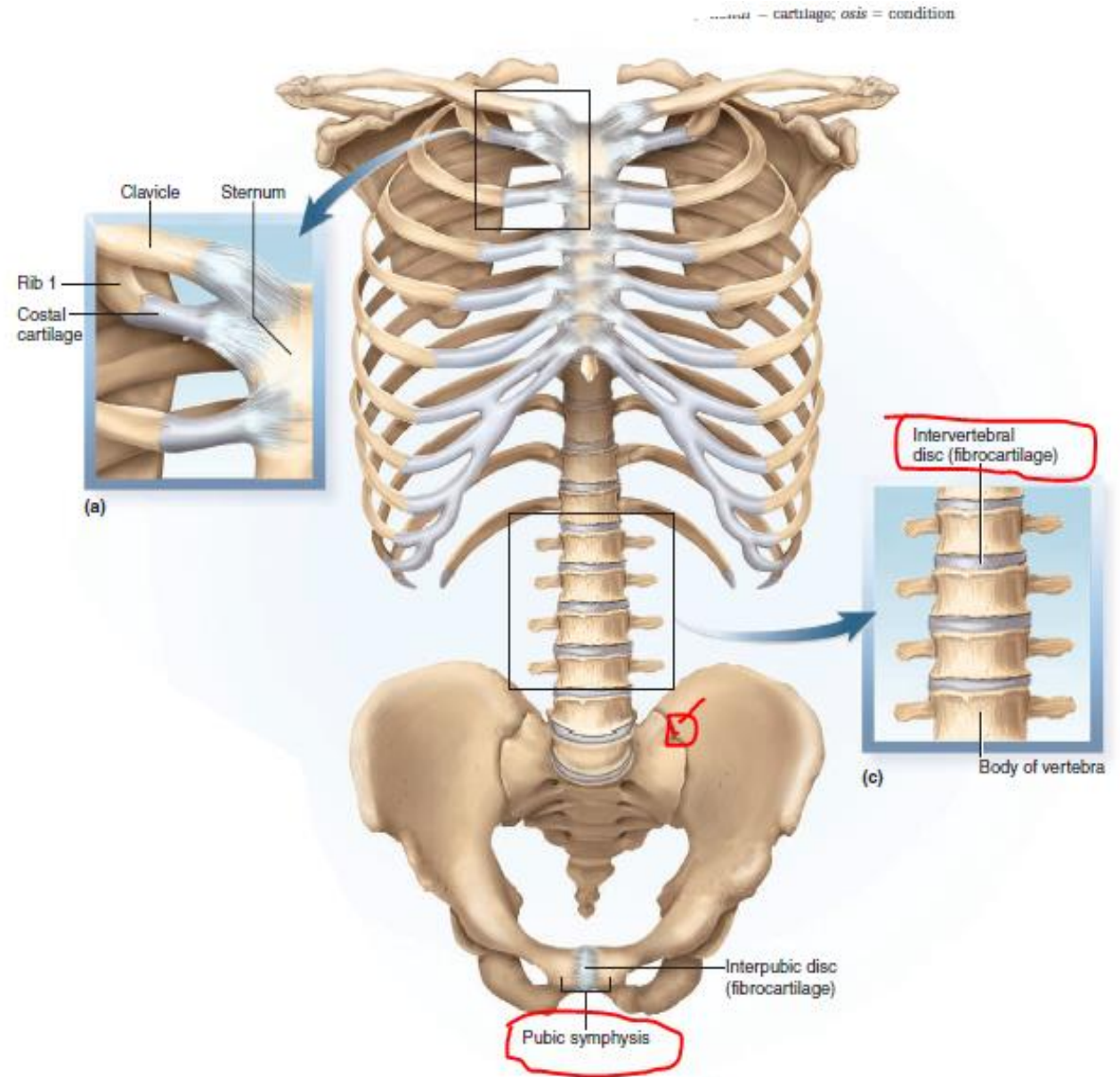
(A) Superior view



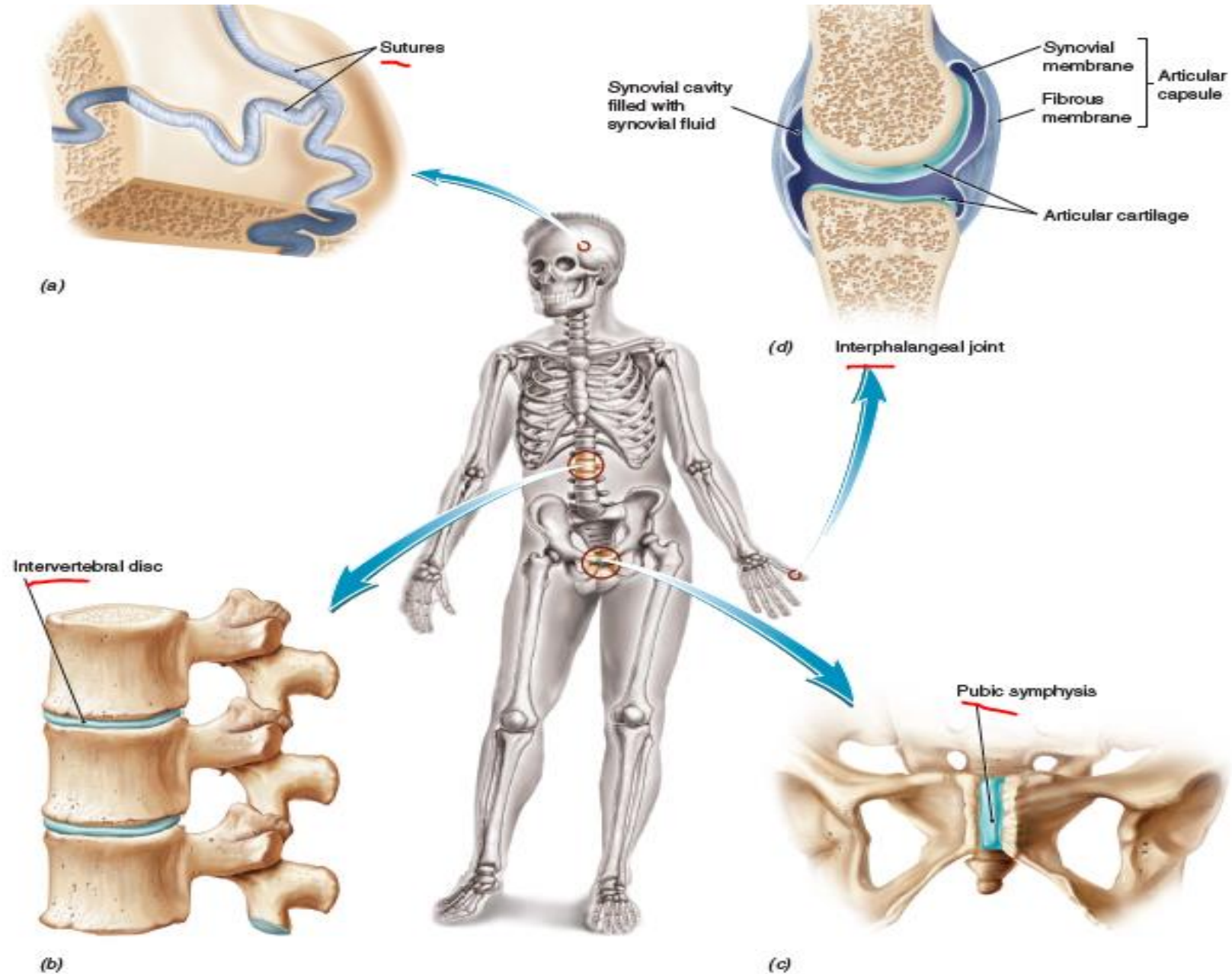
(B) Superior (vertical) aspect

Slightly Movable Joints

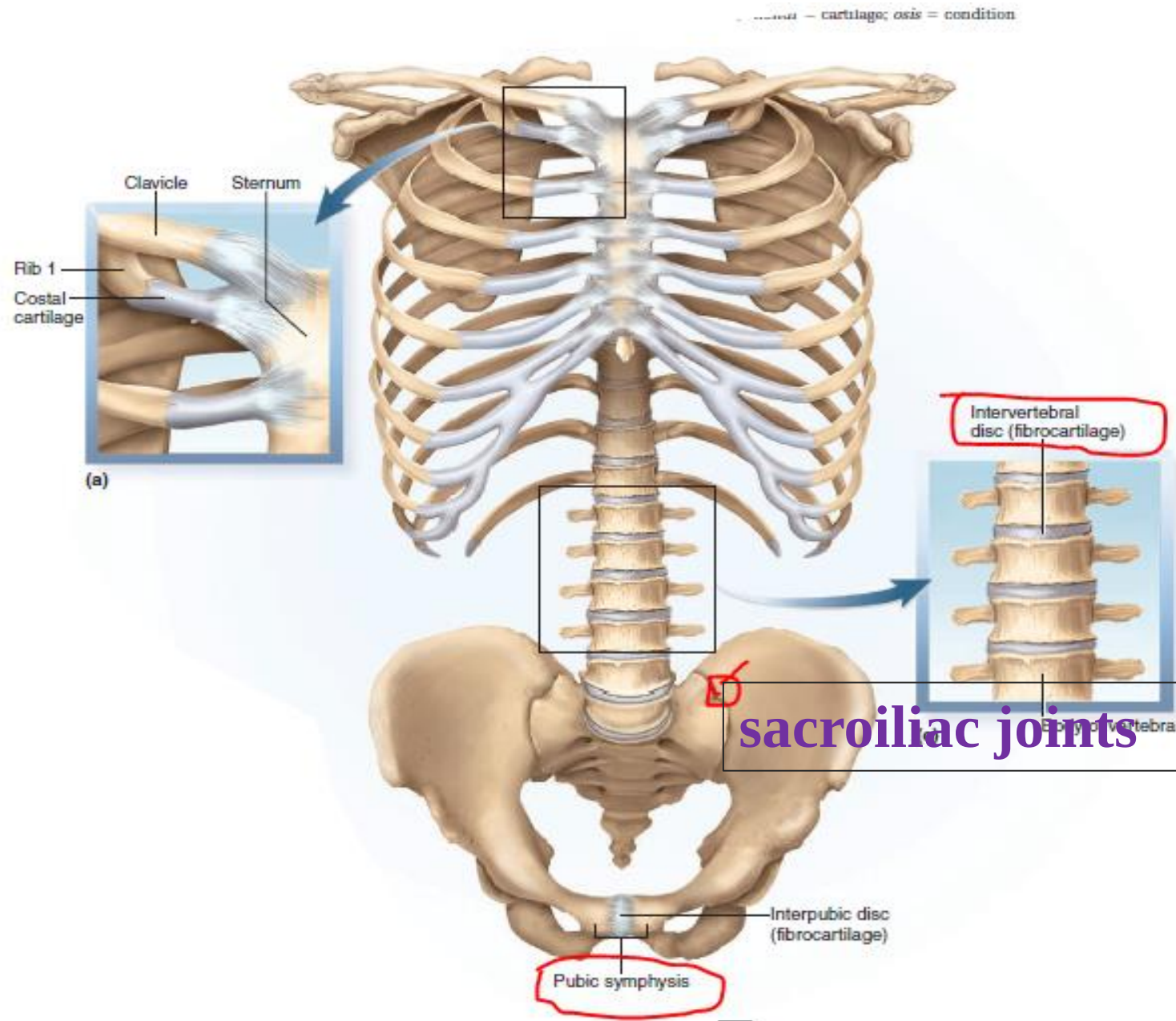
- Bones forming a slightly movable joint, or **amphiarthrosis**, are separated by a layer of cartilage or dense connective tissue.
- For example, the joints formed by adjacent vertebrae contain intervertebral discs formed of fibrocartilage.
- The limited flexibility of the discs allows slight **movement between adjacent vertebrae**.
- Other examples include the pubic symphysis and **sacroiliac** joints.



Slightly Movable Joints



Slightly Movable Joints,,



Freely Movable Joints

- Most articulations are freely movable. These joints are also called **synovial joints**.
- The structure of a freely movable joint, or **diarthrosis**, is more complex.
- The ends of the bones forming the joint are bound together by **an articular, or joint, capsule**.
- The thin internal layer of the capsule, **called *synovial membrane*, secretes *synovial fluid* that lubricates the joint**.
- The ends of the bones are covered with *articular cartilage* , which protects bones and reduces friction.
- **Ligaments**, the cords or bands of dense regular connective tissue that connect bones together, reinforce the joints.

Freely Movable Joints,,,

- ❖ Freely movable joints are categorized into several types based on their structure and types of movements: *Plane Joints, Condylar Joints, Saddle Joint, Hinge Joints, Pivot Joints and Ball-and-Socket Joints.*
- ❖ *Movements at Freely Movable Joints;* Movement at a joint results from the contraction of skeletal muscles that span across the joint.
- The type of movement that occurs is determined by the type of joint and the location of the muscle or muscles involved.

The structural classification of joints

- **The structural classification of joints** is based on whether the articulating surfaces of the adjacent bones are directly connected by fibrous connective tissue or cartilage, or whether the articulating surfaces contact each other within a fluid filled joint cavity.
- These differences serve to divide the joints of the body into three structural classifications.
- ❖ **Structurally**, joints are classified as **fibrous**, **cartilaginous**, or **synovial joints**.

The structural classification of joints,,

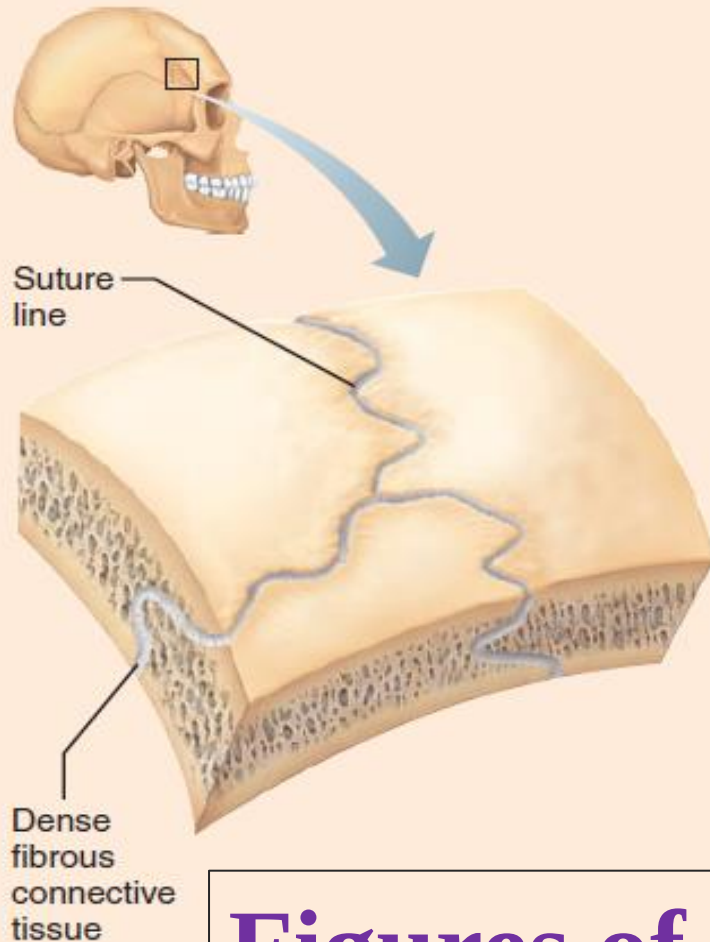
Summary of Joint Classes

Structural Class	Structural Characteristics	Types		Mobility
Fibrous	Adjoining bones united by collagenic fibers	1. Suture (short fibers)		Immobile (synarthrosis)
		2. Syndesmosis (longer fibers)		Slightly mobile (amphiarthrosis) and immobile
		3. Gomphosis (periodontal ligament)		Immobile
Cartilaginous	Adjoining bones united by cartilage	1. Synchondrosis (hyaline cartilage)		Immobile
		2. Symphysis (fibrocartilage)		Slightly movable
Synovial	Adjoining bones separated by a joint cavity, covered with articular cartilage, and enclosed within an articular capsule lined with synovial membrane	1. Plane 2. Hinge 3. Pivot	4. Condylloid 5. Saddle 6. Ball and socket	Freely movable (diarthrosis) movements depend on design of joint

The structural classification of joints,,

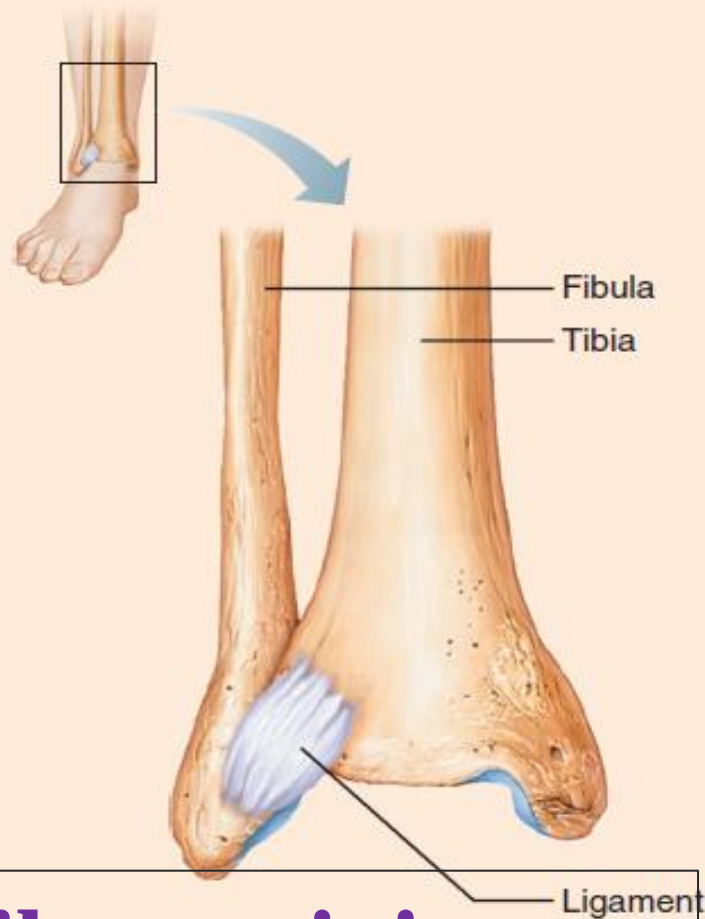
(a) Suture

Joint held together with very short, interconnecting fibers, and bone edges interlock. Found only in the skull.



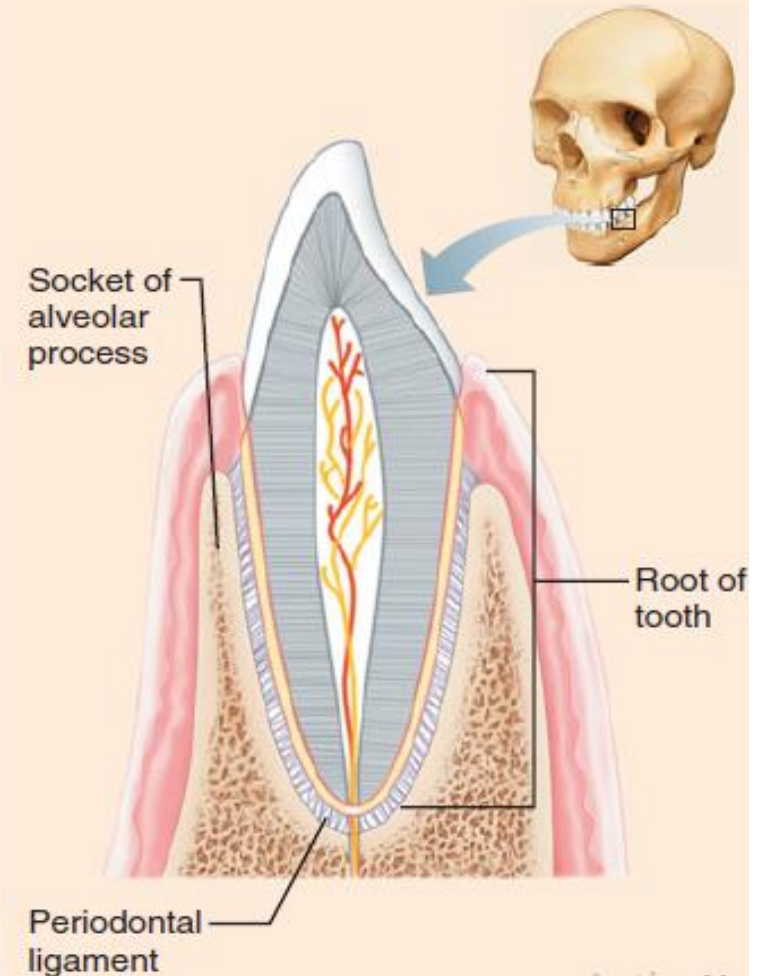
(b) Syndesmosis

Joint held together by a ligament. Fibrous tissue can vary in length but is longer than in sutures.



(c) Gomphosis

Peg-in-socket fibrous joint. Periodontal ligament holds tooth in socket.

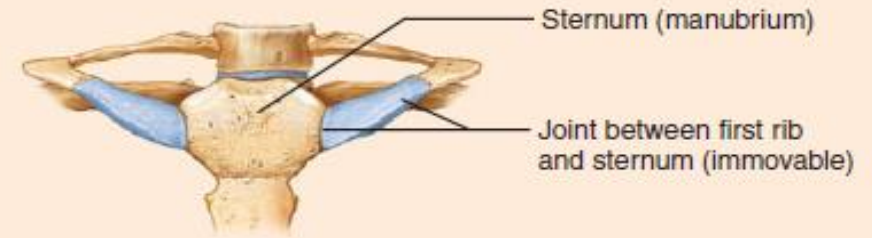
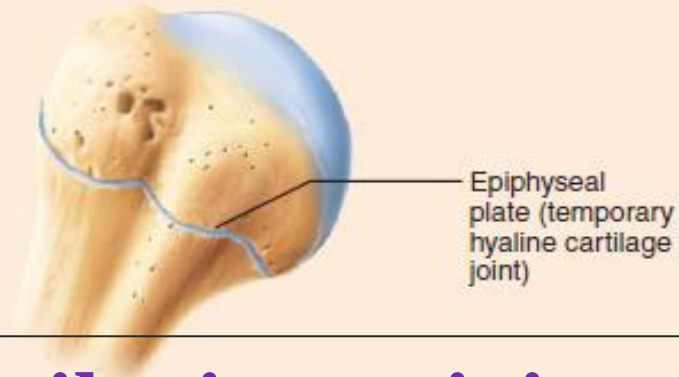


Figures of Fibrous joints

The structural classification of joints,,

(a) Synchondroses

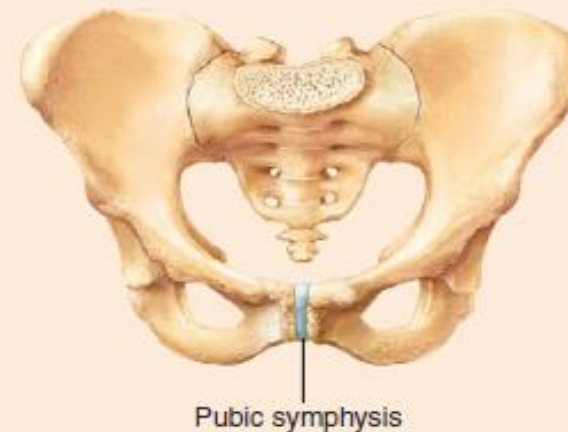
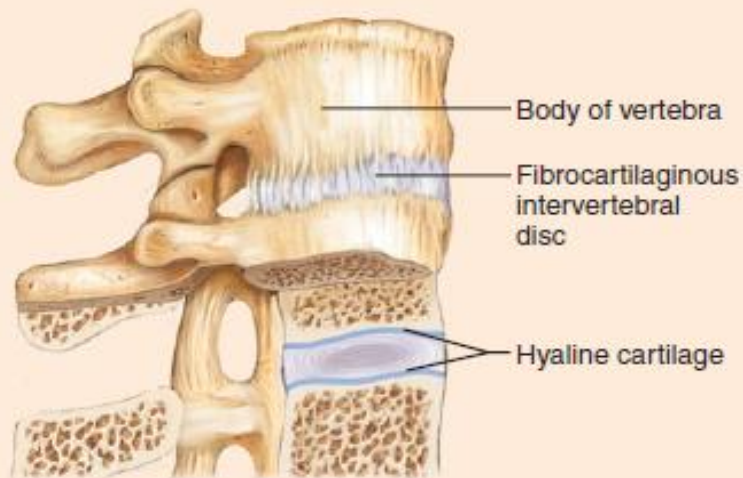
Bones united by hyaline cartilage



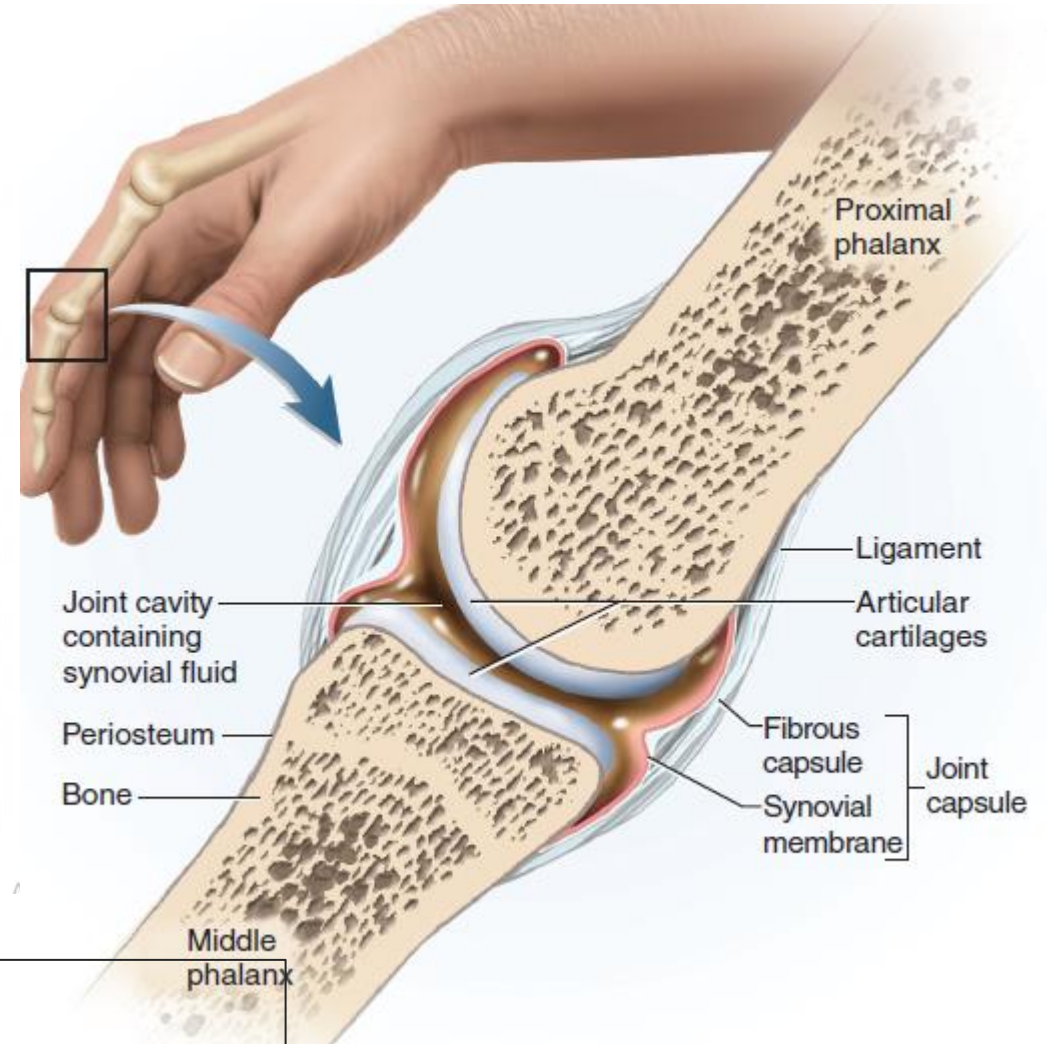
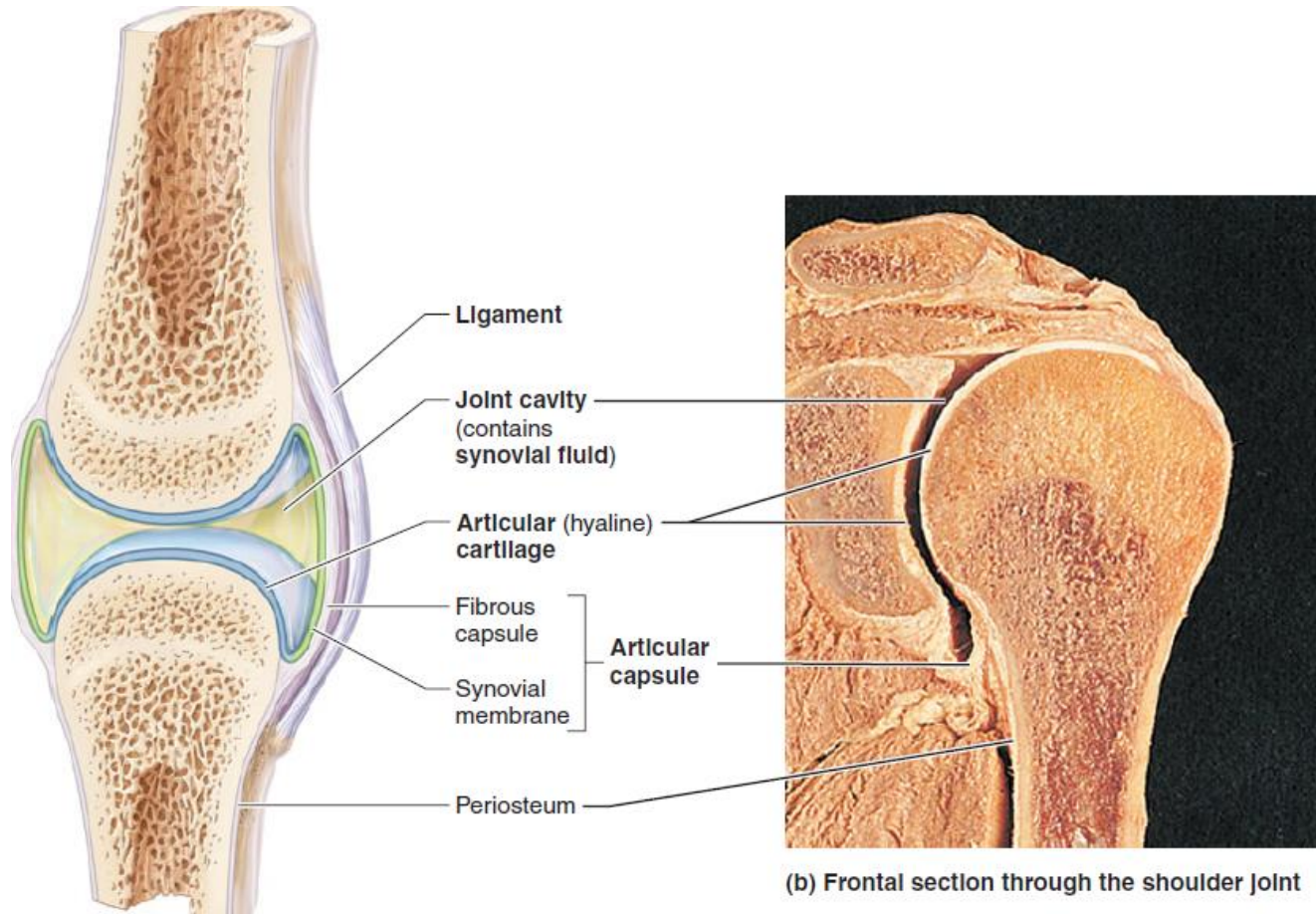
Figures of Cartilaginous joints

(b) Symphyses

Bones united by fibrocartilage



The structural classification of joints,,



Figures of synovial joints

The structural classification of joints,,,

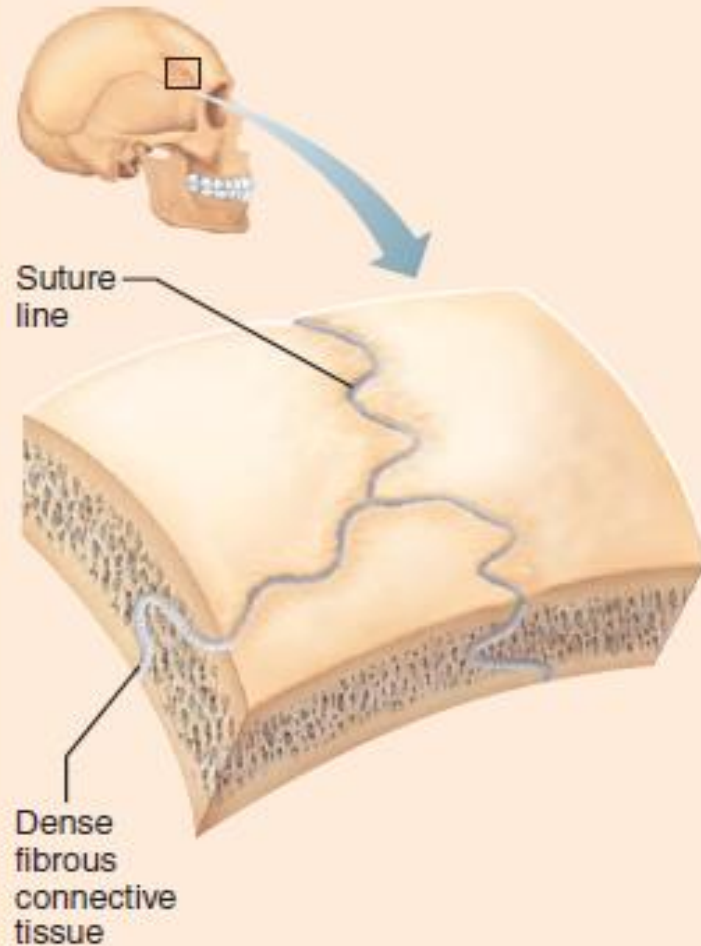
❑ Structurally, joints are classified as one of the following types:

- ❖ **Fibrous joints:** There is no synovial cavity, and the bones are held together by dense irregular connective tissue that is rich in collagen fibers.
 - Joints in this group are further subdivided on the basis of structure as sutures, **syndesmoses, or gomphoses.**
 - **Sutures** are fibrous joints between the bones of the skull
 - **Syndesmoses** are fibrous joints in which the bones are separated by some distance and held together by ligaments. **An example is the fibrous membrane connecting most of the distal parts of the radius and ulna.**
 - **Gomphoses** consist of pegs fitted into sockets and held in place by ligaments. The joint between a tooth and its socket is a gomphosis.

Fibrous joints

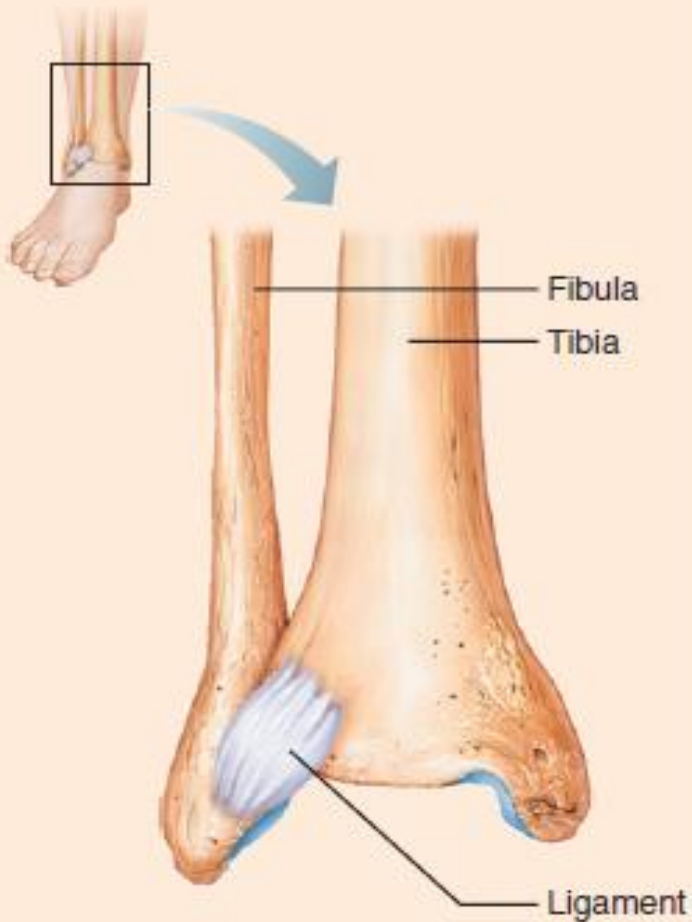
(a) Suture

Joint held together with very short, interconnecting fibers, and bone edges interlock. Found only in the skull.



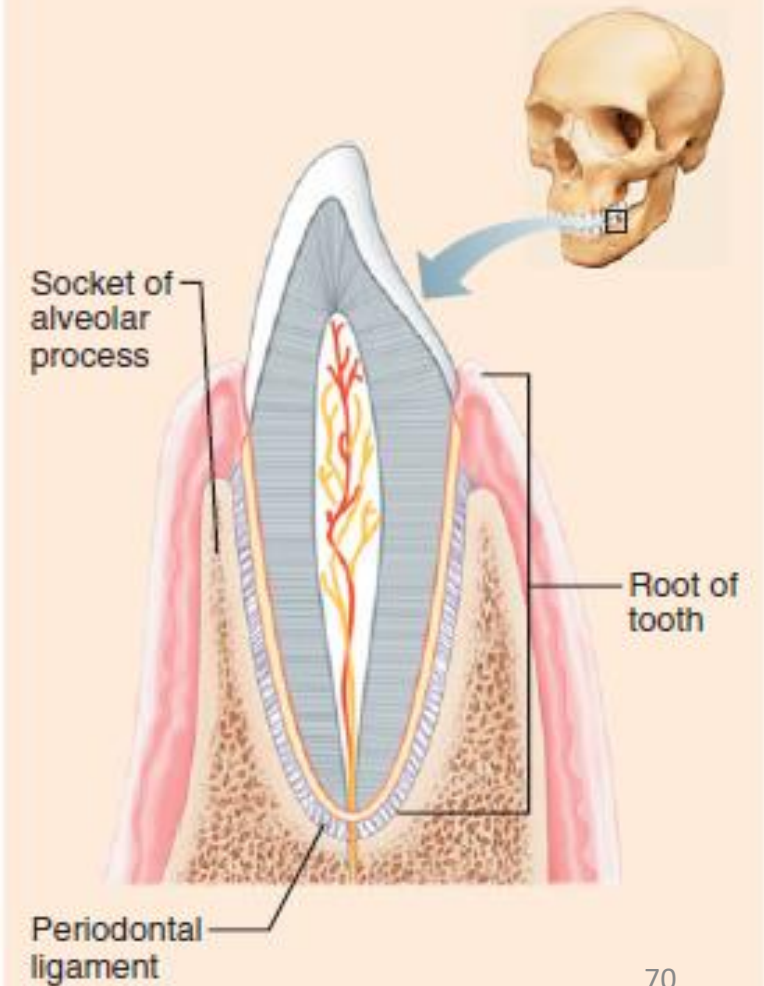
(b) Syndesmosis

Joint held together by a ligament. Fibrous tissue can vary in length but is longer than in sutures.



(c) Gomphosis

Peg-in-socket fibrous joint. Periodontal ligament holds tooth in socket.



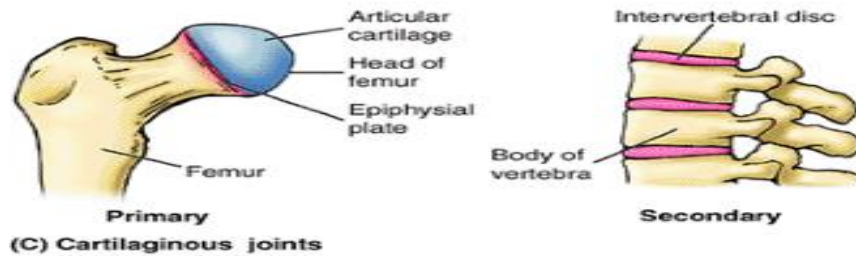
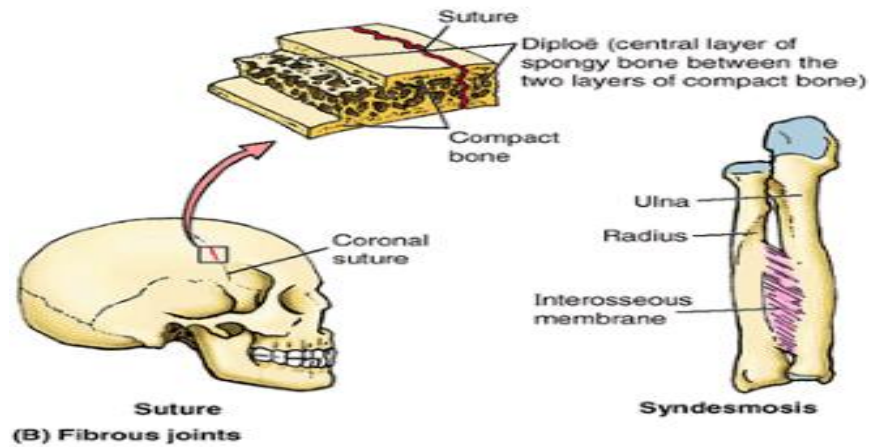
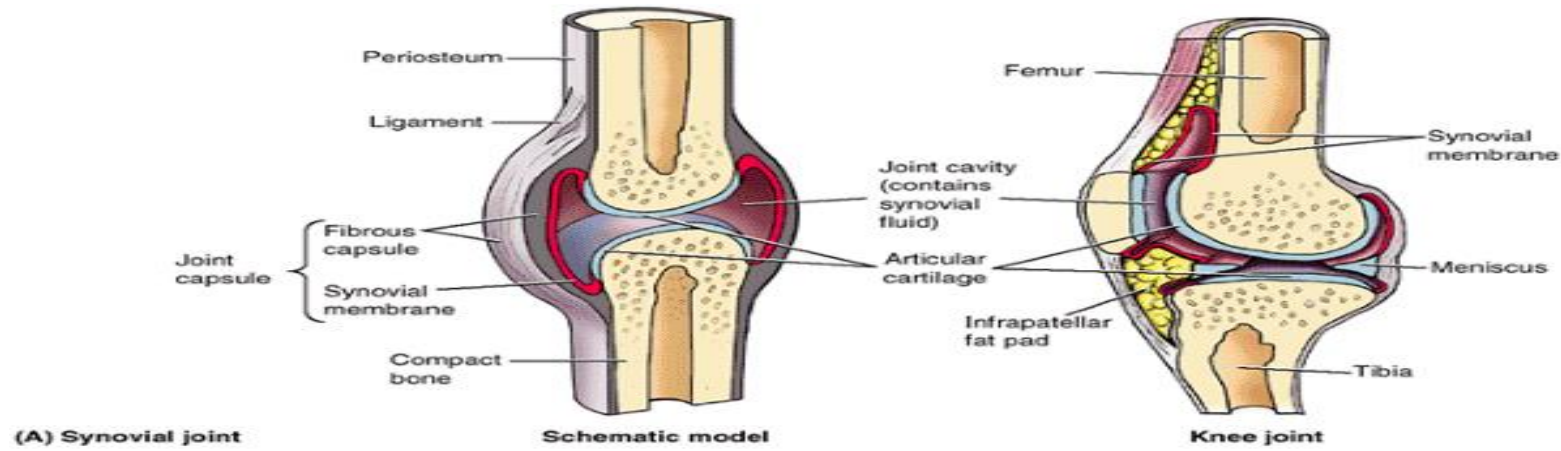
Fibrous joints,,,

- The articulating bones of fibrous joints are united by **fibrous tissue**
- The amount of movement occurring at a fibrous joint depends in most cases *on the length of the fibers uniting the articulating bones*
- *The sutures of the cranium are examples of fibrous joints*
- These bones are close together, either interlocking along a wavy line or overlapping
- **A syndesmosis type of fibrous joint** unites the bones with a sheet of fibrous tissue, either a ligament or a fibrous membrane. Consequently, this type of joint is *partially movable*
- The interosseous membrane in the forearm is a sheet of fibrous tissue that joins the radius and ulna in a syndesmosis

Cartilaginous joints

- The articulating structures of cartilaginous joints are united by ***hyaline cartilage*** or ***fibrocartilage***
- ***In primary cartilaginous joints***, or ***synchondroses***, the bones are united by ***hyaline cartilage***, which permits slight bending during early life
- ***Secondary cartilaginous joints***, or ***symphyses***, are strong, slightly movable joints united by ***fibrocartilage***
- **The fibrocartilaginous** intervertebral discs between the vertebrae consist of binding connective tissue that joins the vertebrae together.

Cartilaginous joints,,



Cartilaginous joints,,

❖ **In cartilaginous joints** there is no synovial cavity, and the bones are held together by cartilage. **Cartilaginous joints** unite two bones by means of cartilage.

❑ **Only slight movement can occur at these joints.**

- Examples are the cartilage in the epiphyseal plates of growing long bones and the cartilages between **the ribs and the sternum**.
- The cartilage of some cartilaginous joints, where much strain is placed on the joint, may be reinforced by additional collagen fibers.
- This type of cartilage, called **fibrocartilage**, forms joints such as the intervertebral disks

Cartilaginous joints,,,

(a) Synchondroses

Bones united by hyaline cartilage



Epiphyseal plate (temporary hyaline cartilage joint)

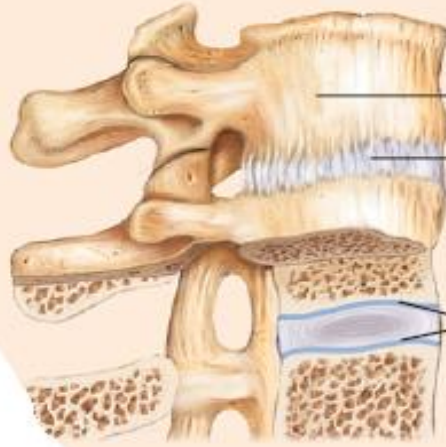


Sternum (manubrium)

Joint between first rib and sternum (immovable)

(b) Symphyses

Bones united by fibrocartilage



Body of vertebra

Fibrocartilaginous intervertebral disc

Hyaline cartilage



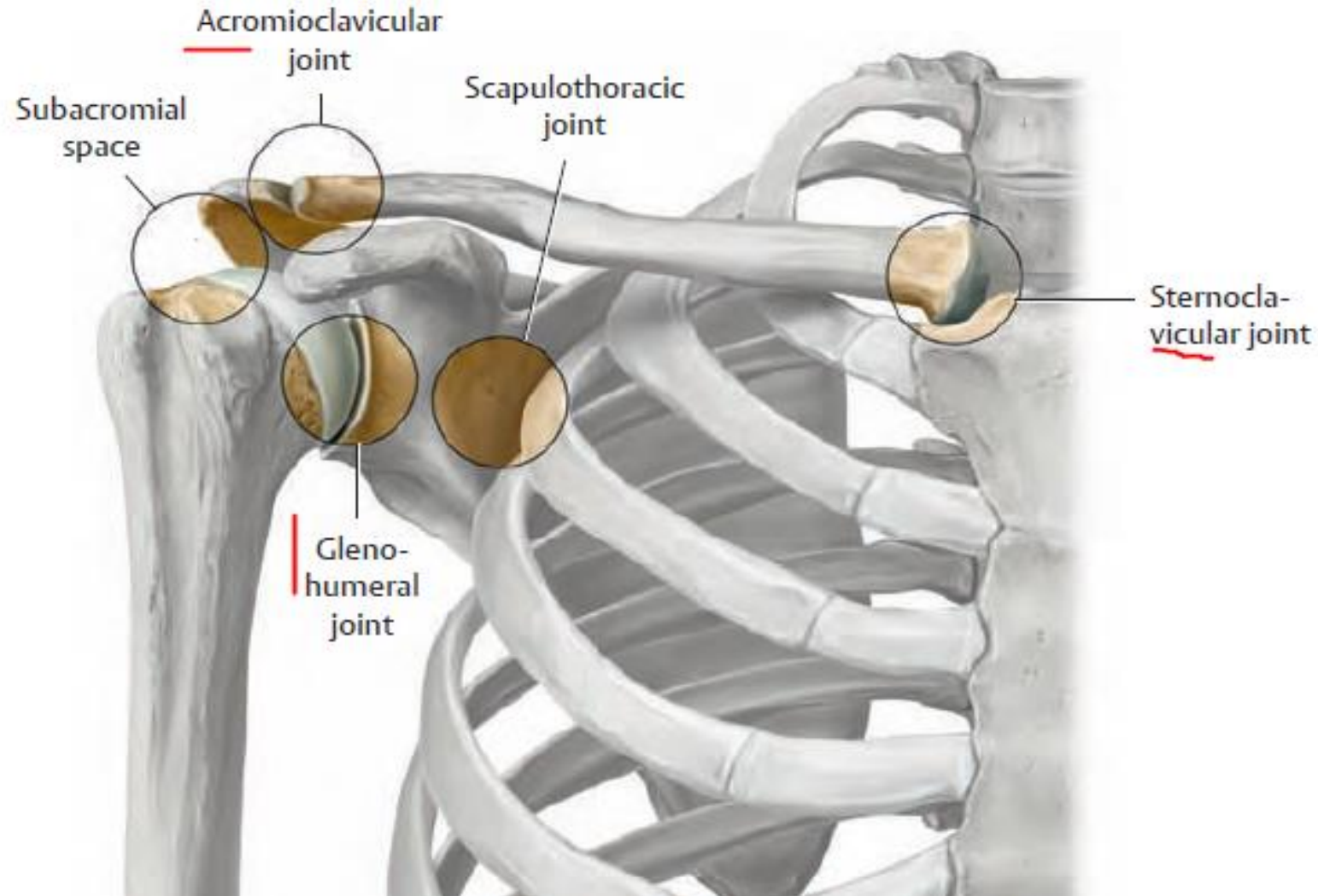
Pubic symphysis

Cartilaginous joints.

Joints of the upper limb

The shoulder joint

- The shoulder joint complex is composed of three joints; **the sternoclavicular, acromioclavicular & glenohumeral** joints, usually all moving simultaneously.

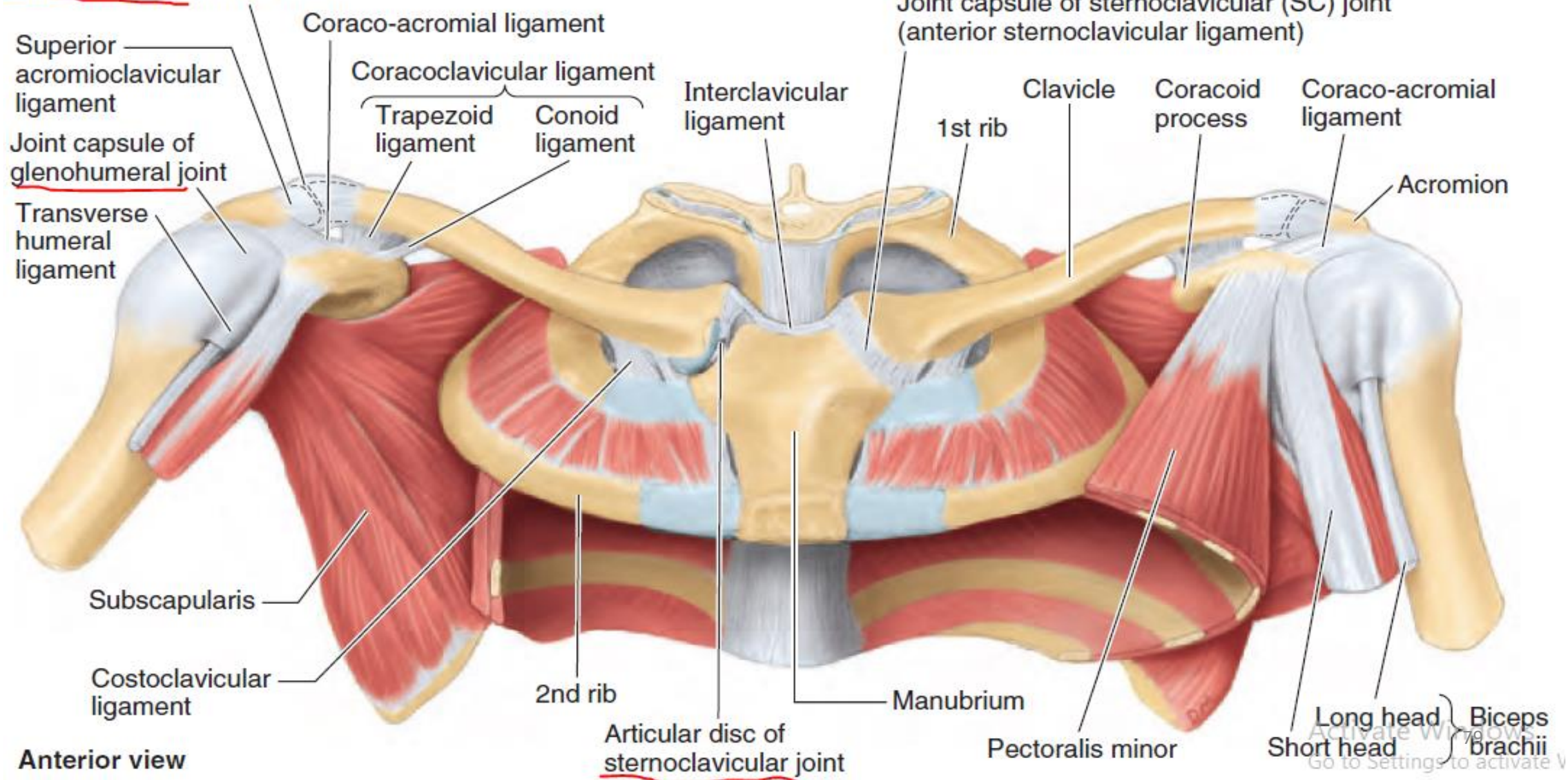


The shoulder joint,,

- The mobility provided by **three** joints working together gives the shoulder complex a wide range of movement.
- The extensive mobility in the shoulder joint complex also results from the flexible support provided by numerous muscles, ligaments, and tendons.
- Functional defects in any of the joints impair movements of the pectoral girdle.
- The bones articulating as part of the shoulder joint complex are the humerus, scapula, clavicle, and sternum.

The shoulder joint,,

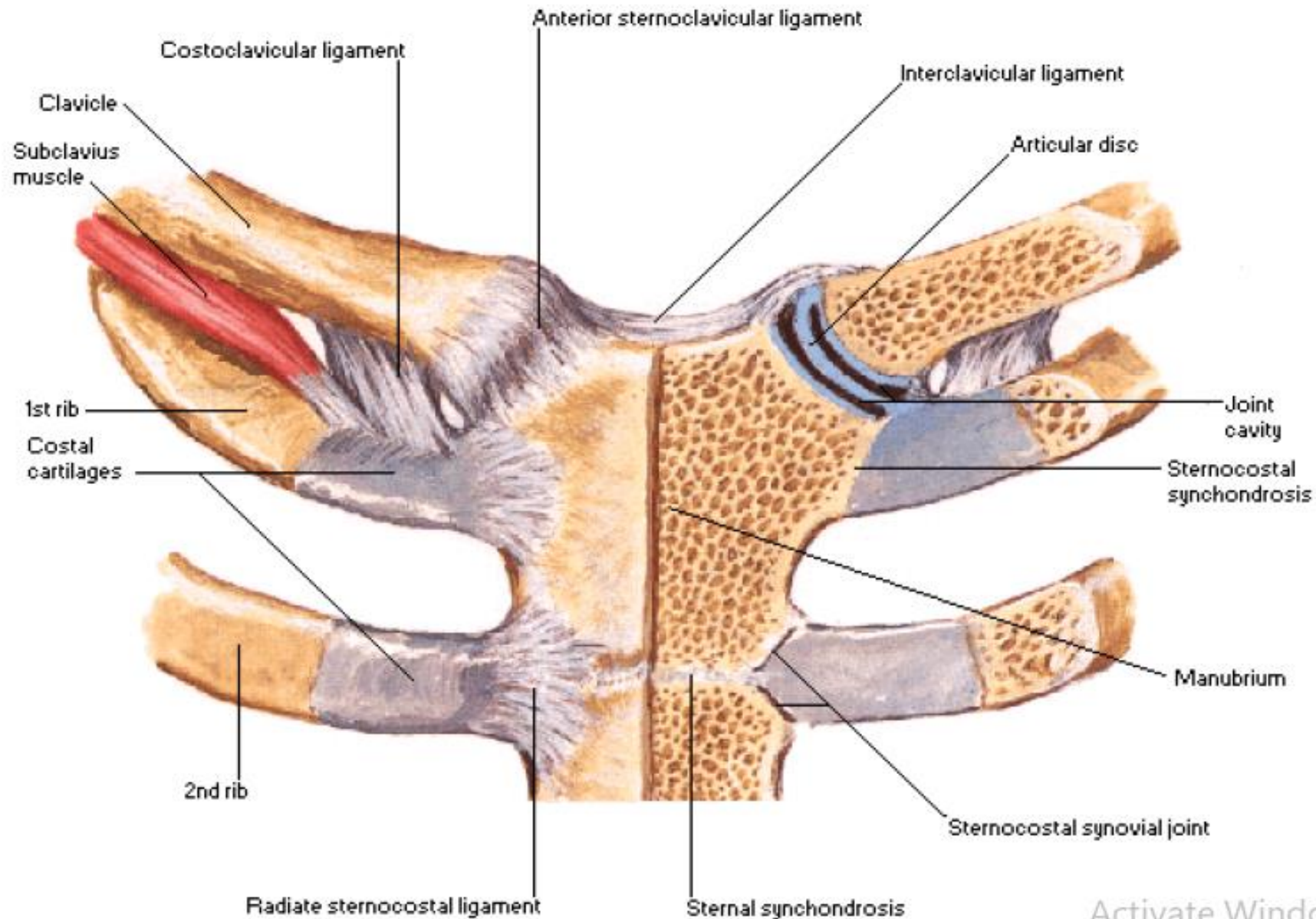
Acromioclavicular (AC) joint



Sternoclavicular Joint

- The presence of an articular disc separates the joint into two cavities, which prevents the clavicle from overriding the sternum.
- ✓ **Type: saddle synovial articulation**
- ✓ It is between sternal end of clavicle and manubrium of sternum and 1st costal cartilage
- ✓ **The SC joint is the only articulation between the upper limb and the axial skeleton**, divided into two compartments by articular disc
- ❖ **Joint capsule** the joint is surrounded by a capsule and is reinforced by ligaments.

Sternoclavicular Joint,,

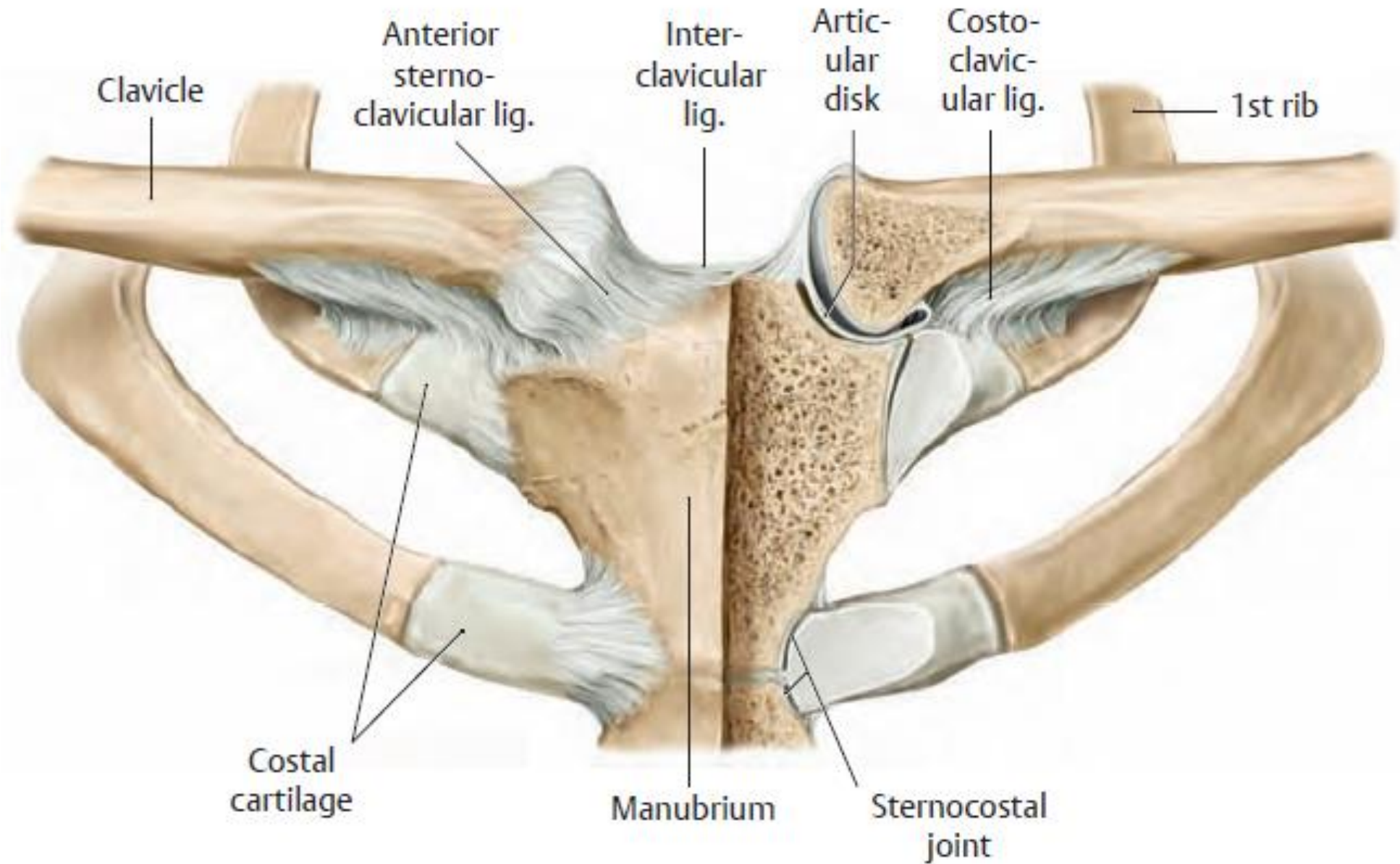


Sternoclavicular Joint ,,

□ Ligaments of Sternoclavicular joint:

- ✓ Anterior and posterior SC ligaments reinforce joint capsule anteriorly & posteriorly
- ✓ Interclavicular ligament strengthens the capsule superiorly
- ✓ Costoclavicular ligament • anchors the inferior surface of the sternal end of the clavicle to the 1st rib and its costal cartilage and limiting elevation of the pectoral girdle
- The strength of the SC joint depends on its ligaments and articular disc.
- ❖ Sternoclavicular joint allows elevation and depression, protraction and retraction, and circumduction of the shoulder.

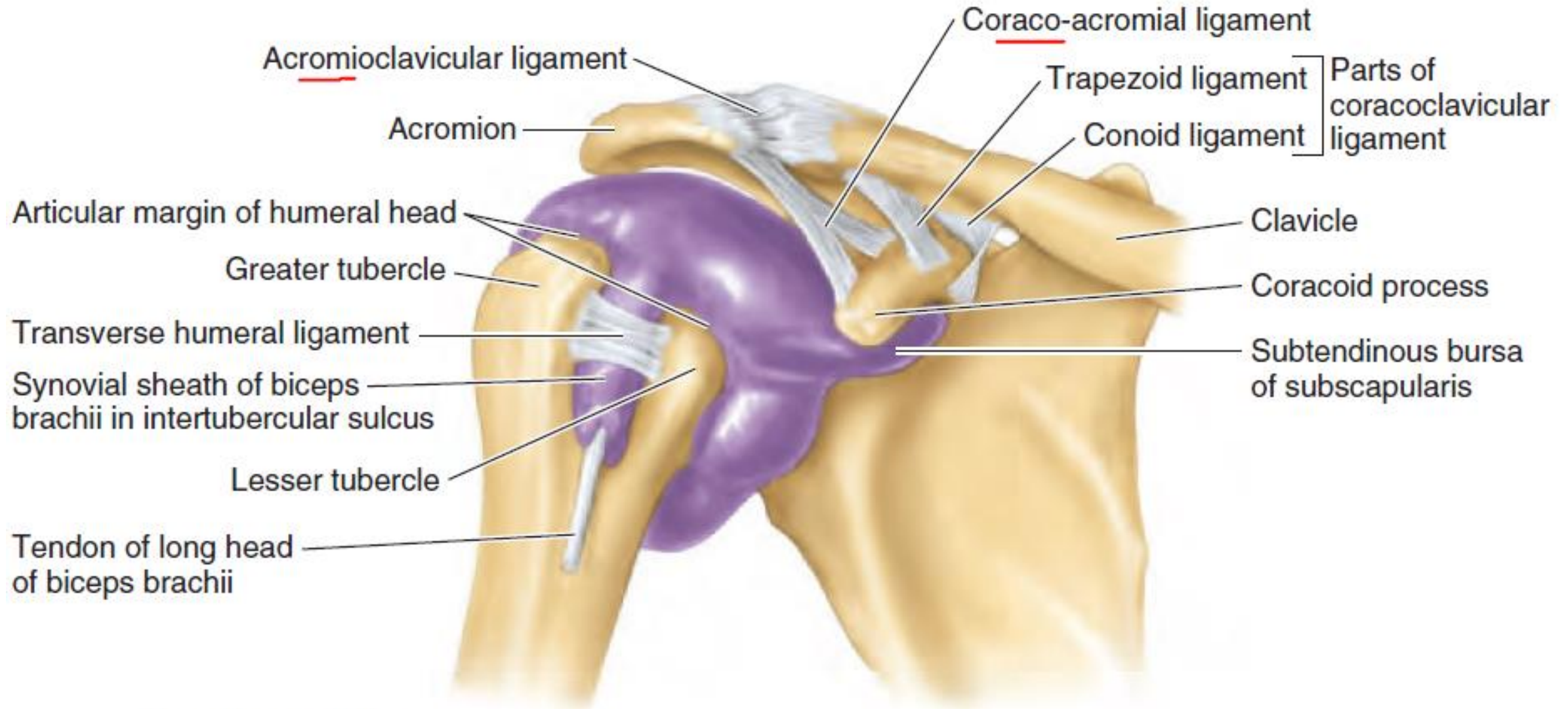
Sternoclavicular Joint ,,



Acromioclavicular Joint

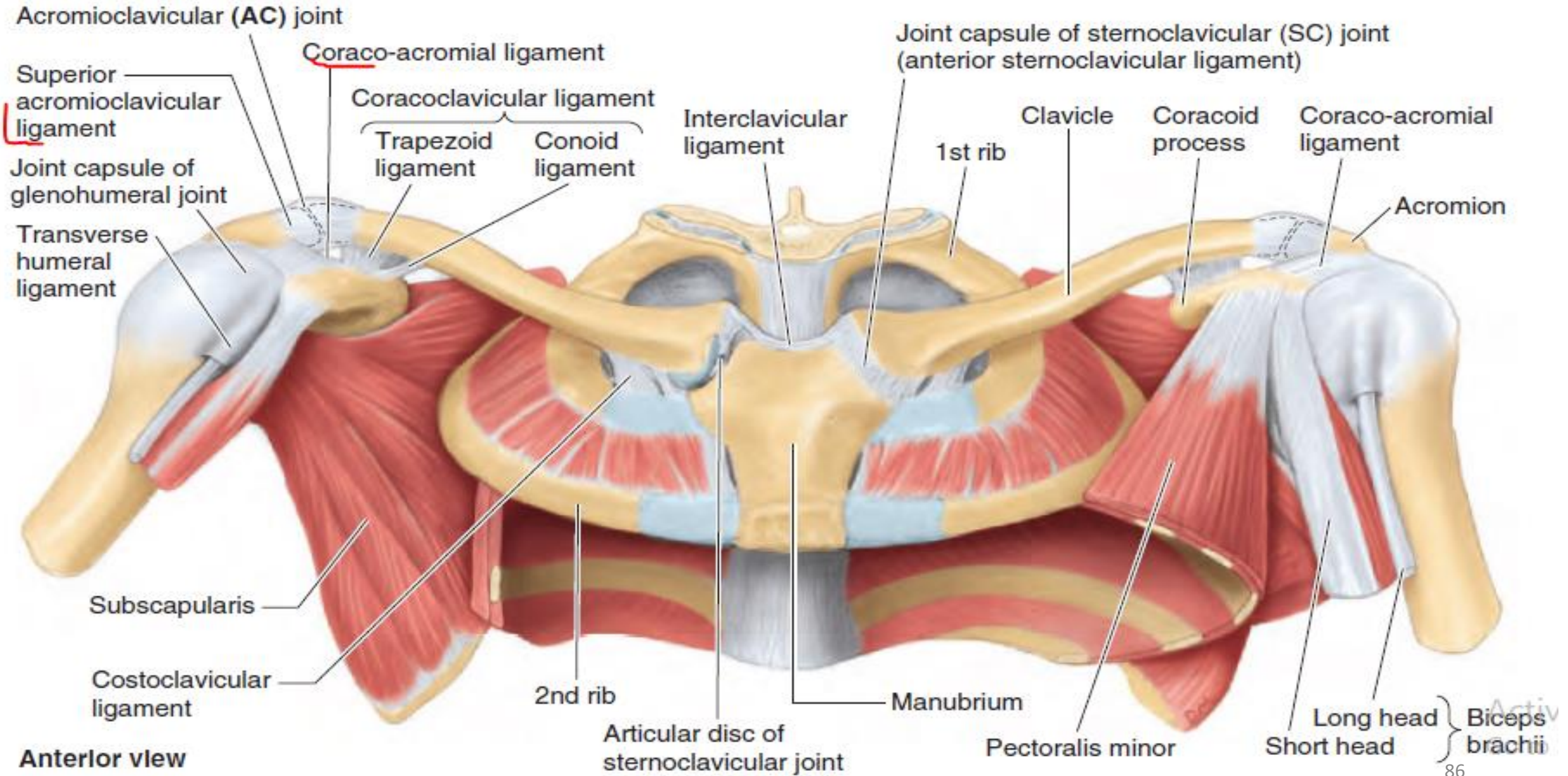
- The **acromioclavicular joint** (AC joint) is a plane **type of synovial joint b/n the** lateral end of the clavicle and the medial surface of the acromion of the scapula.
- **This joint** allows a gliding movement when the scapula rotates.
- The joint is surrounded by a capsule and is reinforced by several ligaments.
- Superiorly, the capsule is thickened to form the **acromioclavicular ligament**.
- However, the major support for the joint is the **coracoclavicular ligament**.
- This ligament is composed of **2 parts o the trapezoid & conoid ligaments**, w/c connect the inferior aspect of the clavicle to the coracoid process of the scapula

Acromioclavicular Joint,,



(A) Anterior view

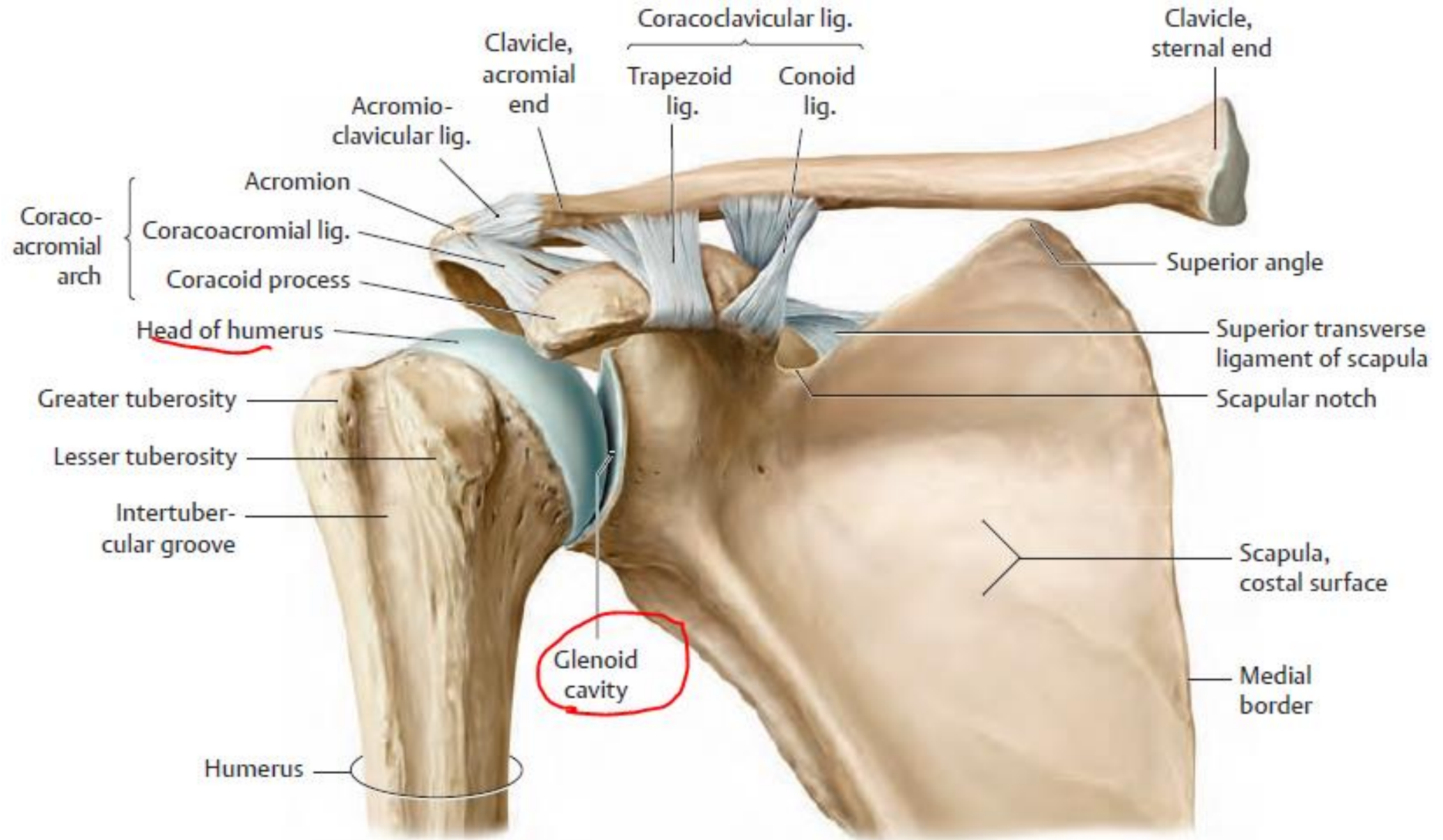
Acromioclavicular Joint,,



Glenohumeral joint

- The **glenohumeral (shoulder) joint** is a **multiaxial, ball-and socket synovial joint** which allows abduction & adduction, flexion & extension, and circumduction & rotation.
- This joint has more freedom of movement than any other joint in the body.
- The head of the humerus is large compared with the shallow glenoid cavity, this joint is relatively unstable but several factors help to increase the its stability.
- Is surrounded by the **fibrous capsule** attached to the margin of the glenoid cavity.
- **The capsule** is reinforced by the **rotator cuff muscles**, the **glenohumeral ligaments**, and the **coracohumeral ligaments**.

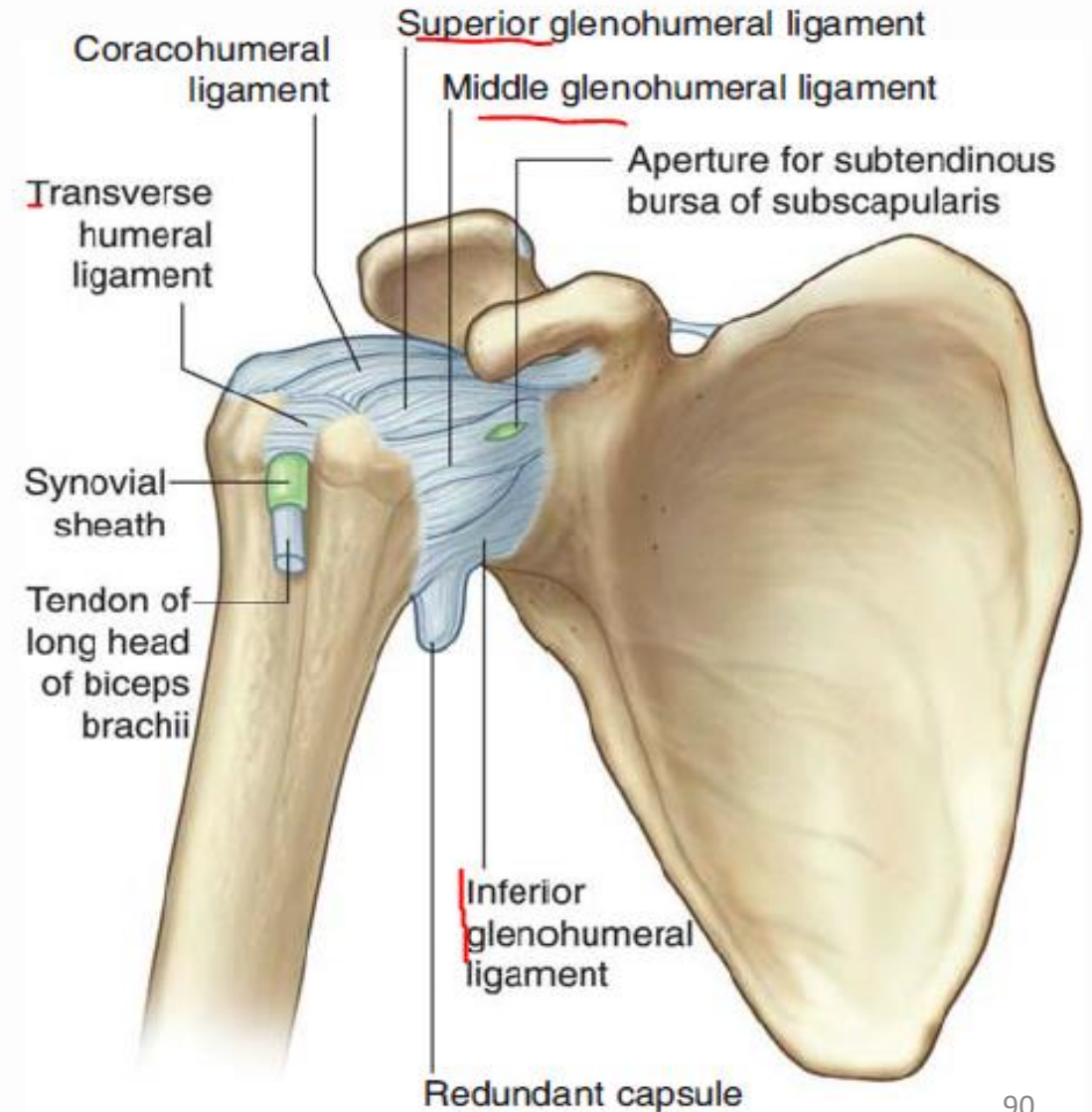
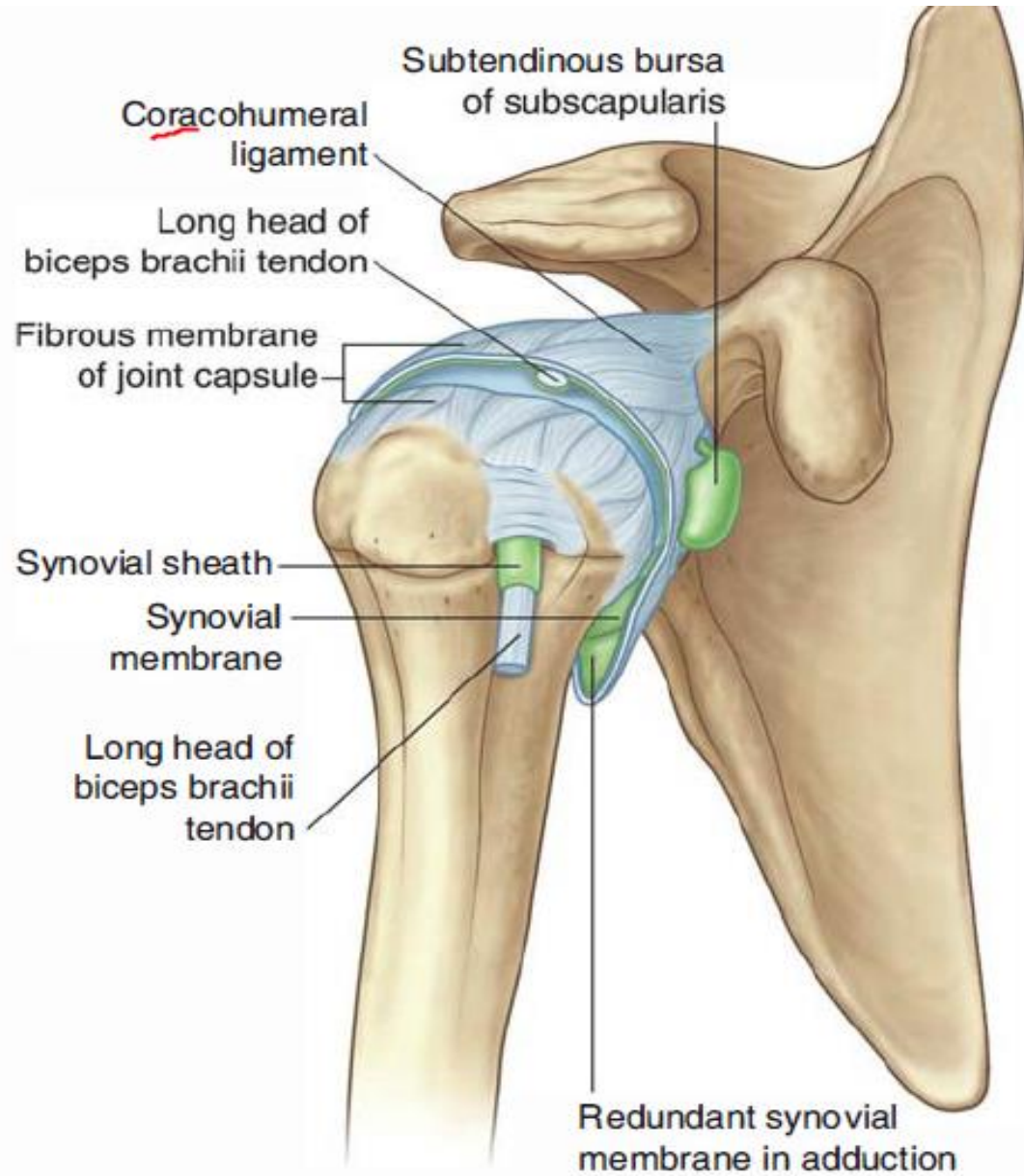
Glenohumeral joint,,



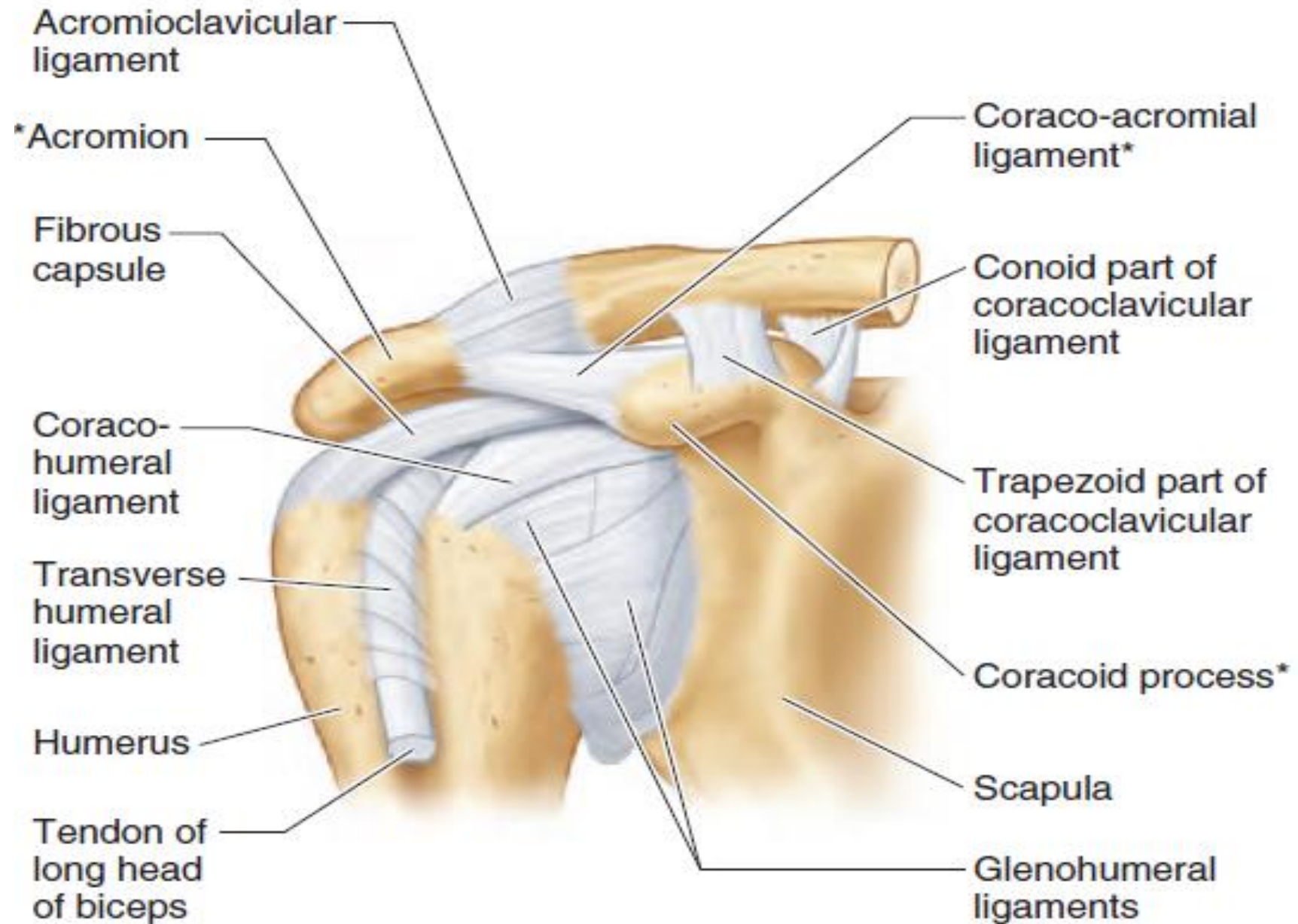
Glenohumeral joint,,

- The joint capsule is reinforced **anteriorly** by the **superior, middle and inferior glenohumeral ligaments**.
- These are referred to as **intrinsic ligaments** due to the fact that they are part of the joint capsule.
- Superiorly, the capsule is reinforced by a further intrinsic ligament – the **coracohumeral** ligament.
- Also reinforcing the superior aspect of the joint is the **coracoacromial** ligament, which forms part of the **coraco-acromial** arch (along with the acromion and coracoid process).

Glenohumeral joint,,

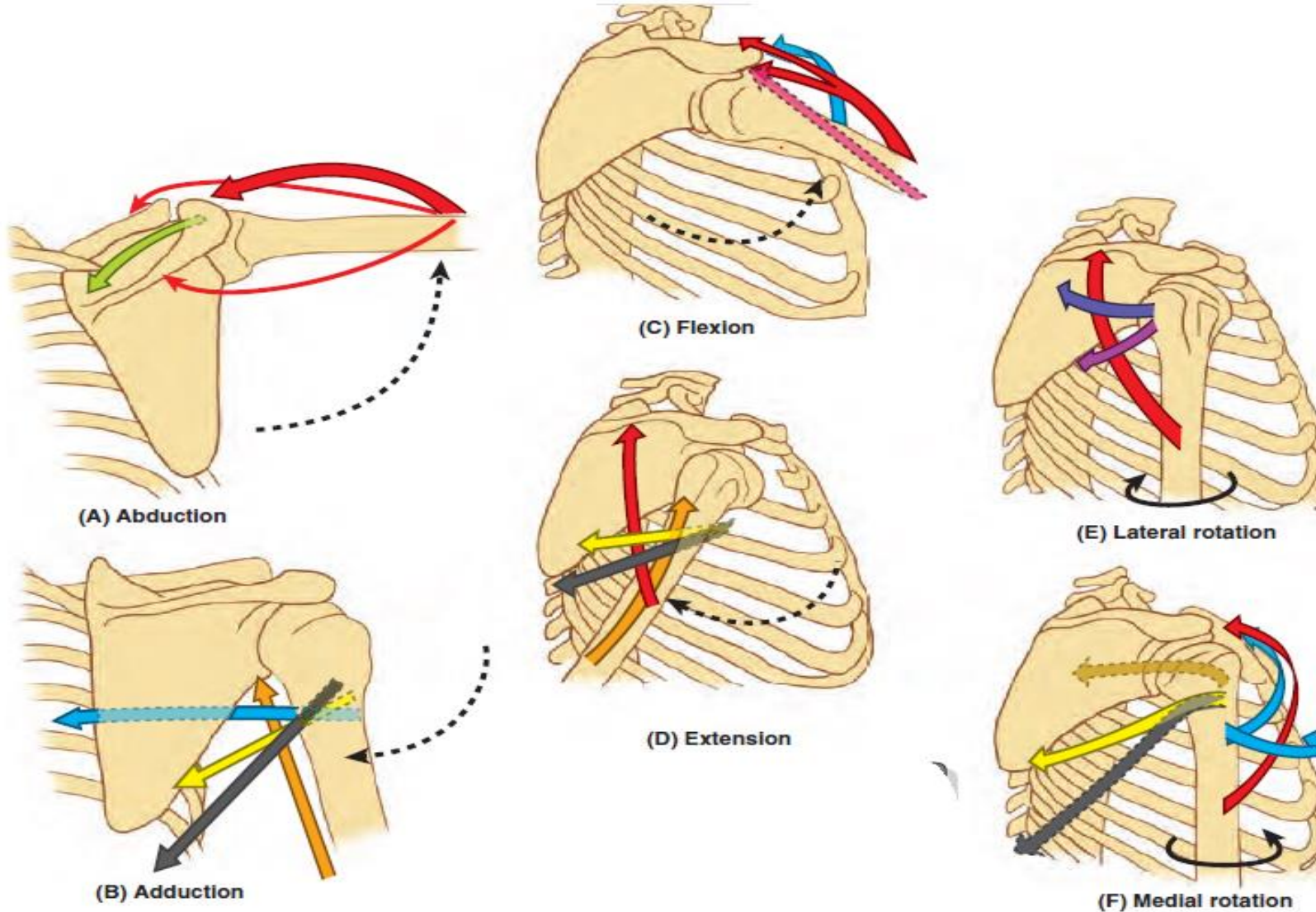


Glenohumeral joint,,,



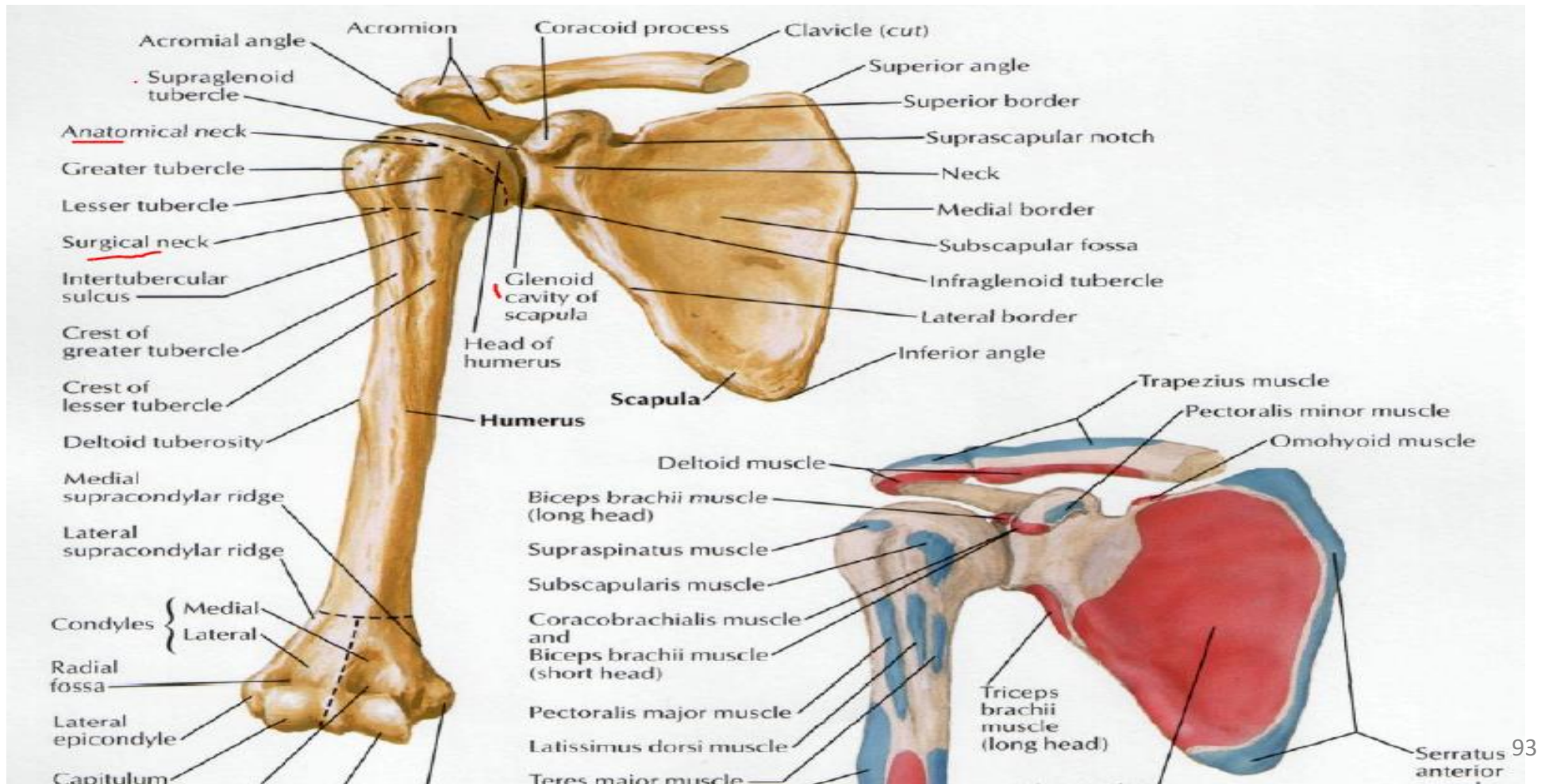
Glenohumeral joint,,

□ Movements of glenohumeral joint.



Glenohumeral joint,,

- Its position prevents superior dislocation of the joint.
- Muscles that act on the **glenohumeral joint** include *axio-appendicular muscles, and scapulohumeral muscles,*

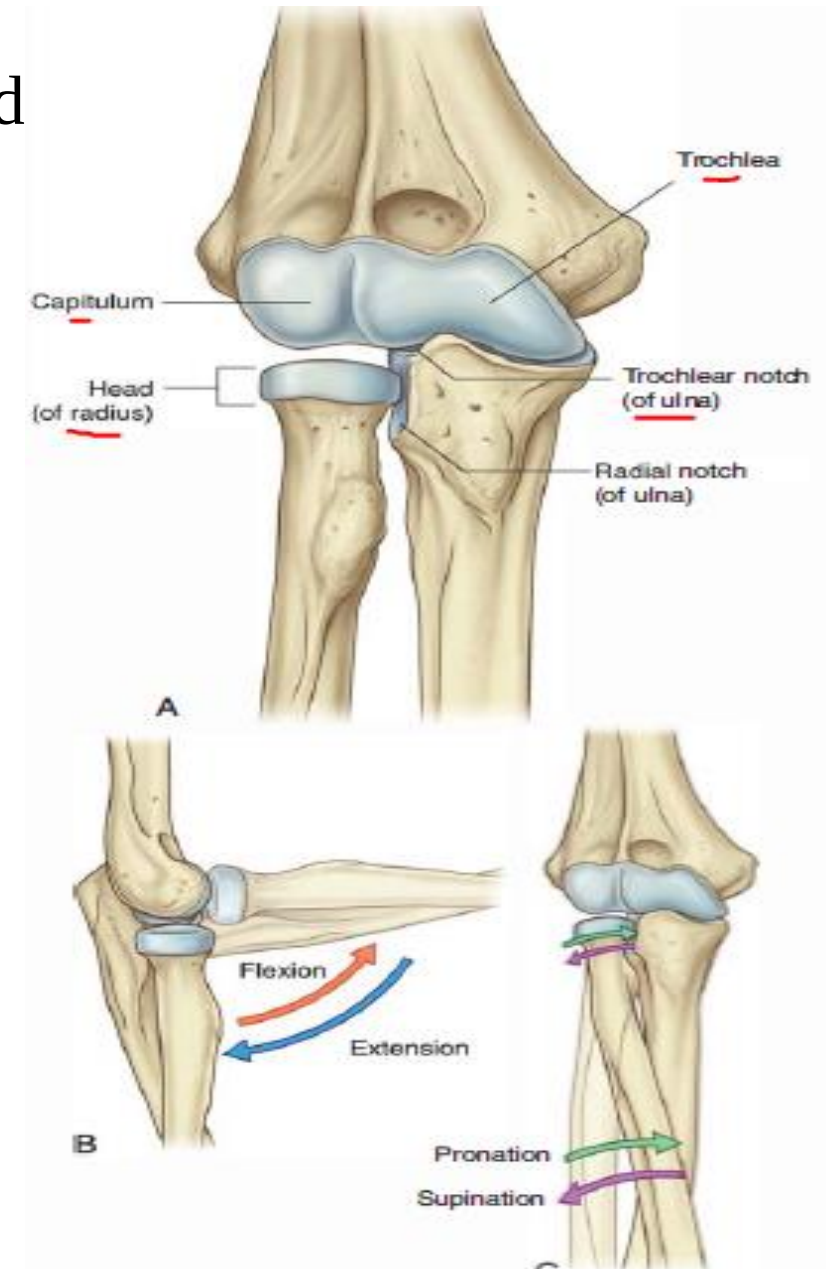


Elbow Joint

- The elbow is a synovial joint at the junction of the arm and forearm.
- It joins the distal end of the humerus to the proximal ends of the radius & ulna.

□ The elbow joint comprises 3 distinct articulations:

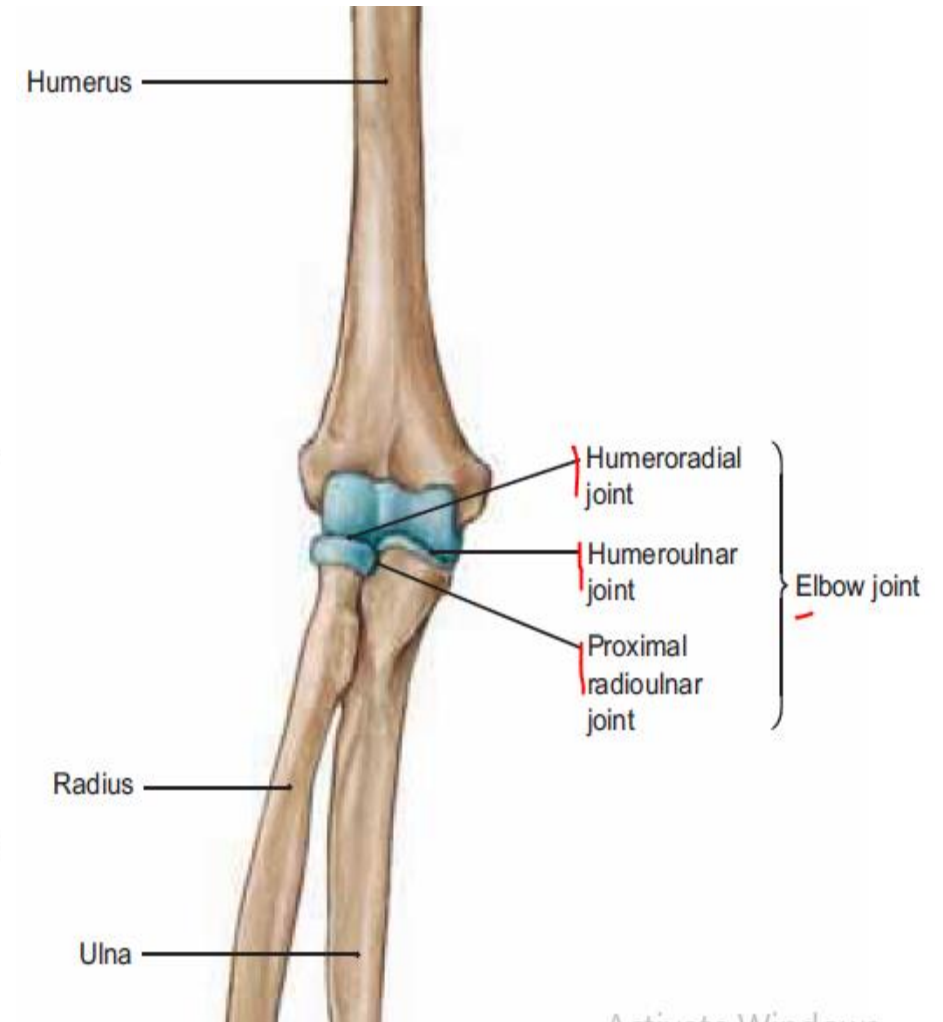
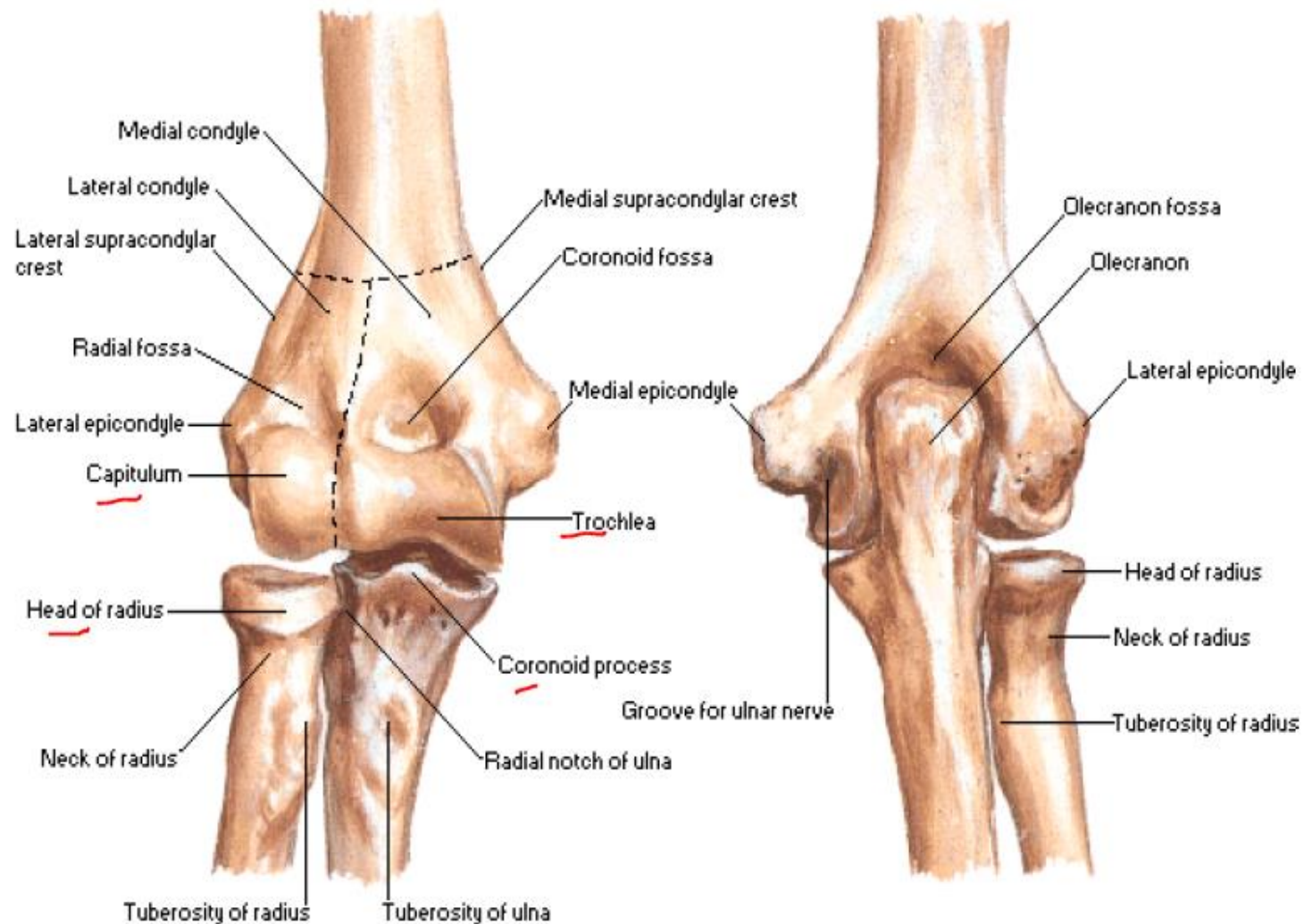
- I. The **humero-ulnar joint** b/n the trochlea of the humerus and the trochlear notch of the ulna (a hinge joint)
- II. The **humero-radial joint** b/n the capitulum of the humerus & the head of the radius (a ball-and-socket joint)
- III. The **proximal radio-ulnar joint** between the head of the radius and the radial notch of the ulna



Elbow Joint,,

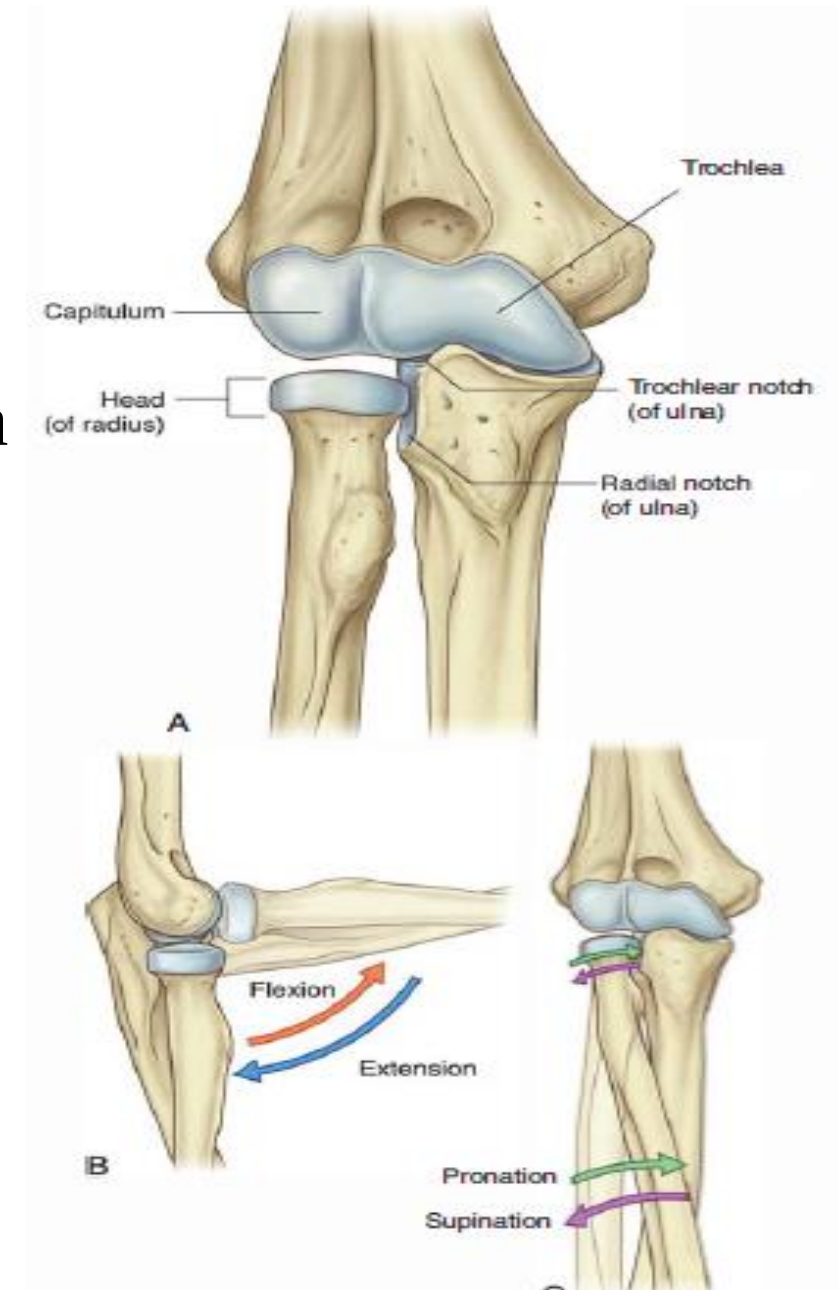
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Anterior and Posterior Views



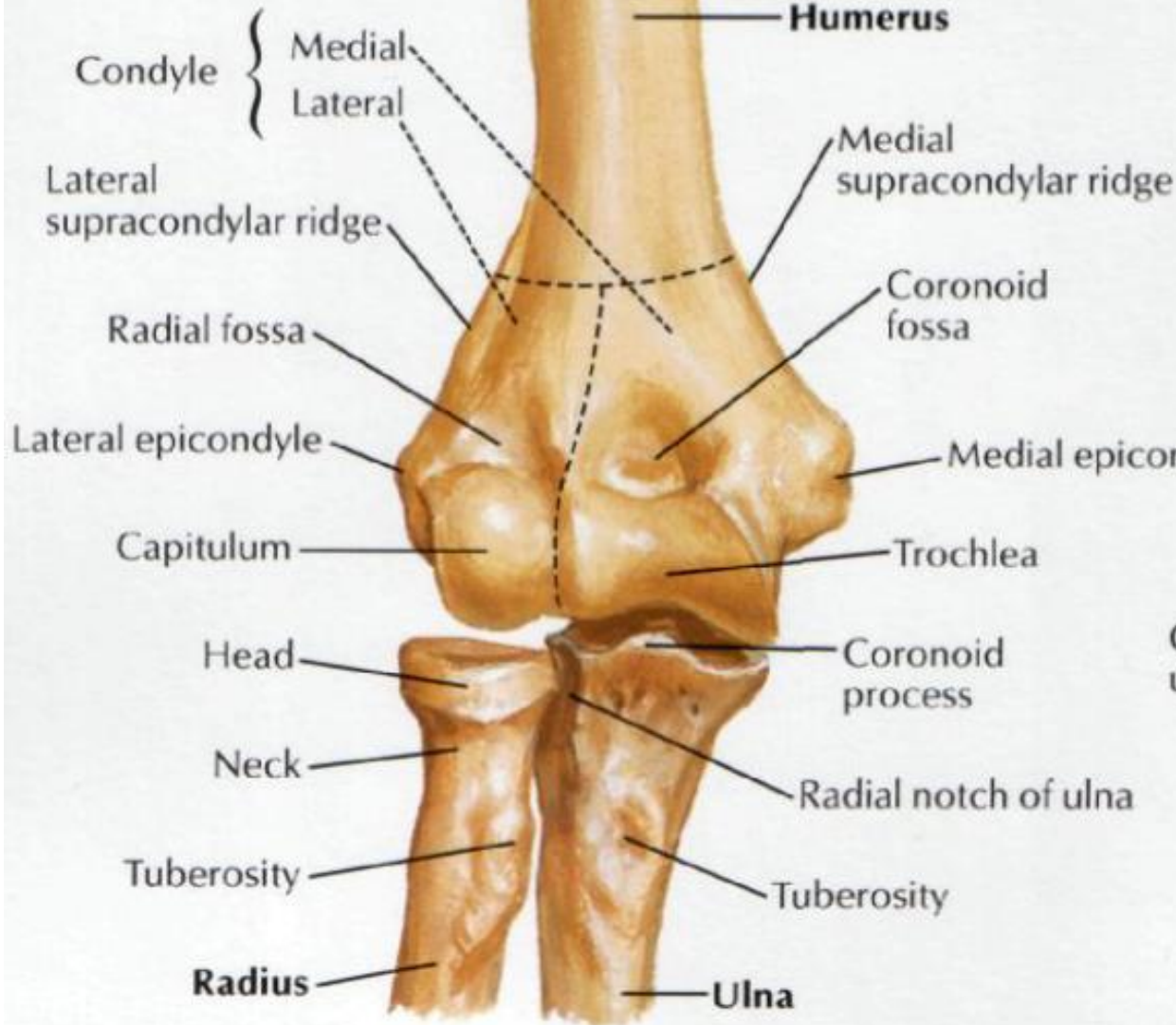
Elbow Joint,,

- The elbow joint primarily mobile in only one axis, which allows active **flexion and extension**
- The **humero-ulnar and humeroradial joints** act as a single hinge joint during **extension and flexion** of the forearm; the **proximal radio-ulnar joint** allows rotation of the head of the radius during **pronation & supination** of the forearm.
- The **proximal radio-ulnar joint** is a pivot type of synovial joint.

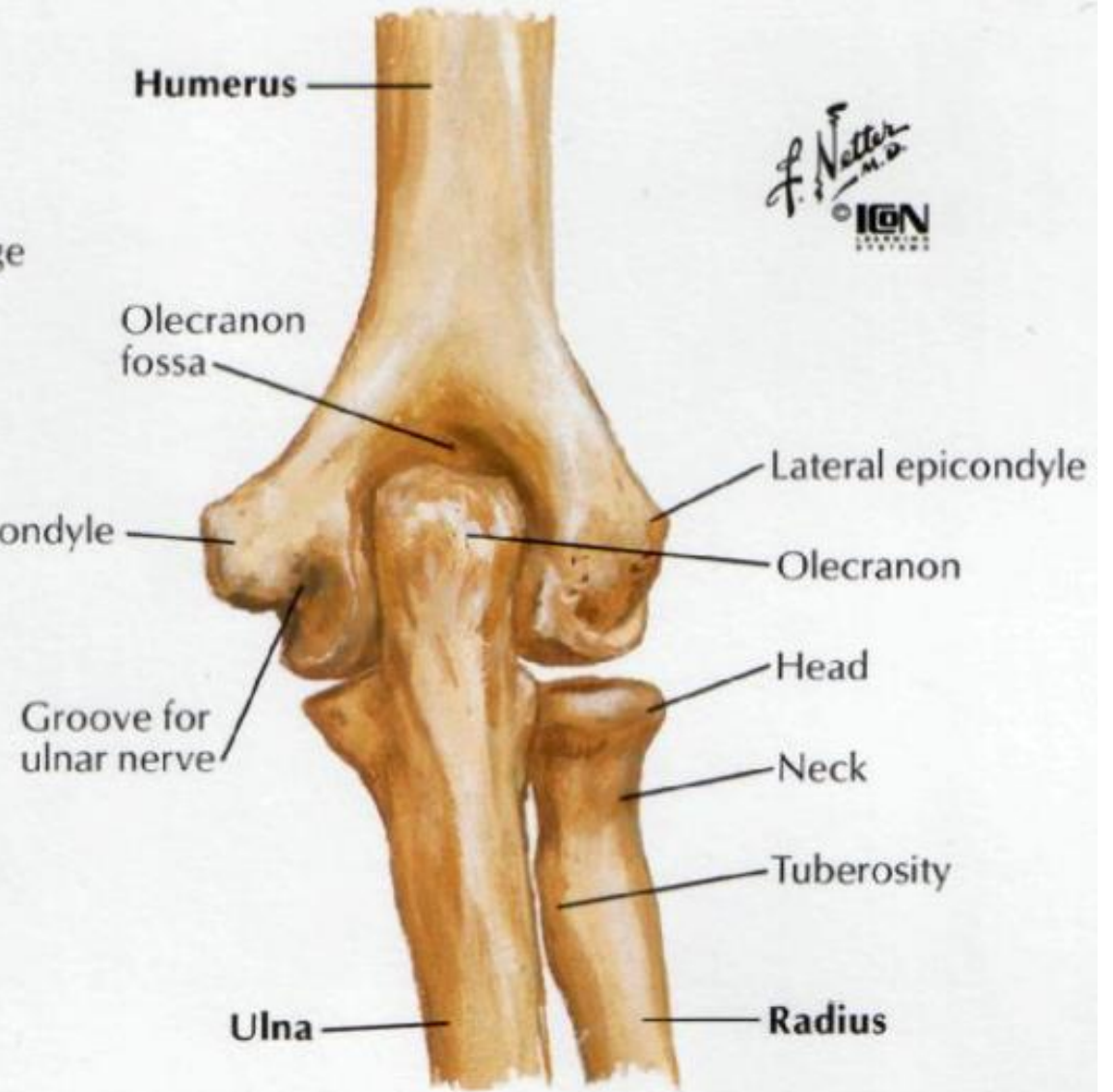


Elbow Joint,,

Right elbow



In extension: anterior view



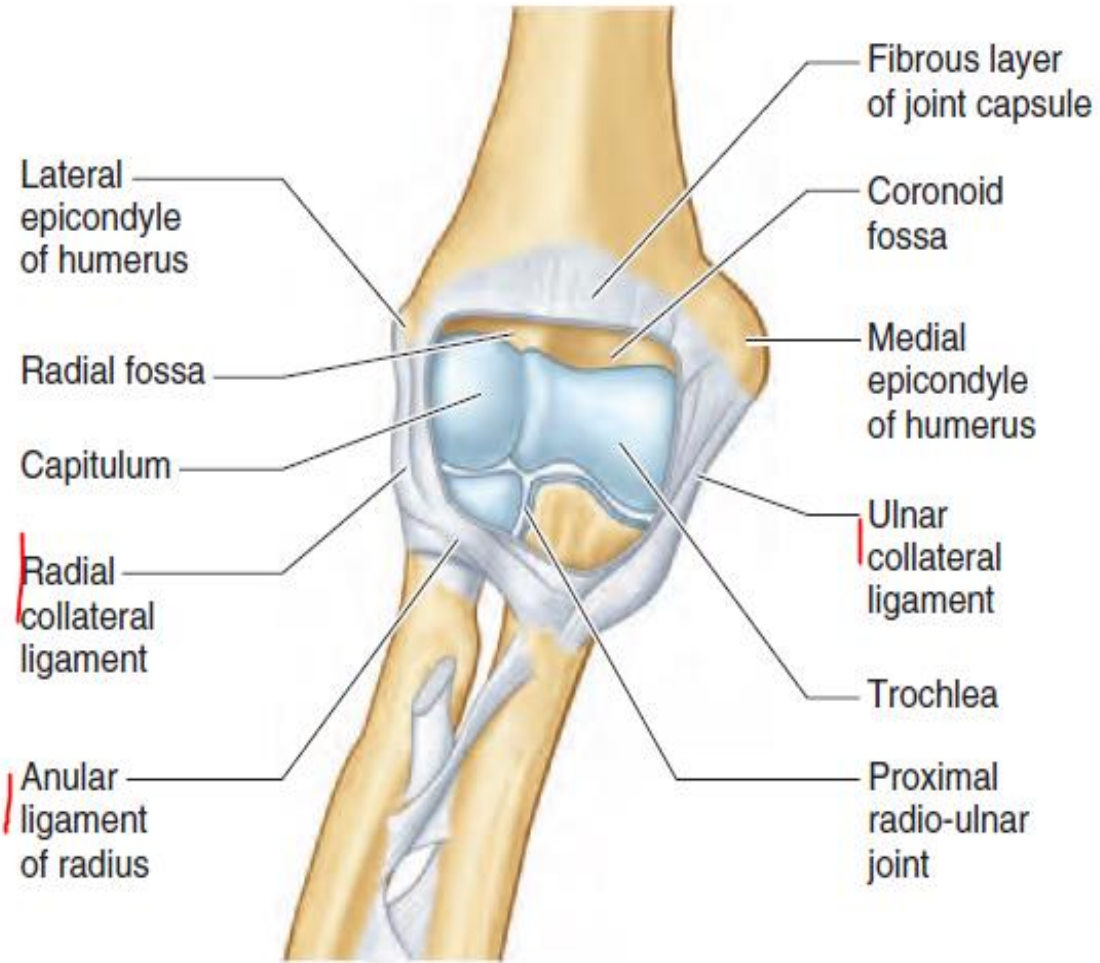
In extension: posterior view

Elbow Joint,,

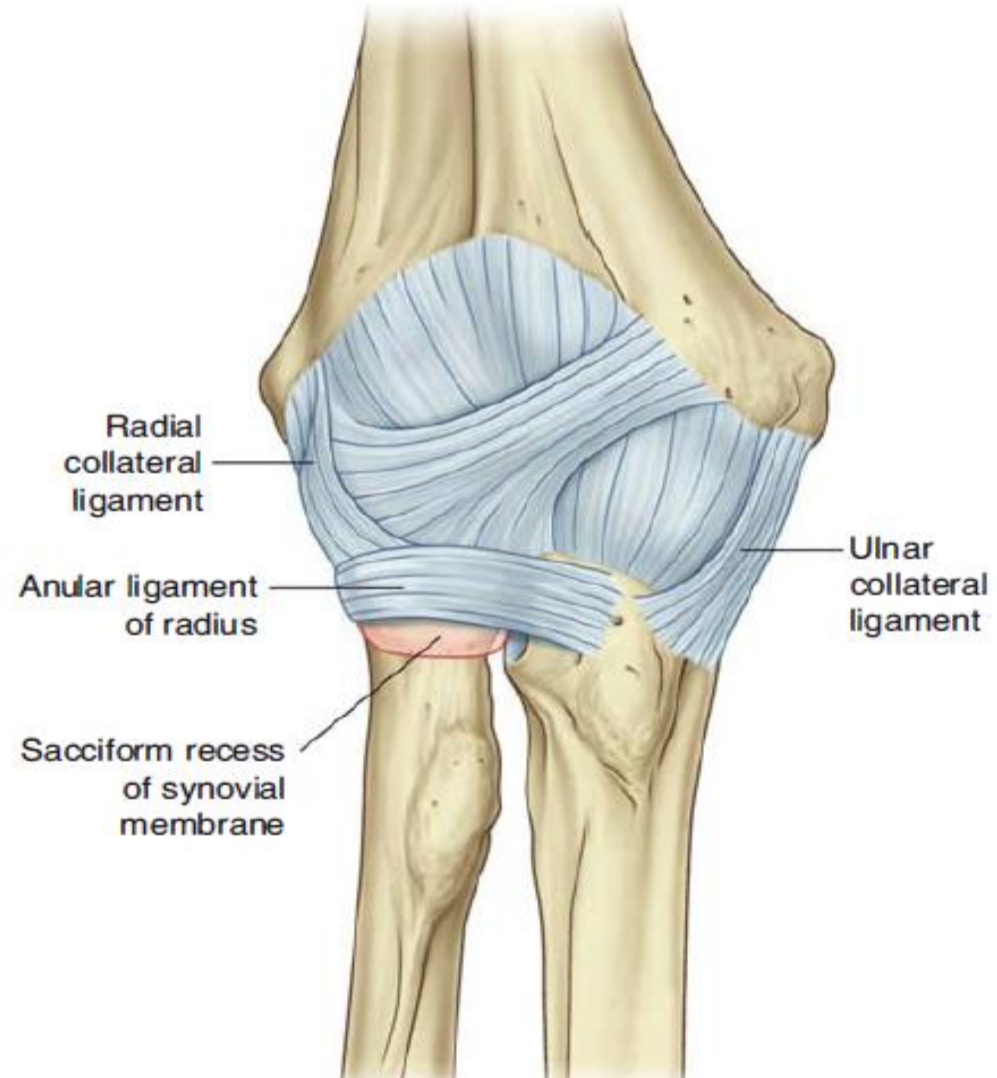
❖ These three articulations of the elbow joint are enclosed within the joint capsule of the elbow.

- The joint capsule thickened to form medially and laterally collateral ligament.
- ✓ The radial collateral ligament is fan-like and joins the lateral epicondyle of the humerus to the anular ligament of the radius.
- ✓ The ulnar collateral ligament is triangular and extends from the medial epicondyle of the humerus to the coronoid process and olecranon of the ulna.
- ❖ **The anular ligament** of the radius is attached to the margins of the radial notch of the ulna. It holds the head of the radius in the radial notch of the ulna.

Elbow Joint,,

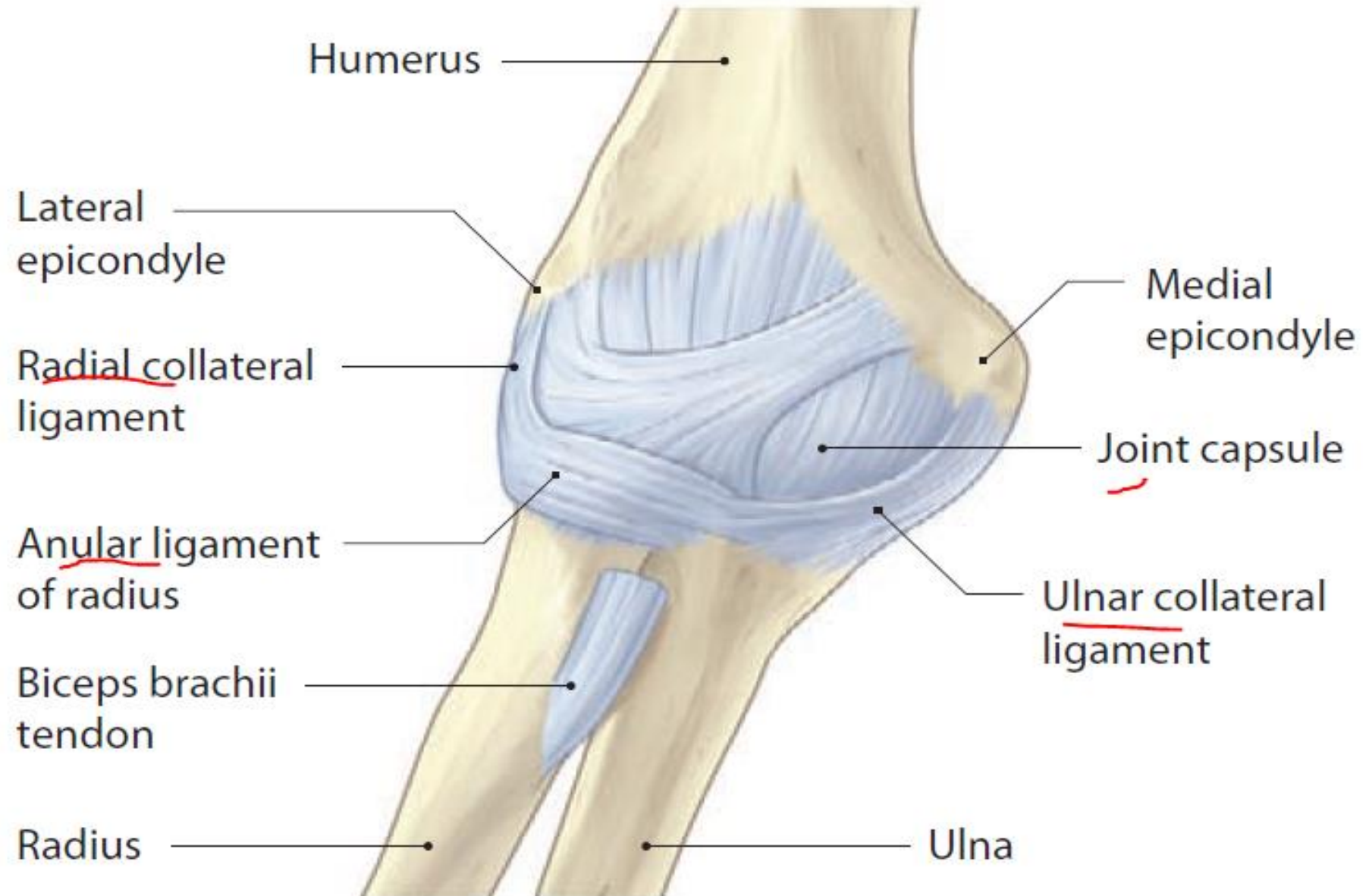


(A) Anterior view



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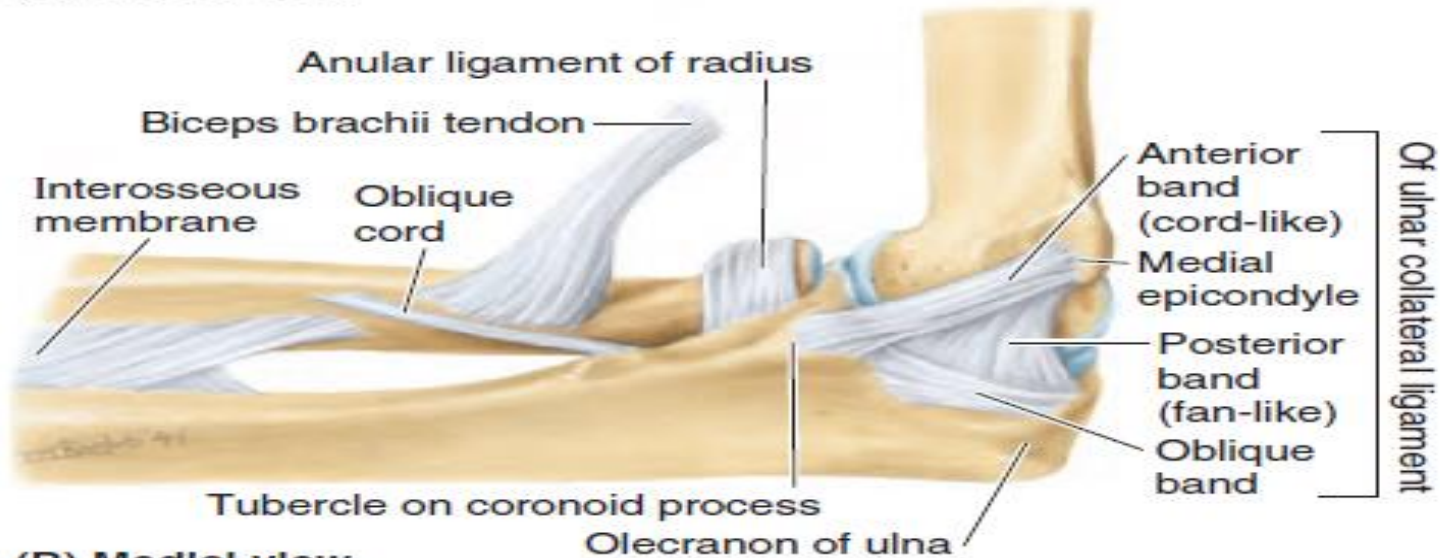
Elbow Joint,,



Elbow Joint,,



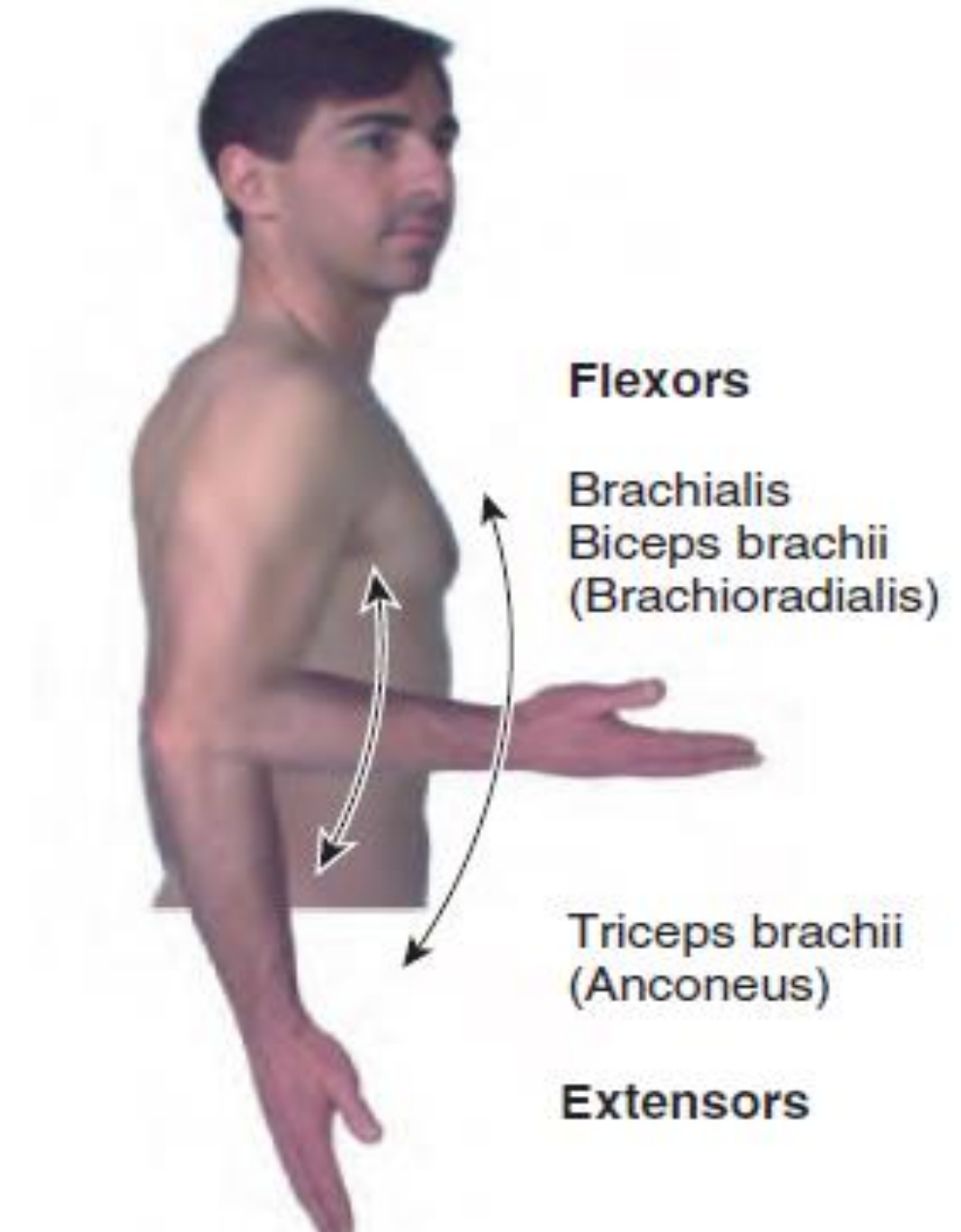
(A) Lateral view



(B) Medial view

Elbow Joint,,

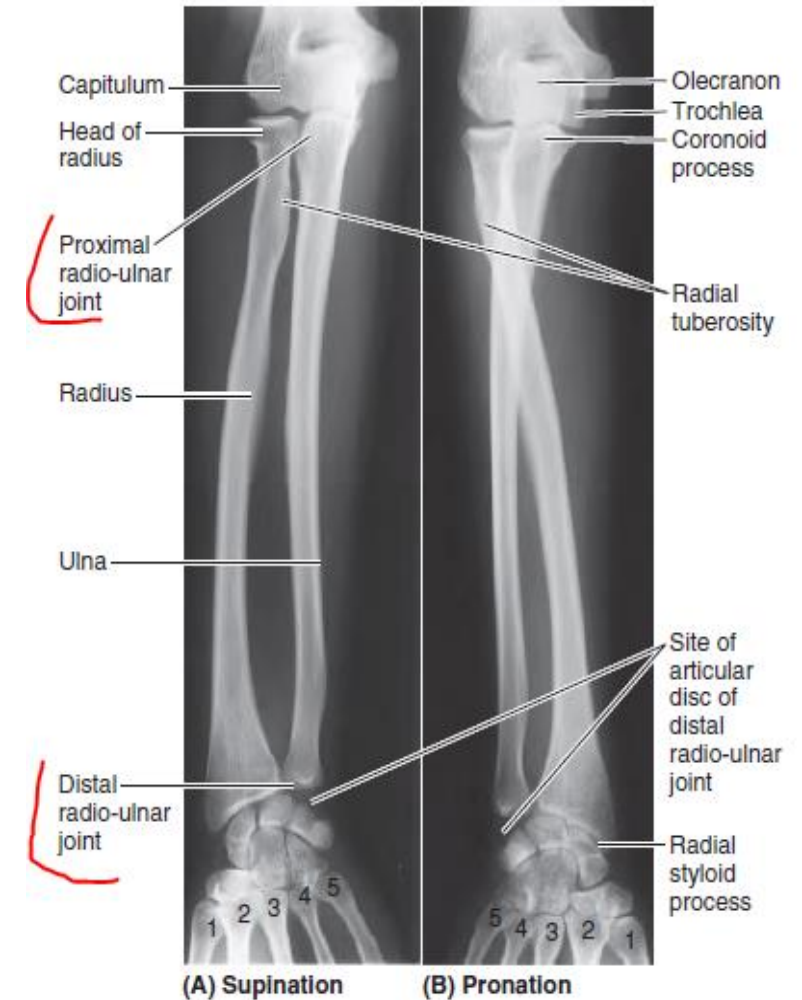
- There are many muscles that affect elbow joint movement.
- *The chief flexors of the elbow joint are the brachialis and biceps brachii.*
- *The chief extensor of the elbow joint is the triceps brachii*, especially the medial head, weakly assisted by the anconeus.



Radioulnar joints

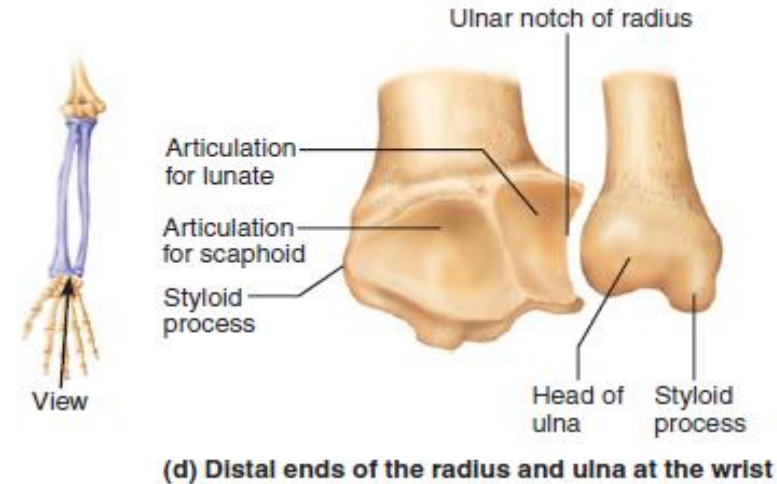
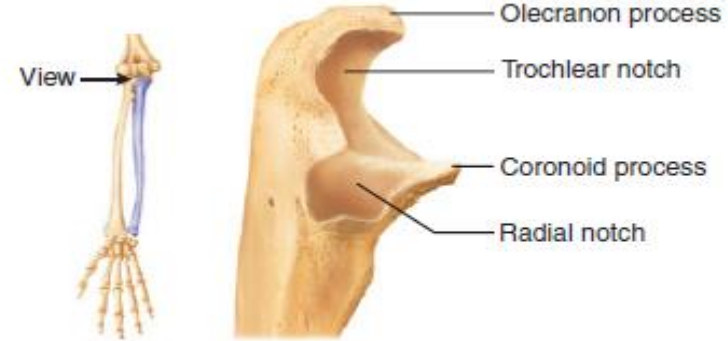
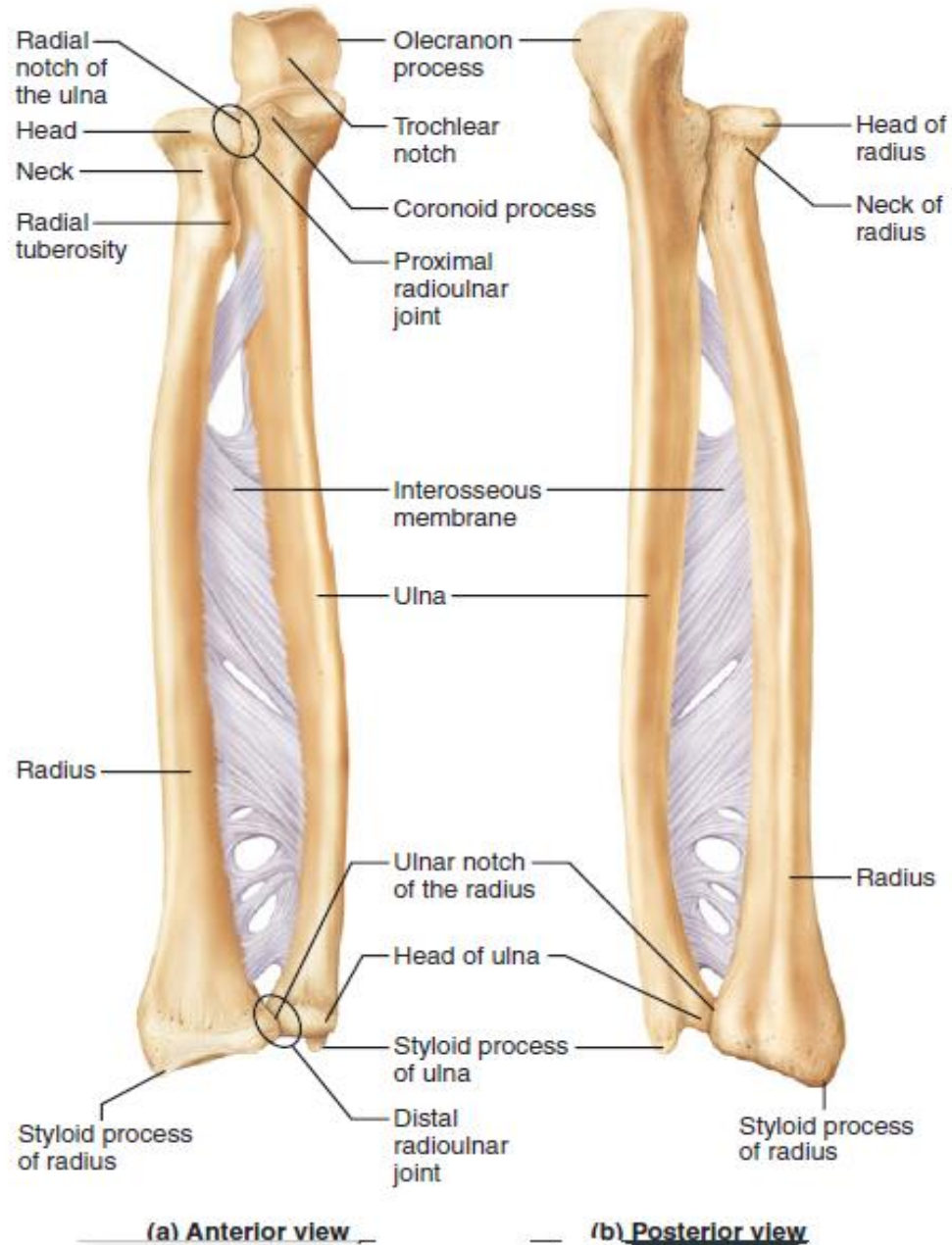
- Pronation and supination occur at the proximal and distal radioulnar joints.

- ❑ **At the proximal radioulnar joint**, the head of the radius is held within the anular ligament, which is attached to the anterior and posterior edges of the radial notch of the ulna.
- The head of the radius rotates within the anular ligament during pronation and supination.

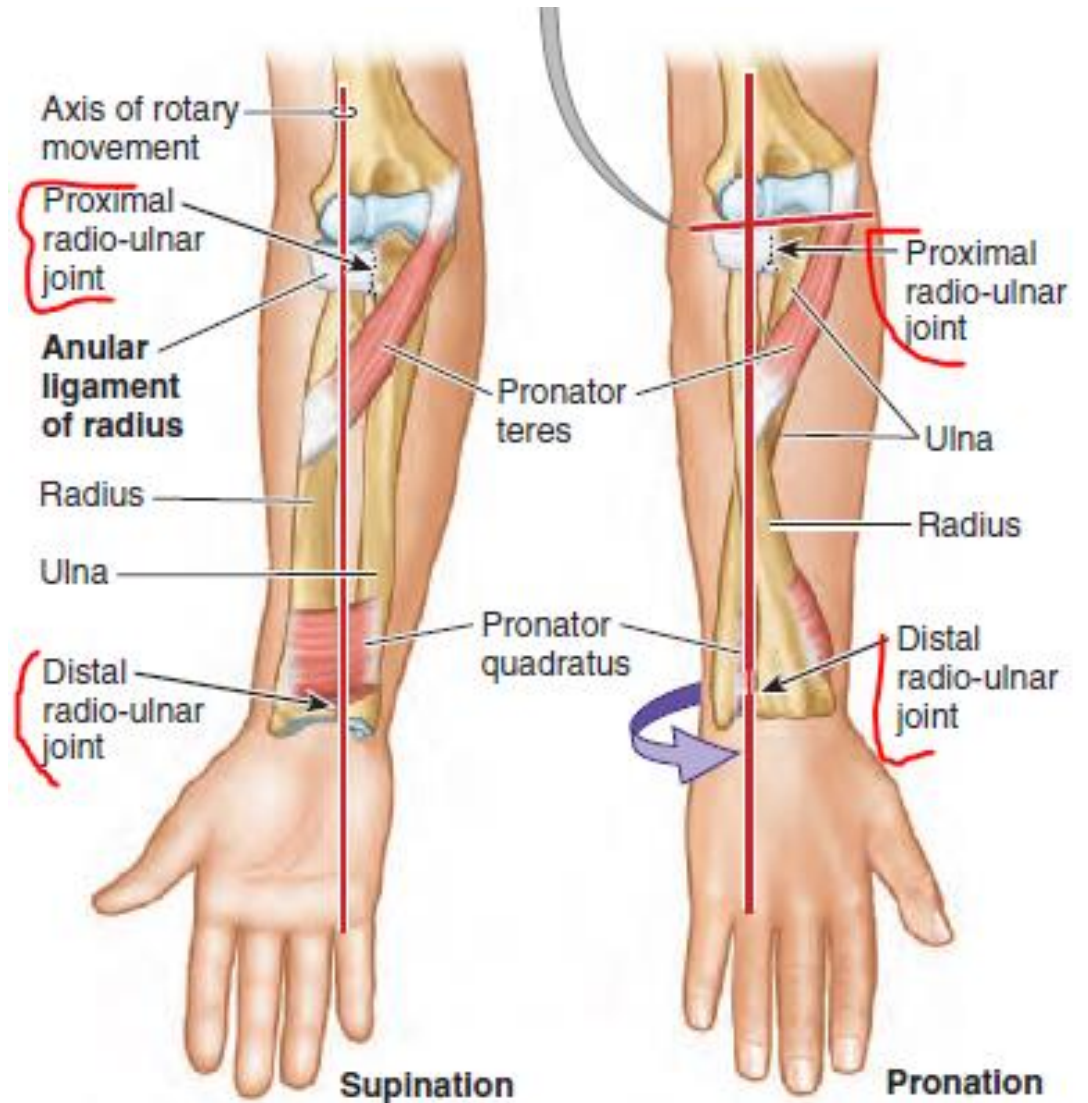
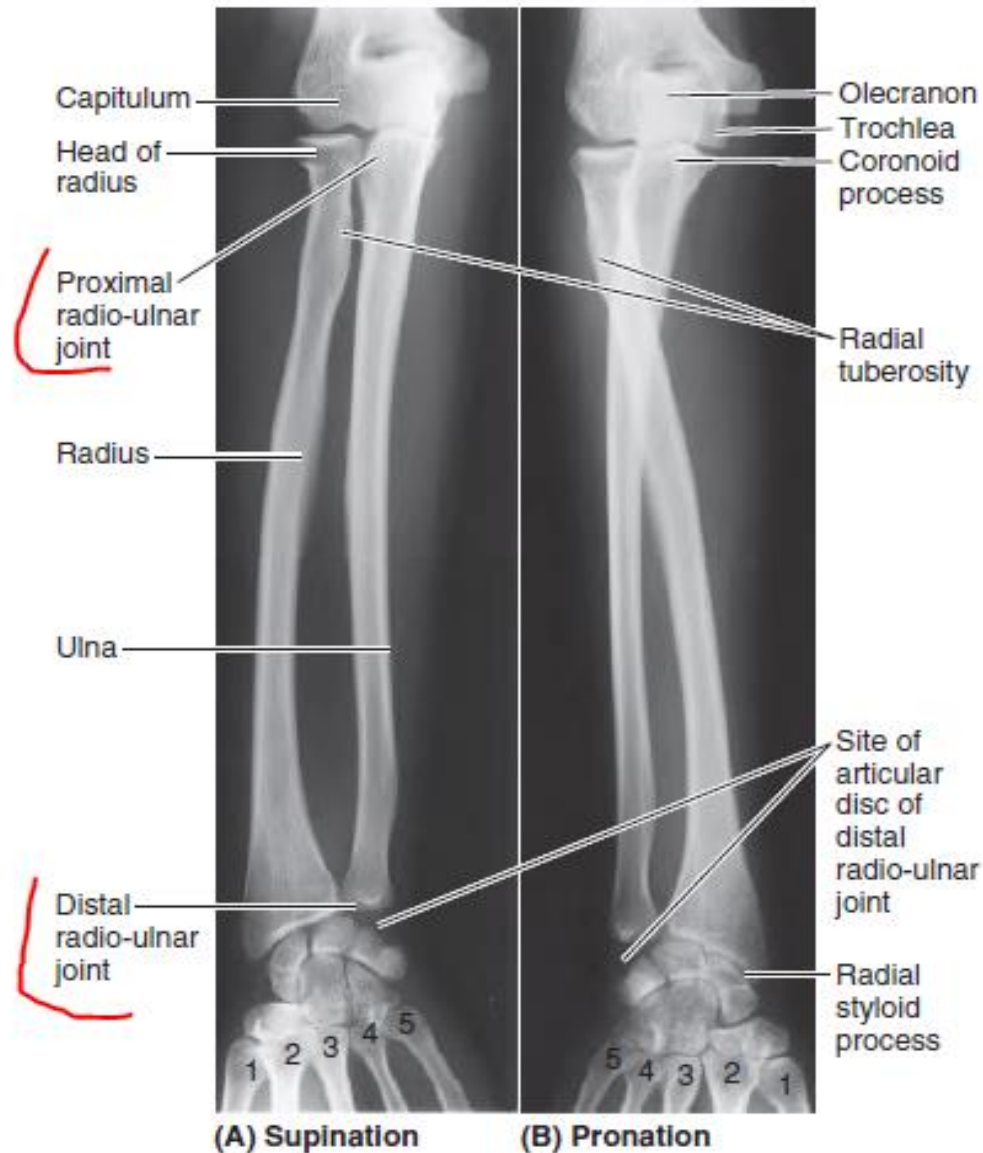


- ❑ **At the distal radioulnar joint the head** of the ulna articulates with the ulnar notch of the radius.
- The distal ends of the radius and ulna are held together by an articular disc (known as the triangular ligament).

Radioulnar joints,,,



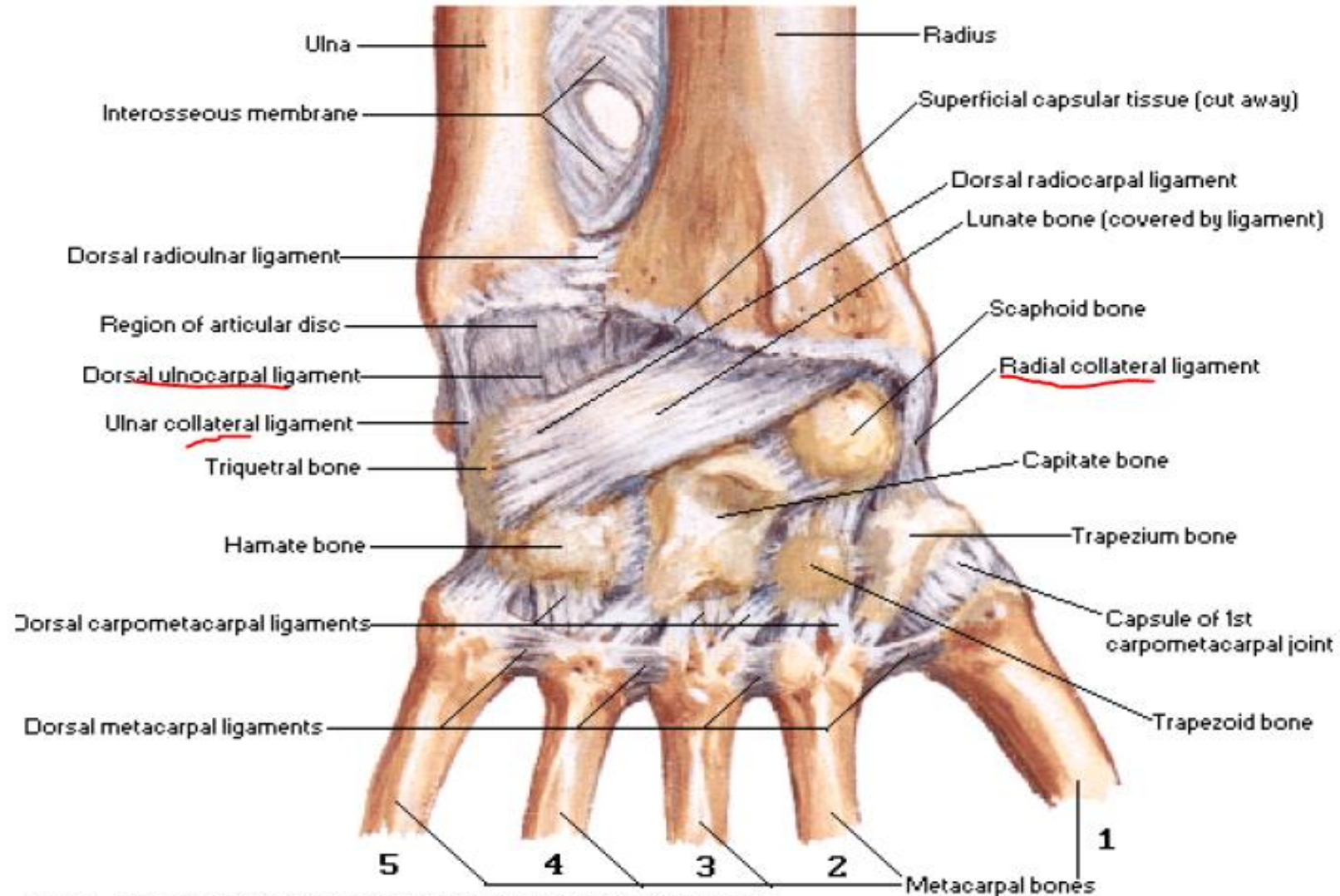
Radioulnar joints,,



Radioulnar joints,,

≡

Posterior [Dorsal] View



Note: dorsal ligaments weaker than palmar ligaments

The wrist (radiocarpal) joint

- The anatomy of wrist joint is probably the most complex joints of the body that connects hand and forearm.

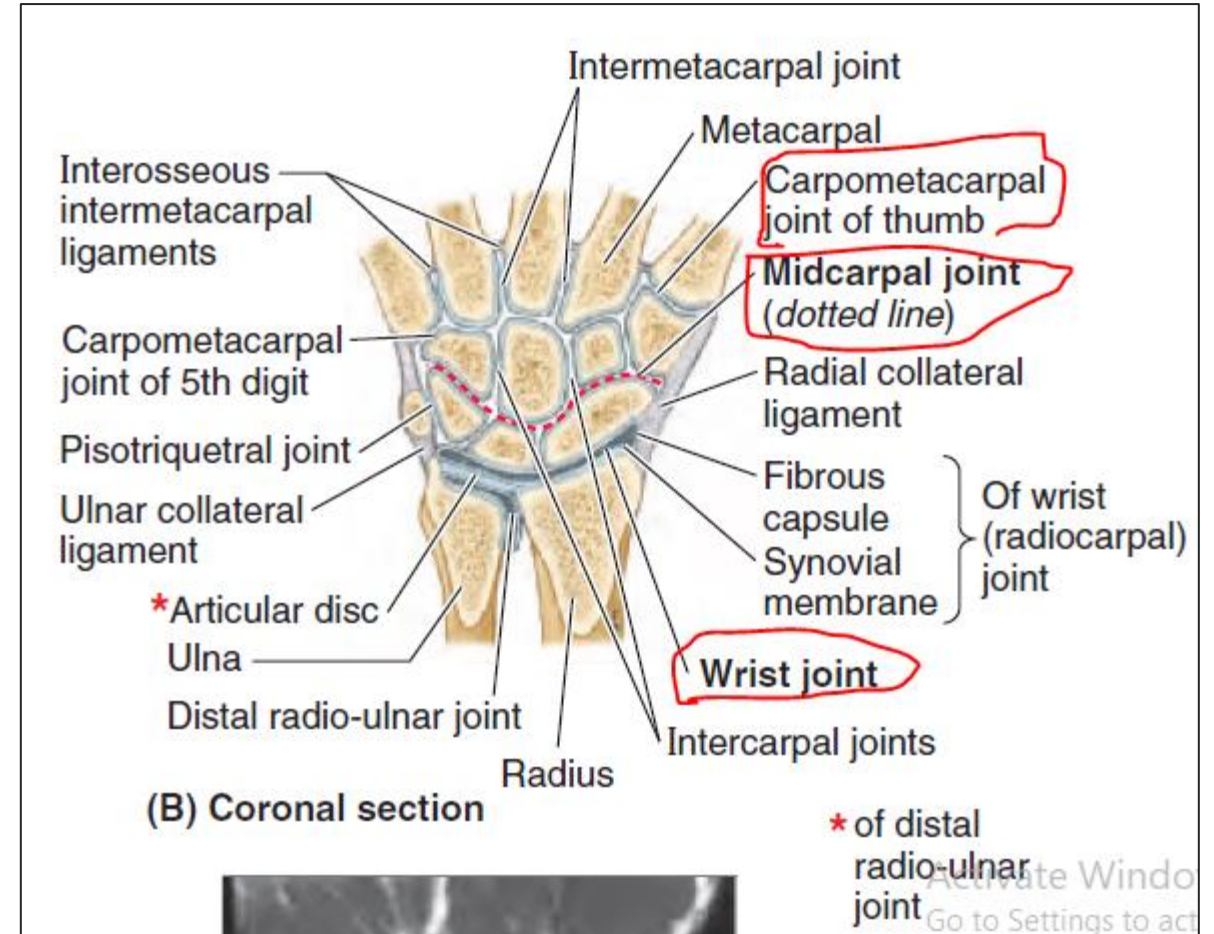
❑ It is actually a collection of multiple bones and joints.

- The bones comprising the wrist include the distal ends of the radius and ulna, eight carpal bones, and the proximal portions of the five metacarpal bones (see the images below).
- The **wrist (radiocarpal) joint** is a condyloid (ellipsoid) type of synovial joint.
- **Note! The ulna does not participate in the wrist joint.**

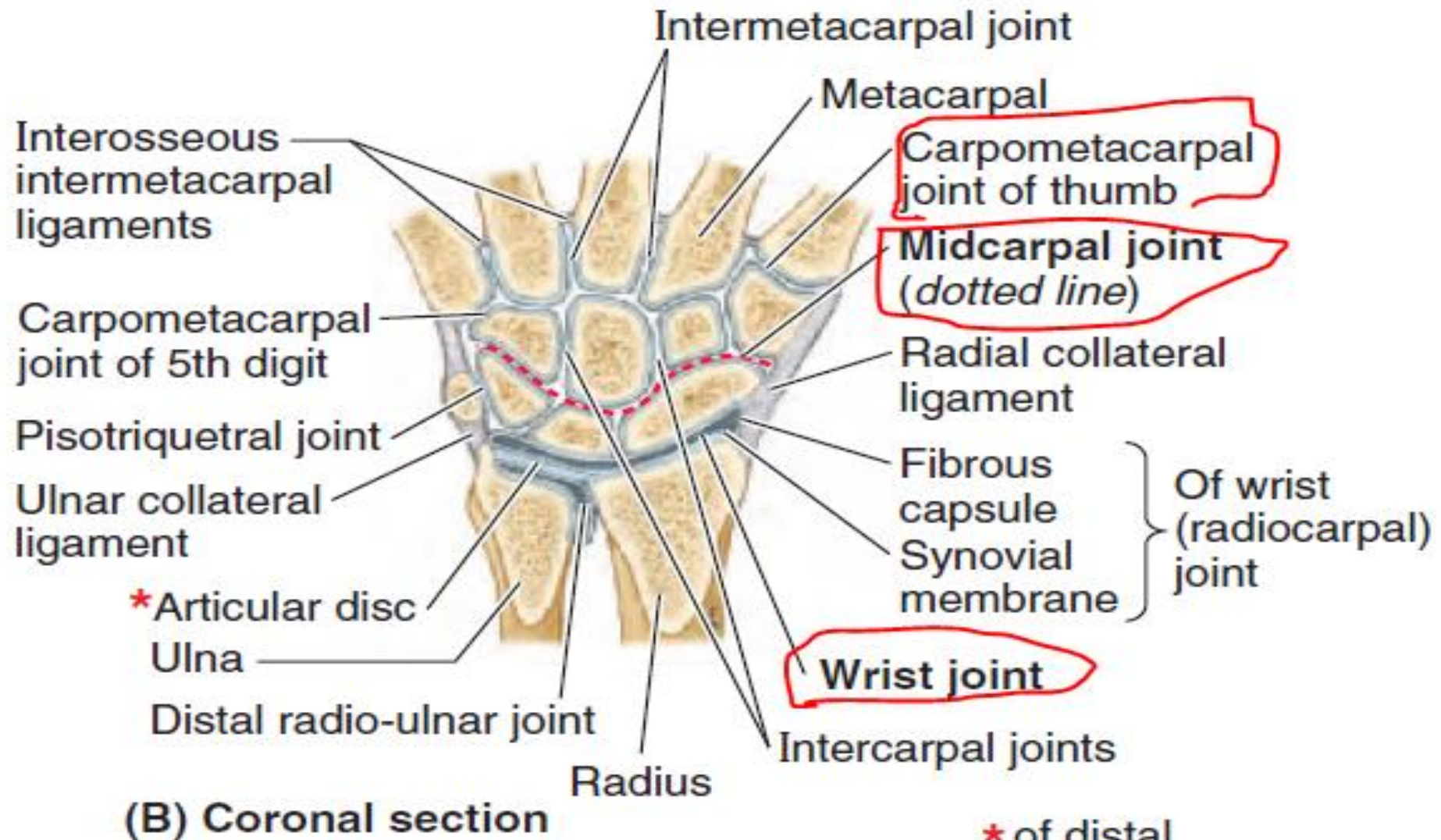
The wrist (radiocarpal) joint,,

❖ The carpal bones form three primary articulations:

- ✓ wrist joint (**radiocarpal joint**) between the forearm and wrist
- ✓ **midcarpal joints** between the proximal and distal rows of carpal bones
- ✓ **carpometacarpal joints** between the distal row of carpal bones and the hand



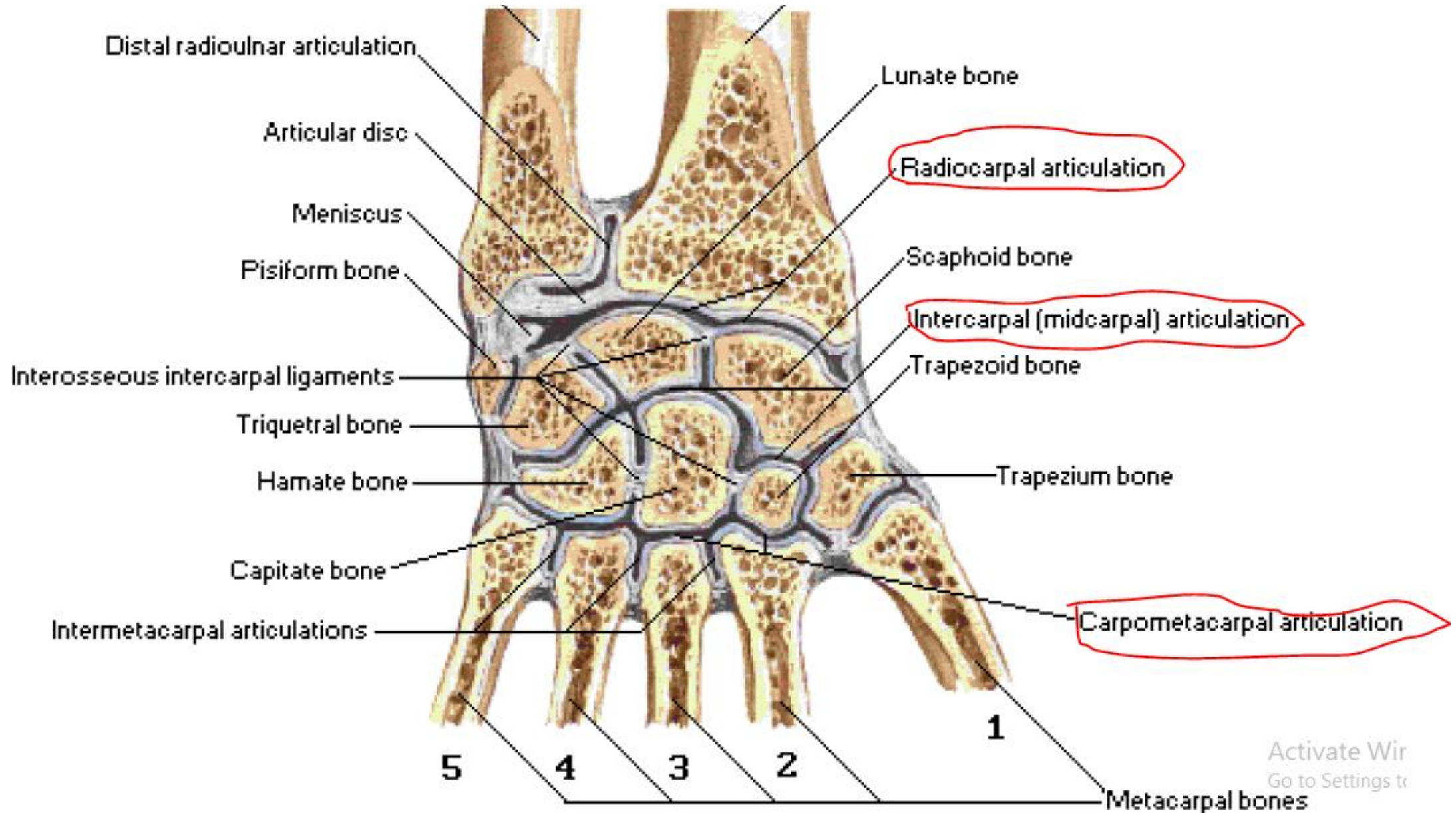
The wrist (radiocarpal) joint,,



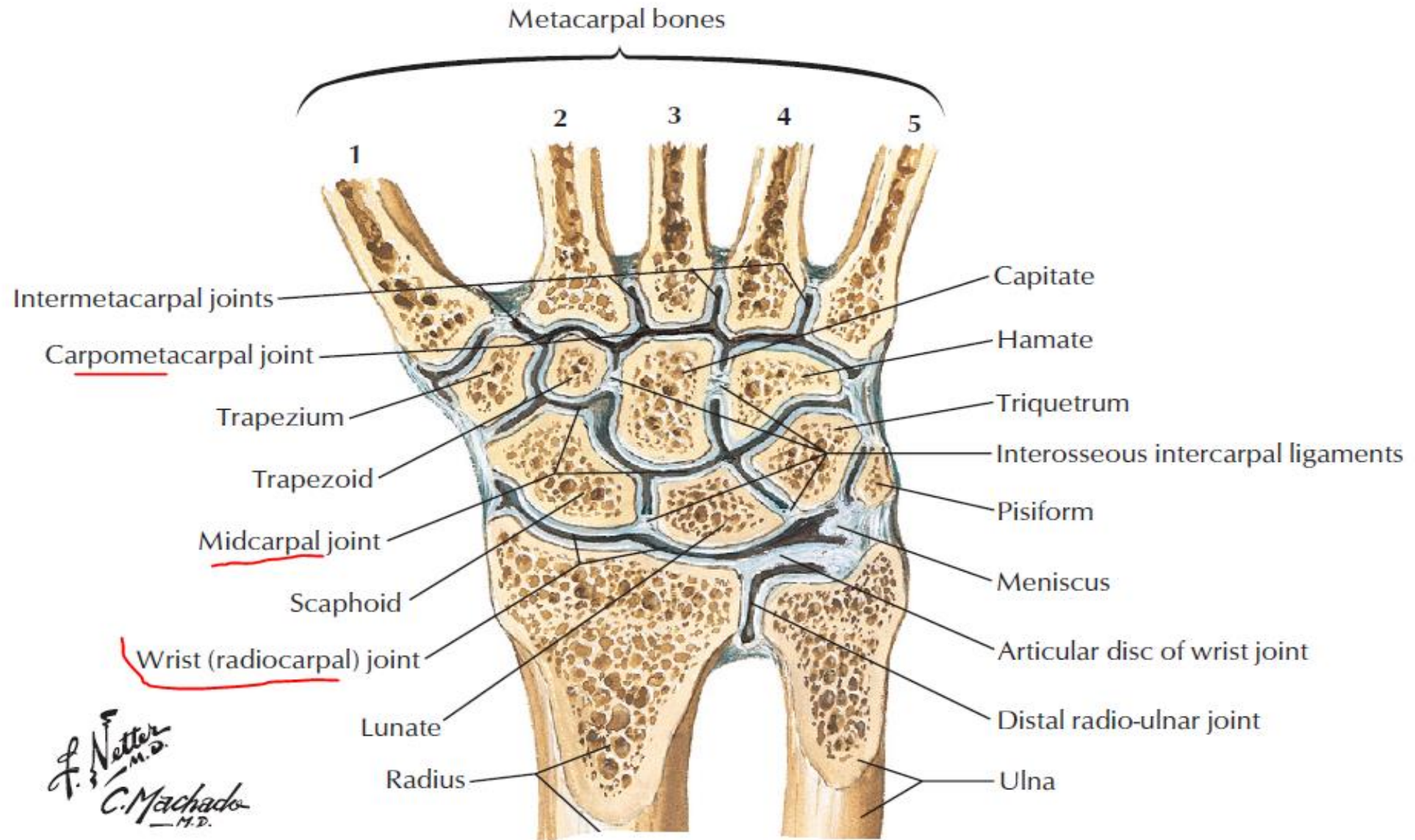
* of distal
radio-ulnar
joint

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The wrist (radiocarpal) joint,,



The wrist (radiocarpal) joint,,

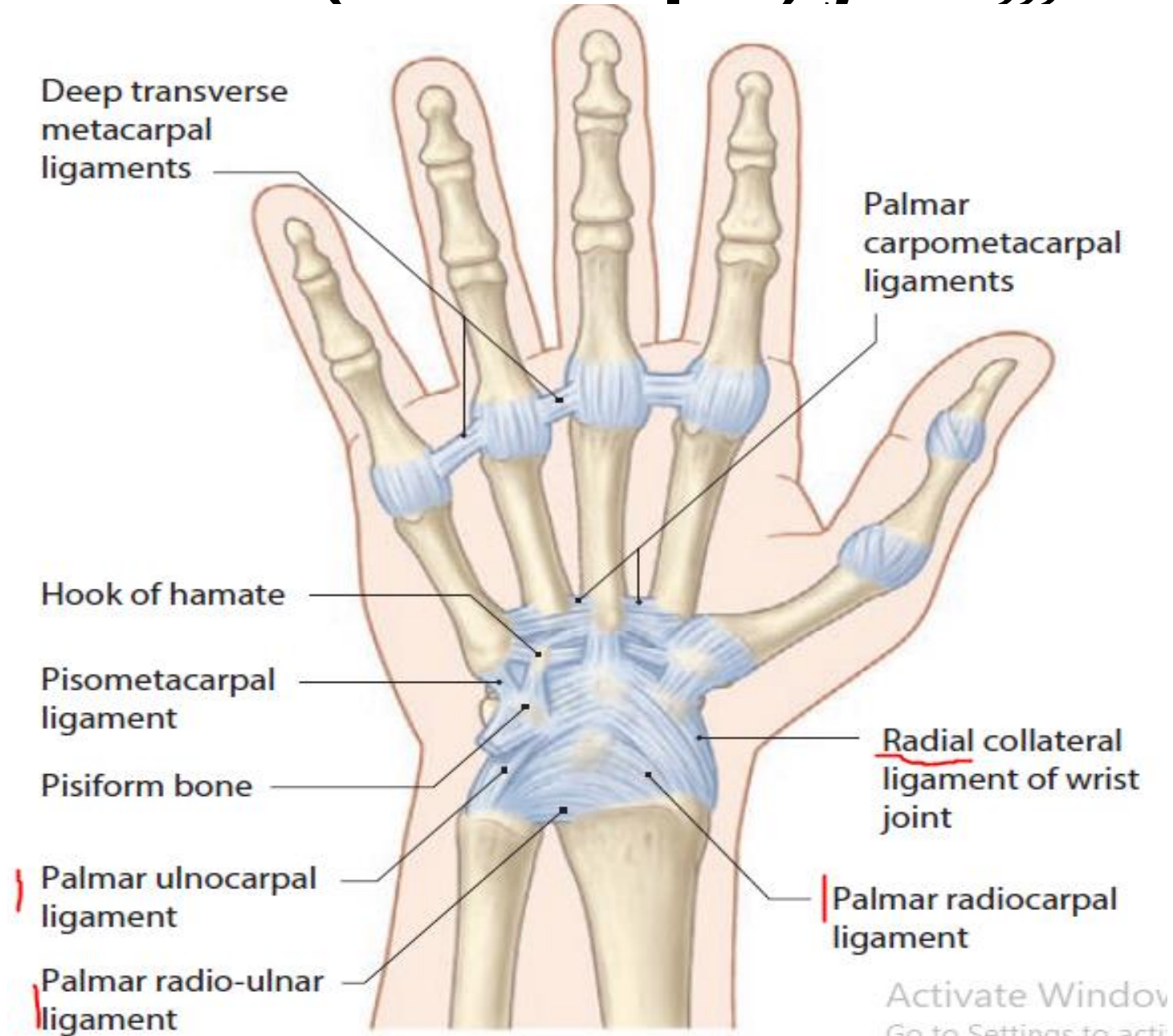


The wrist (radiocarpal) joint,,

❑ **Ligaments of wrist joint:** *fibrous layer of the joint capsule* surrounds the wrist joint and is attached to the distal ends of the radius & ulna and the proximal row of carpals.

- The joint capsule has many thickenings, which are **named as ligaments** (e.g., the **palmar radiocarpal** and **ulnocarpal**, and **dorsal radiocarpal** and **ulnocarpal ligaments** and the **ulnar** and **radial collateral ligaments of the wrist joint**).
- ✓ The joint capsule is also strengthened medially by the **ulnar collateral ligament**, which is attached to the ulnar styloid process and triquetrum.
- ✓ The joint capsule is also strengthened laterally by the **radial collateral ligament**, which is attached to the radial styloid process and scaphoid.

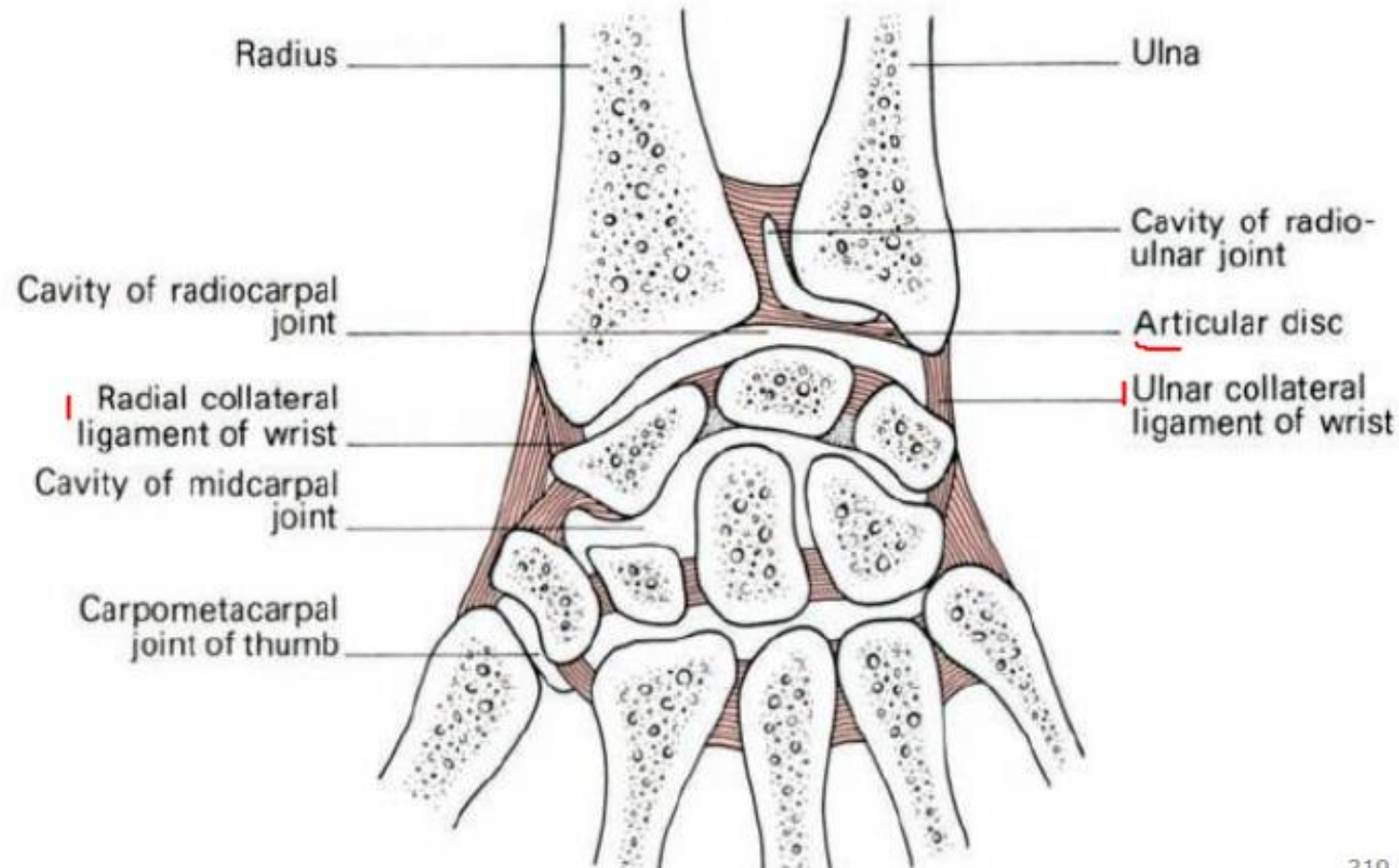
The wrist (radiocarpal) joint,,,



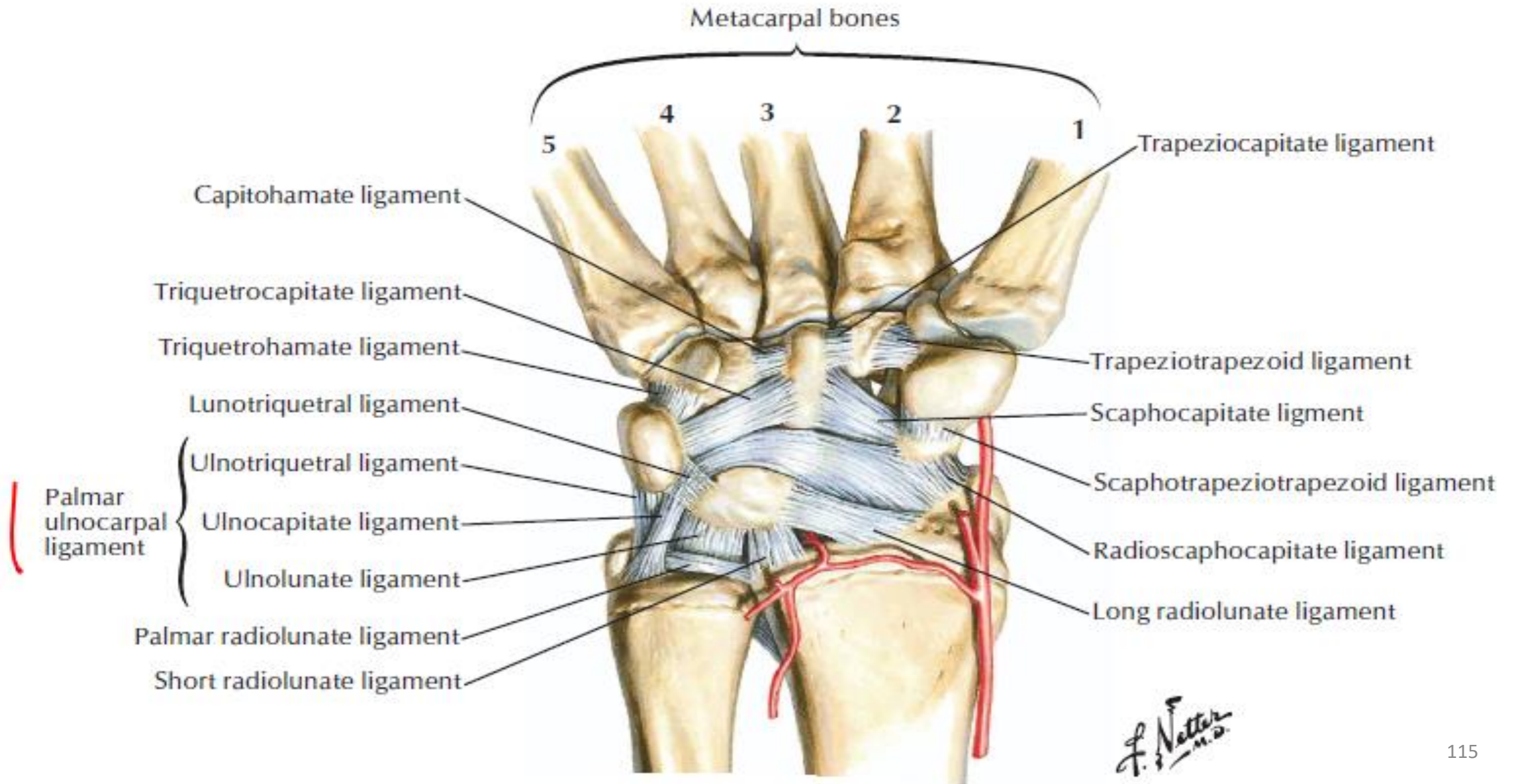
The wrist (radiocarpal) joint,,

□Note!

- ✓ Anterior and posterior ligaments strengthen fibrous capsule
- ✓ Ulnar collateral ligament attaches to styloid process of ulna and triquetrum
- ✓ Radial collateral ligament attaches to styloid process of radius and scaphoid

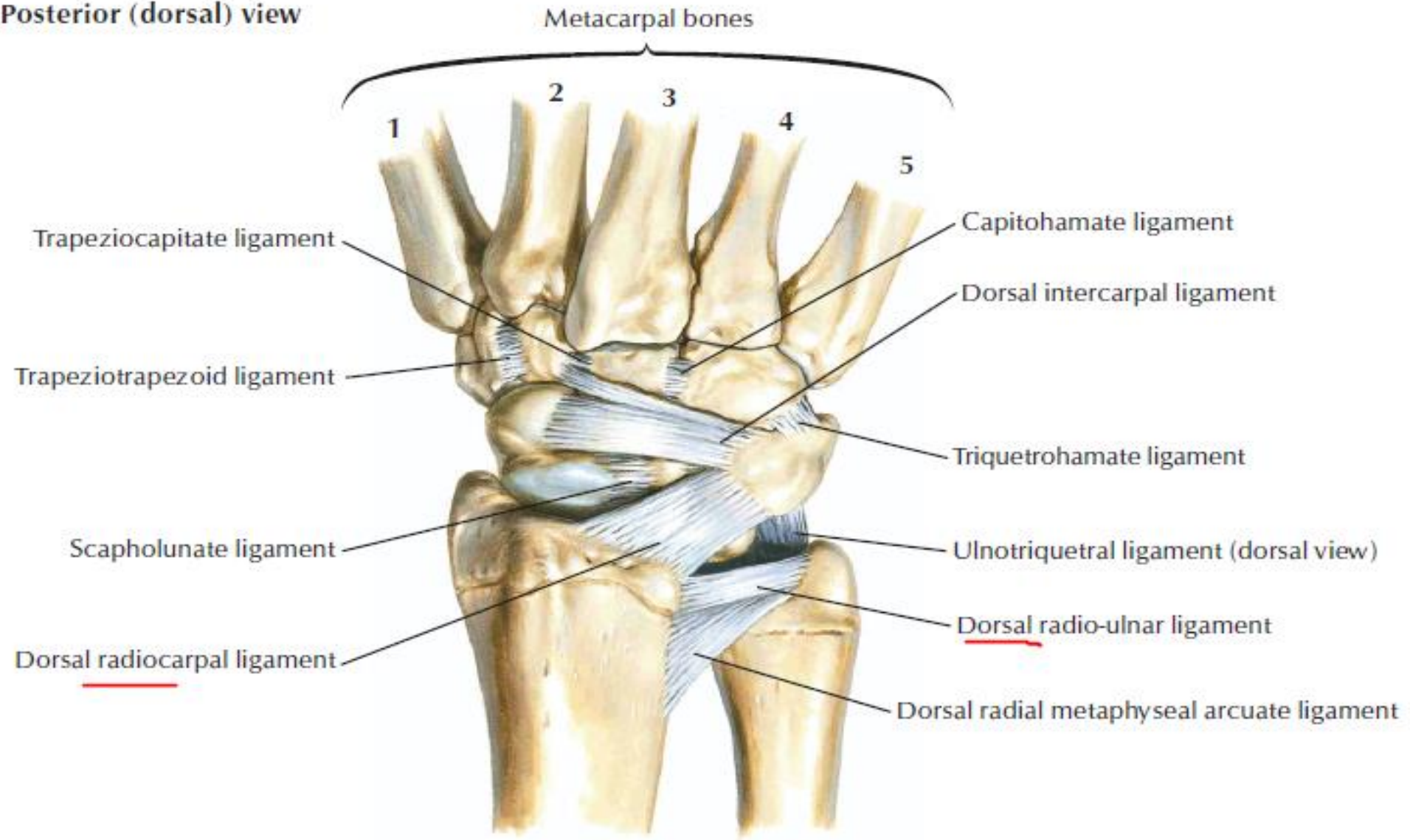


The wrist (radiocarpal) joint,,



The wrist (radiocarpal) joint,,

Posterior (dorsal) view

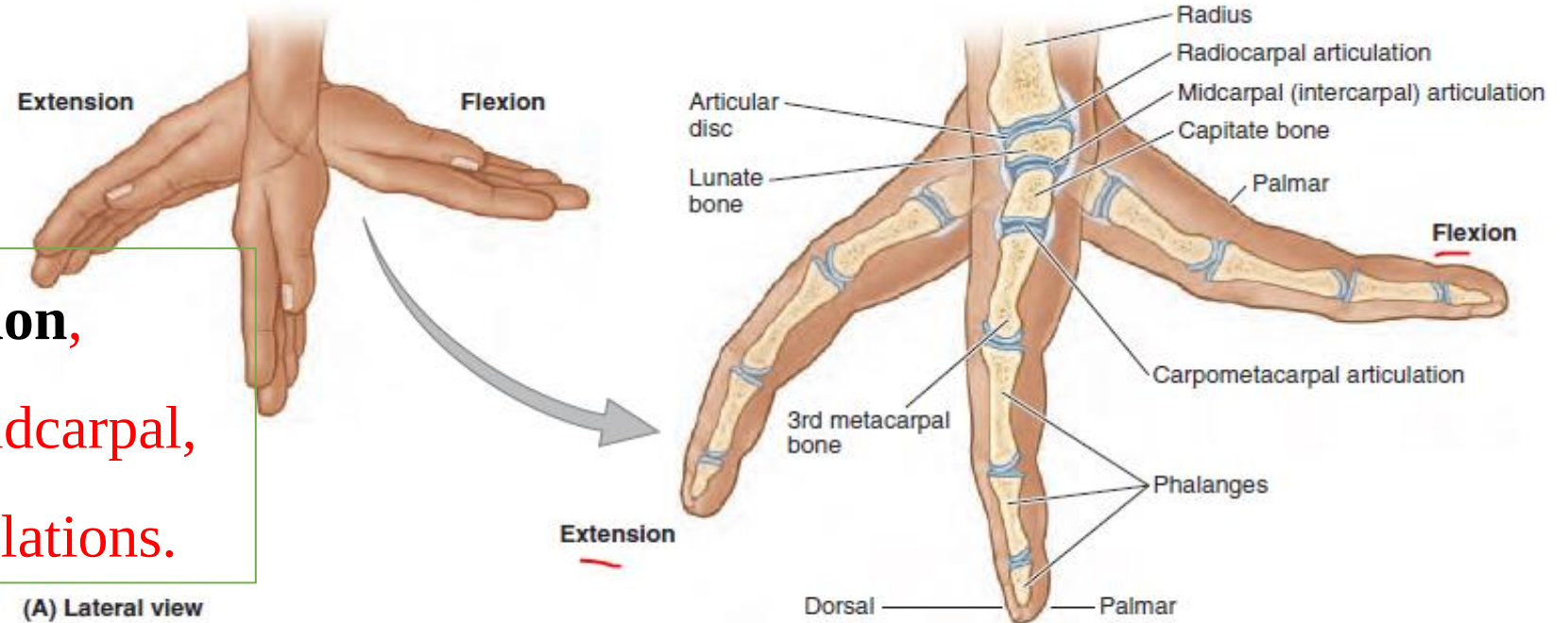


Muscles moving wrist joint

- Movement at the wrist is produced primarily by the “carpi” muscles of the forearm, (carpi ulnaris, flexors & extensors) and (carpi radialis, flexors & extensors).
- *Flexion of the wrist joint* is produced by flexors FCR and FCU which are assisted by flexors of the fingers and thumb, the palmaris longus and the APL.
- *Extension of the wrist joint* is produced by the ECRL, ECRB, and ECU, with assistance from the extensors of the fingers and thumb.
- *Abduction of the wrist joint* is produced by the APL, FCR, ECRL, and ECRB.
- *Adduction of the wrist joint* is produced by simultaneous contraction of the ECU and FCU.

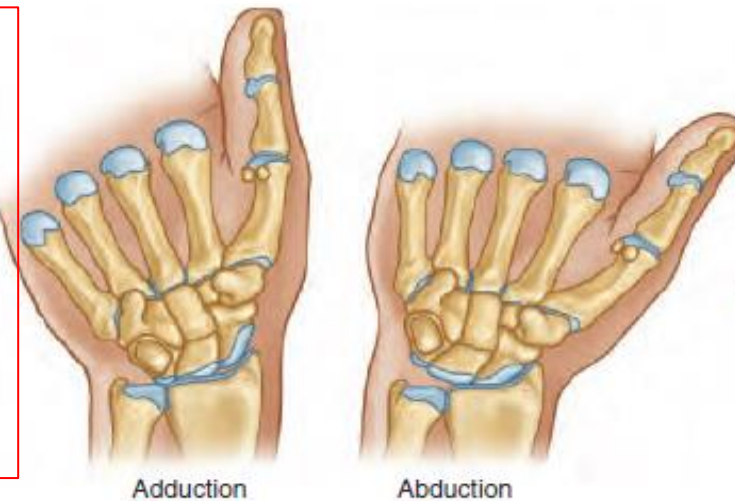
Muscles moving wrist joint,,

- During **extension** and **flexion**, observe the radiocarpal, midcarpal, and carpometacarpal articulations.

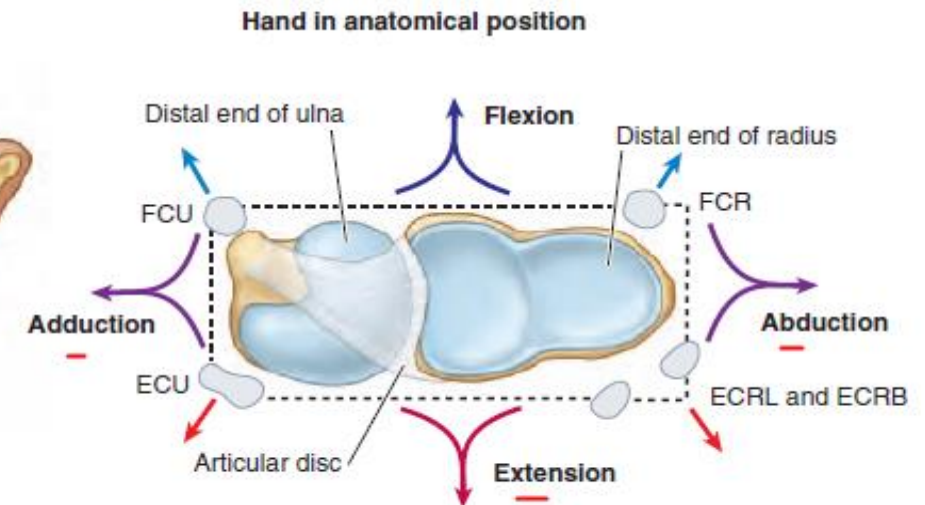


(A) Lateral view

- Movement at the radiocarpal and midcarpal joints during **adduction** and **abduction**.



(B) Anteroposterior view



(C) Inferior view

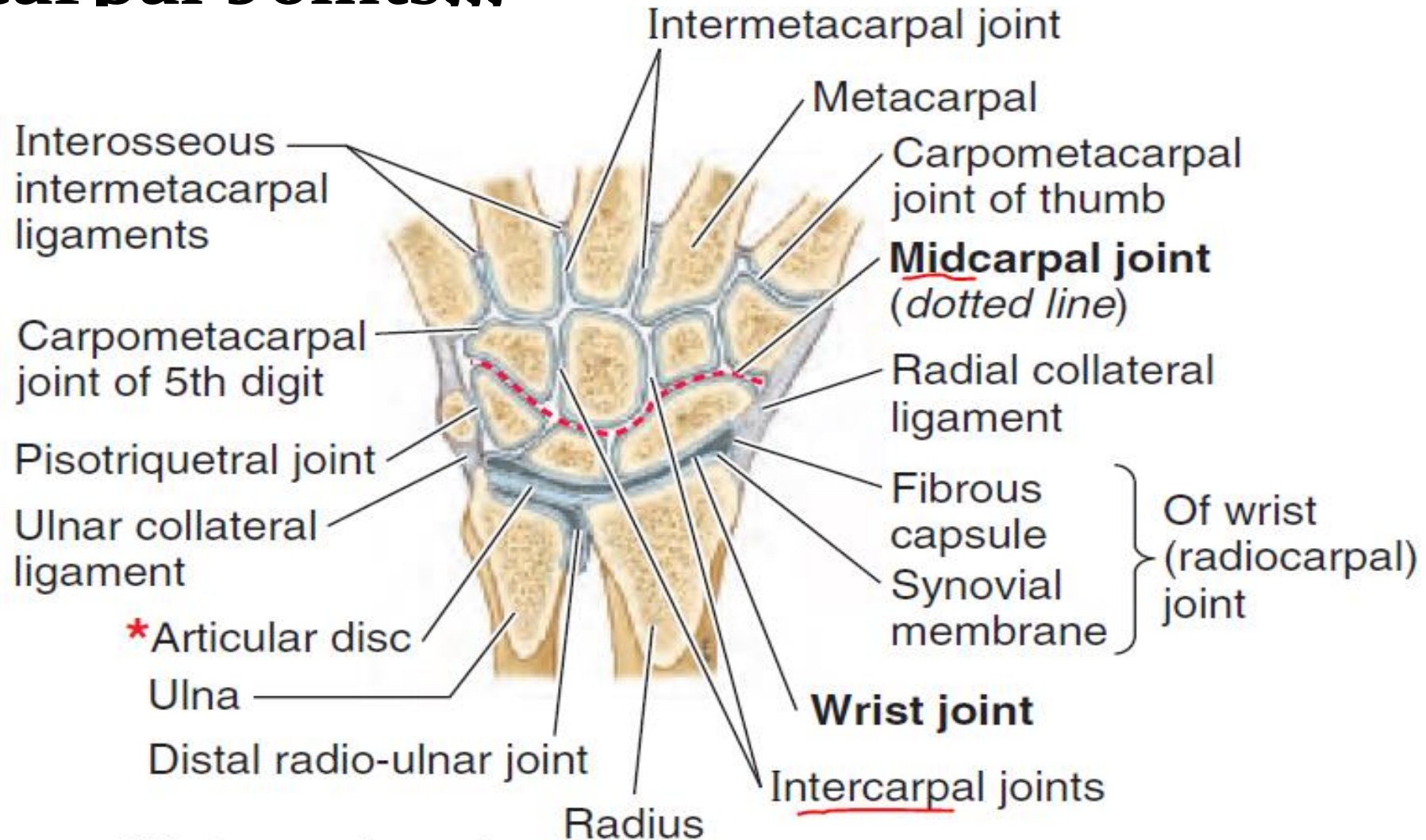
Intercarpal Joints

- The **intercarpal joints**, interconnecting the carpal bones, are plane synovial joints.

❑ **In the intercarpal joints articulation sites can be summarized as follows:**

- ✓ Joints between the carpal bones of the proximal row.
- ✓ Joints between the carpal bones of the distal row.
- ✓ The **midcarpal joint**, a complex joint between the proximal and distal rows of carpal bones.
- ✓ The **pisotriquetral joint**, formed from the articulation of the **pisiform** with the palmar surface of the **triquetrum**.

Intercarpal Joints...



(B) Coronal section

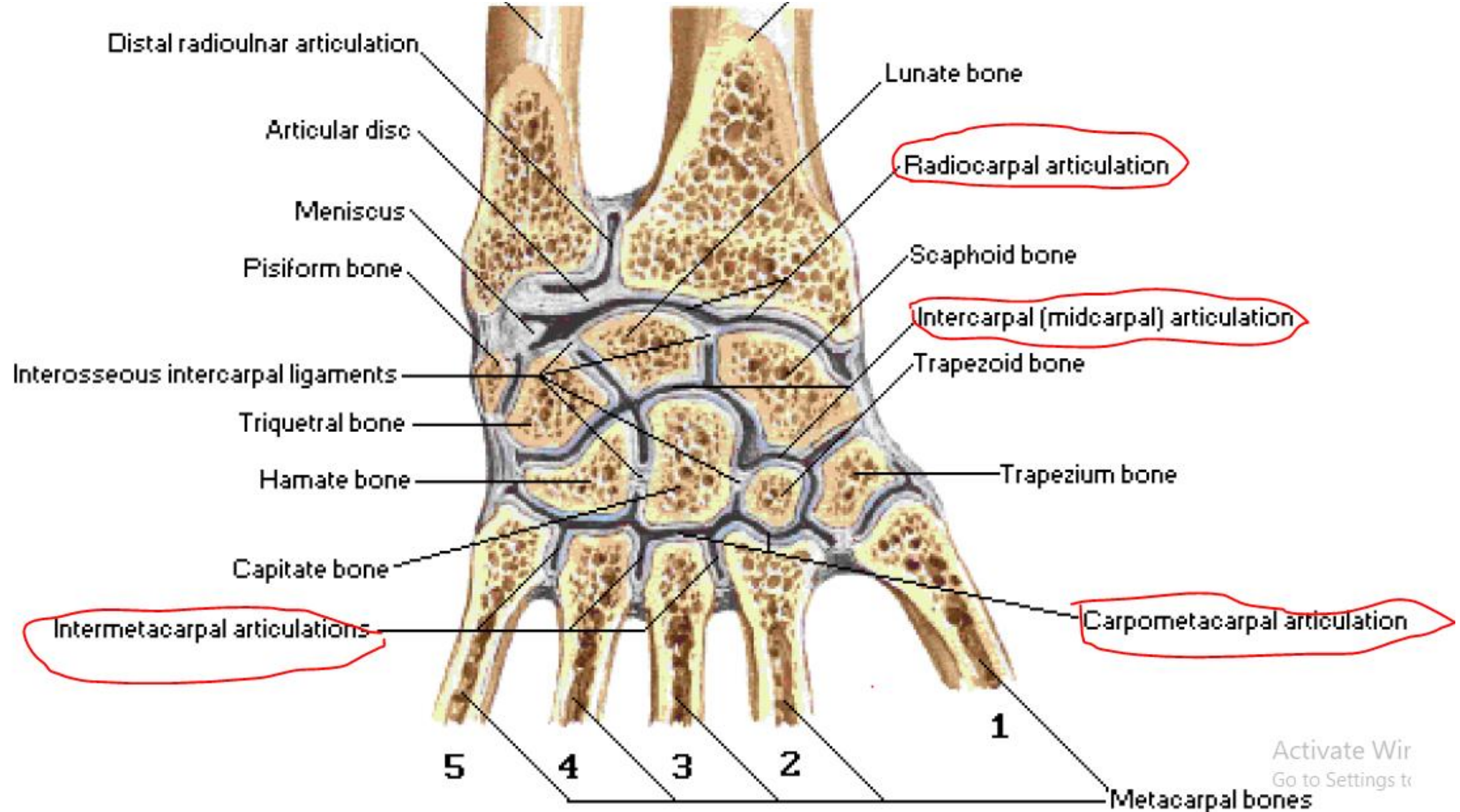


- * of distal radio-ulnar joint

Carpometacarpal and Intermetacarpal Joints

- The **carpometacarpal (CMC)** and **intermetacarpal (IM)** joints are the plane type of **synovial** joint, except for the **CMC joint of the thumb**, which is a **saddle** joint, which is a saddle joint
- The distal surfaces of the carpals of the distal row articulate with the carpal surfaces of the bases of the metacarpals at the CMC joints.
- The important CMC joint of the thumb is between the trapezium and the base of the 1st metacarpal; it has a separate articular cavity.
- Like the carpals, adjacent metacarpals articulate with each other; IM joints occur between the radial and ulnar aspects of the bases of the metacarpals.¹²¹

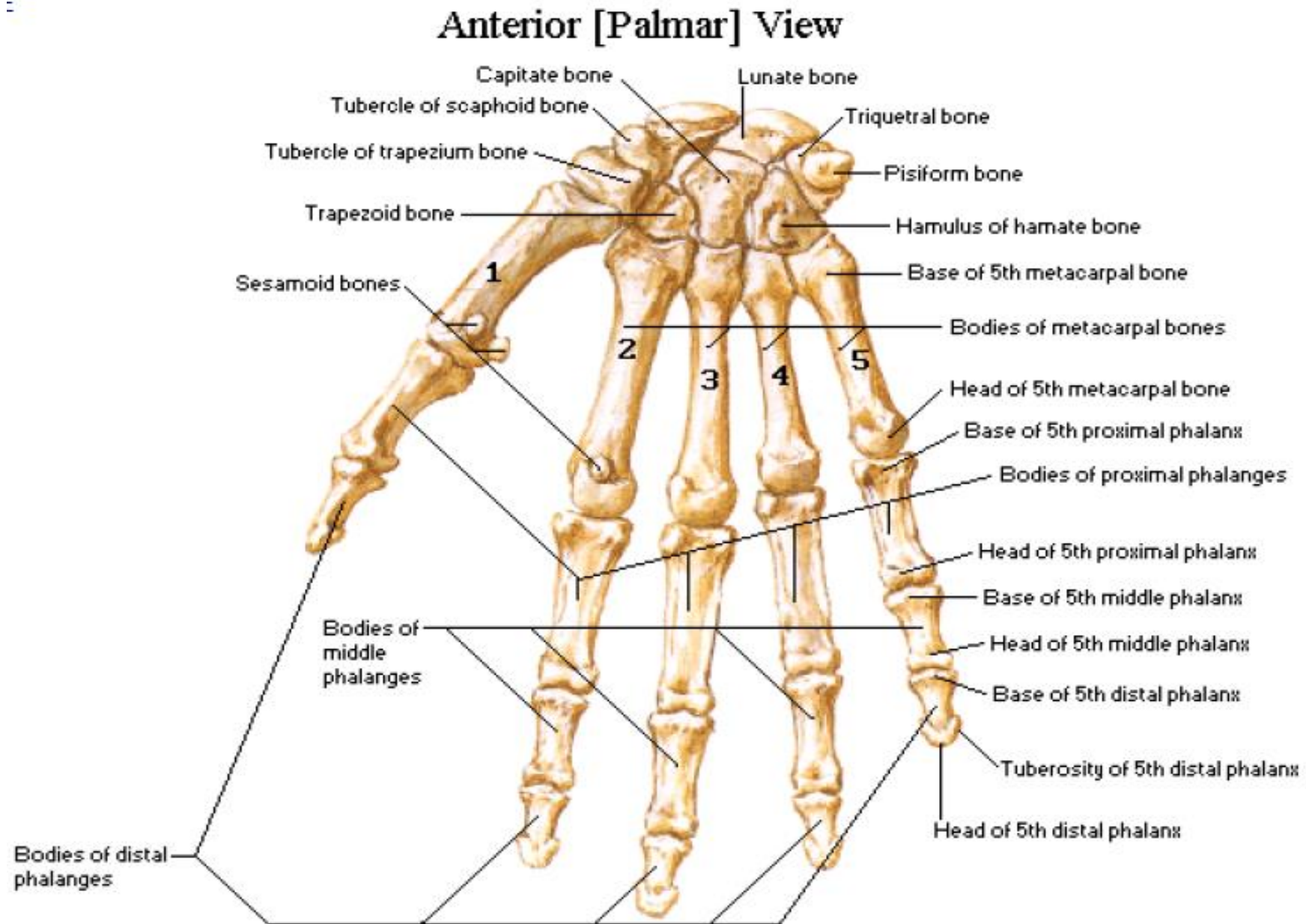
Carpometacarpal and Intermetacarpal Joints,,



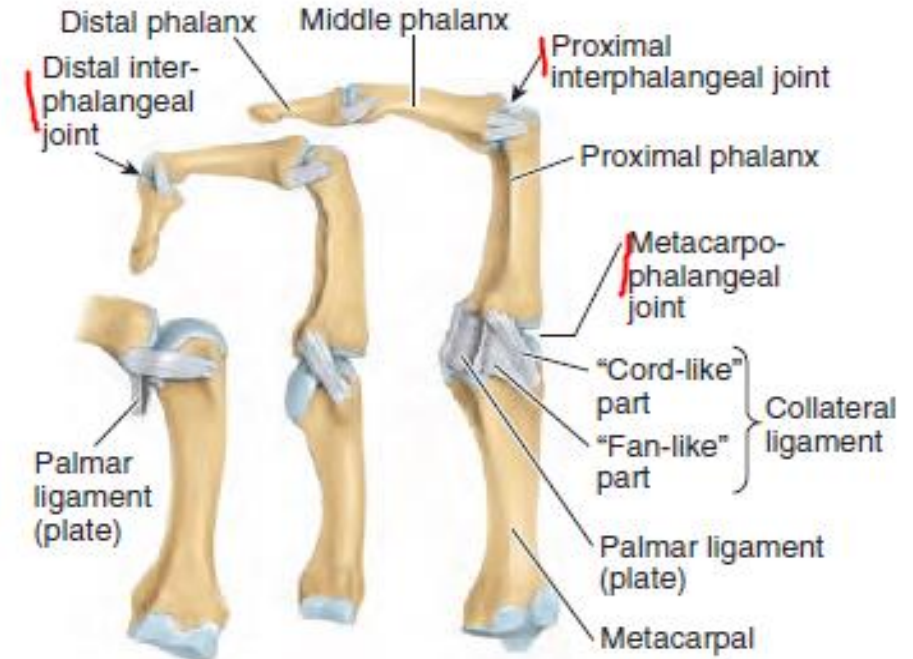
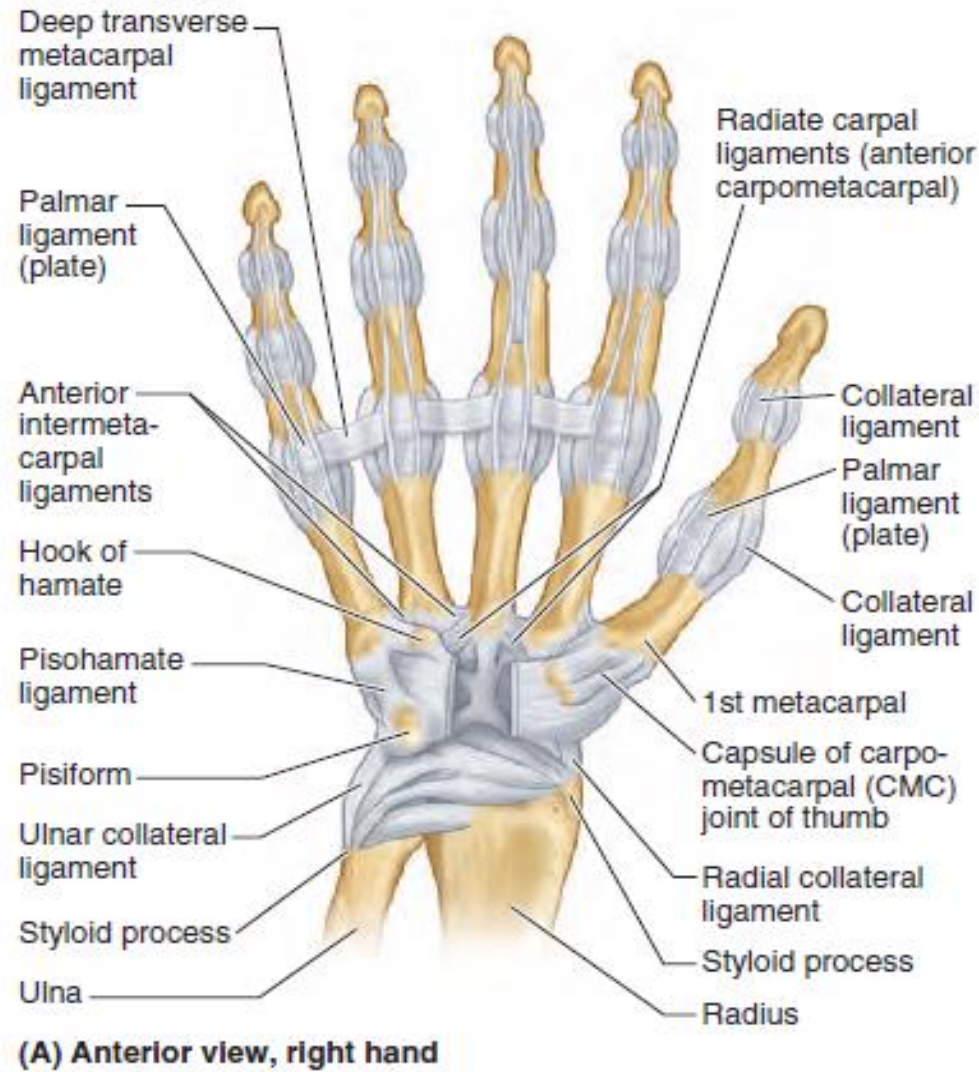
Metacarpophalangeal and Interphalangeal Joints

- The **metacarpophalangeal joints** are the condyloid type of synovial joint that permit movement in two planes: **flexion–extension** and **adduction–abduction**.
- The **interphalangeal joints** are the hinge type of synovial joint that permit **flexion–extension only**.
- The heads of the metacarpals articulate with the bases of the proximal phalanges in the MP joints, and the heads of the phalanges articulate with the bases of more distally located phalanges in the IP joints.

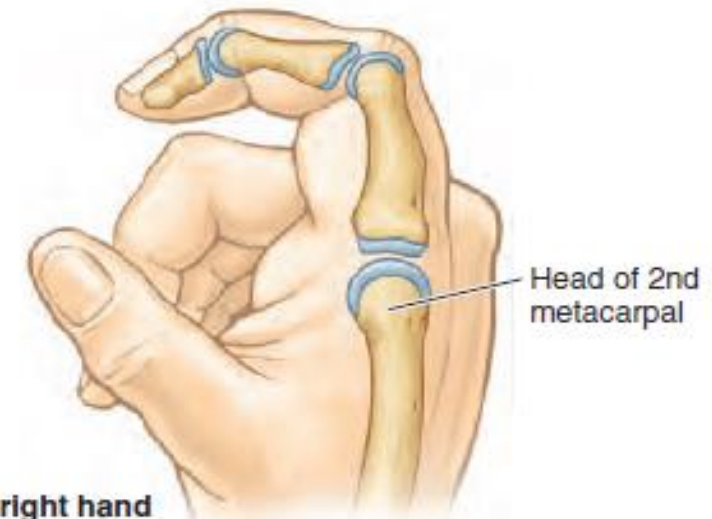
Metacarpophalangeal and Interphalangeal Joints,,,



Metacarpophalangeal and Interphalangeal Joints,,,

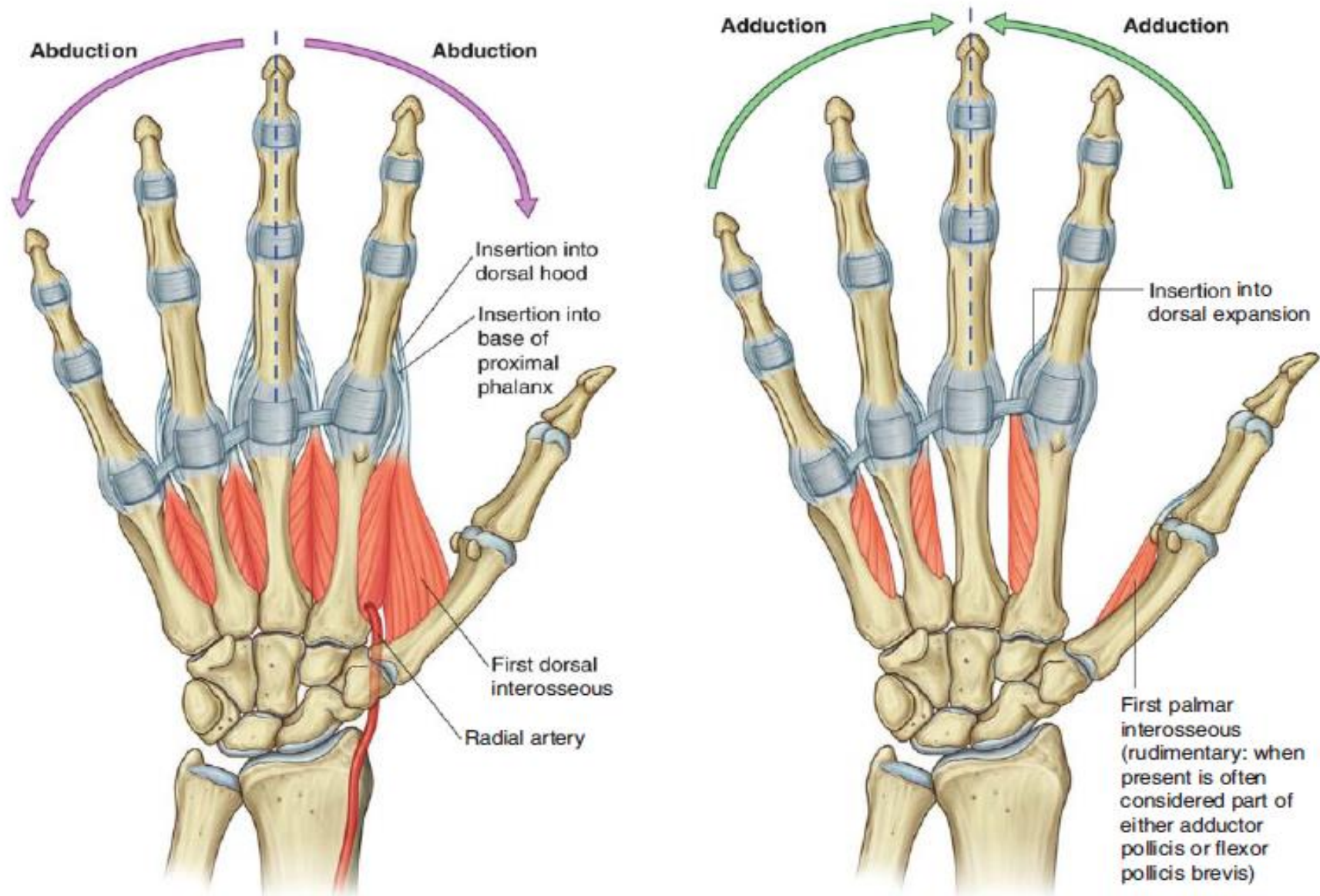


(B) Lateral view, right hand



(C) Lateral view, right hand

Metacarpophalangeal and Interphalangeal Joints,,,





**Thank you
for your attention**